Introduction to Computational Finance and Financial Econometrics with R

Eric Zivot

2026-08-31

Table of contents

Preface

I am a Professor of Economics at the University of Washington (UW) where I teach the class *Introductional to Computational Finance and Financial Econometrics* (ECON 424). This course started as an experiment. In the winter quarter of 1998 I was on leave visiting the economics department at the University of California Los Angeles (UCLA). To help pay for my leave I, I arranged to teach a standard undergraduate econometrics course. My research at the time was moving toward financial econometrics and I thought it would be fun to teach an econometrics course with most of the examples motivated by financial applications. I thought that combining econometrics and statistics topics with financial applications would be a very effective and popular way of getting students interested in learning and applying econometrics. My intuition was right. My students at UCLA loved the version of the course I offered. After returning to the University of Washington, I created the upper division undergraduate course *Introduction to Computational and Financial Econometrics*. The course was an immediate hit. This book evolved from my lecture notes for the course.

Who is the book written for? What background is required?

My ECON 424 course is taken mainly by senior economics majors who are pursuing the more quantitative Bachelor of Science Degree in Economics at UW. These students typically have at least one year of calculus, intermediate microeconomics with some calculus (so they are exposed to optimization under constraints using Lagrange multipliers), and an introductory course in probability and statistics for economics majors. Some students have taken a course in linear algebra, some have taken an introductory econometrics course, and some have taken a financial economics course. I find that the typical introductory probability and statistics course is at a lower level than is needed for much of the material in my course. Ideally, students should have an intermediate probability and statistics course that uses calculus[1]. As a result, I spend a week at the beginning of the course doing a review of random variables and probability that uses basic calculus. A key part of the review is the manipulation of random variables, computing expectations of functions of random variables, etc., so that the students have the necessary tools to work with simple probabilistic models for asset returns. After the review of probability theory I introduce some basic time series concepts (e.g., stochastic processes and stationarity), as these concepts are often not taught in introductory probability and statistics classes, and I also do a quick review of basic matrix algebra that I use throughout the course. The use of matrix algebra is important for two reasons. First, most introductory textbook treatments of portfolio theory show calculations for the simple case of two assets

[1] Insert reference to recent MIT Press book here

and then talk about how the calculations extend to the general case of many assets but do not show how this is done. Hence, students are not capable of implementing the theory for real world portfolios. I emphasize that with knowledge of just a little bit of matrix algebra - essentially matrix addition, multiplication, inversion, and some matrix differential calculus - the general analysis of portfolio theory can be easily learned and implemented. Second, R is a vectorized and matrix programming language and so basic knowledge of matrix algebra allows students to easily understand and implement the R code for portfolio analysis.

Main topics covered in the book

After the review material, I jump into descriptive statistical analysis of historical financial time series data. I emphasize that my approach to building models for asset returns is based on first looking at the historical data and studying its properties. I start by telling students to pretend that observed returns on assets are realizations from some unknown covariance stationary stochastic process. I review common graphical and numerical descriptive statistics (e.g., histograms, QQ-plots, sample mean, variance, correlation, etc.). The goal of descriptive statistical analysis is to compile a collection of "stylized facts" that can be used as building blocks for probabilistic models for asset returns. For example, I show that monthly asset returns are typically uncorrelated over time, have empirical distributions (histograms) that look like normal bell curves, have means close to zero, are positively correlated with other assets, etc. I also emphasize that some "stylized facts" of asset returns appear to stay the same for different investment horizons (daily, weekly, and monthly returns are uncorrelated) and some do not (e.g., daily and weekly returns have fatter tails than the normal and show evidence of nonlinear time dependence in higher moments)

Why us R?

My early versions of *Introduction to Computational and Financial Econometrics* used Microsoft Excel as the main software tool as R was not yet around then. My decision to use Excel was for the students' benefit. Students having experience with Excel was, and still is, in high demand by employers in the finance industry. It is often said, "the financial world runs out of an Excel spreadsheet". However, Excel is not a good tool for doing statistics. Even though you *can* do statistical analysis in Excel, does not mean that you *should* do statistical analysis in Excel. I searched for Excel add-ins to make doing statistical analysis in Excel easier but I did not like what was available. Everything changed when I met Professor Doug Martin from the UW statistics department. Doug founded the company who created S-PLUS, the commercial version of the S statistical programming language developed by John Chambers at Bell Labs (R became the open source version of S in 2000). S is a beautiful language for exploratory data and statistical analysis and S-PLUS made it easily available to students on windows PCs. In the early 2000s I used Excel together with S-PLUS for my course. This hybrid approach worked well but the difficulties of getting student licenses at the beginning of the quarter was always a headache. As R matured in the mid to late 2000s, I made the switch from S-PLUS to R and never looked back. As the course evolved and R became more commonly used in the financial world, and Rstudio became the integrated developer environment of choice for R, I eventually dropped the use of Excel and focused solely on R for the applications and examples in the book.

Many economics students lack experience with programming languages and some find R a difficult language to learn and use. My approach to dealing with this reality is to provide all of the R code to do calculations in the class lectures, in the textbook examples, and on homework assignments. In this way students do not have to worry about the coding aspect of the class and can more easily concentrate on the concepts of computational finance and financial econometrics. I find that students learn to use R quite well this way – learning by doing – and most students are quite proficient in R by the end of the course.

There are a lot of free resources on the web for learning R. In my Coursera course I partnered with DataCamp to provide a web-based learning environment for R tailored to the course material. I have found that this has substantially lowered the learning curve for many students. Student mostly use either a MAC or PC for their work (a few geeks use UNIX). I found that some things didn't always work well across platforms (usually issues with the file system). In 2014, I required students to use Rstudio as the environment for using R. This makes learning and working with R even easier. Another advantage of using Rstudio is that my lecture notes and homework assignments are be created using Rmarkdown.

Longer print book

(Shorter ebook). This book is based on my University of Washington sponsored Coursera course Introduction to Computational Finance and Financial Econometrics that has been running every quarter on Coursera since 2013. This Coursera course is based on the Summer 2013 offering of my University of Washington advanced undergraduate economics course of the same name. At the time, my UW course was part of a three course summer certificate in Fundamentals of Quantitative Finance offered by the Professional Masters Program in Computational Finance & Risk Management that was video-recorded and available for online students. An edited version of this course became the Coursera course. The popularity of the course encouraged me to convert the class notes for the course into a short book.

The book was typeset using Lyx with all of the R code examples embedded into the document and evaluated using knitr. I am extremely grateful to Yihui Xie for writing his wonderful book Dynamics Documents with R and knitr as this book would not have been possible without it. I am a firm believer in the "reproducible research paradigm".

The book's accompanying R package `introcompfinr` contains all of the financial data used for the examples presented in the book as well as a number of R functions written by me for estimating some simple models of asset returns, and doing portfolio and risk analysis. The package is freely available on GitHub at https://github.com/ezivot/introcompfinr and can be installed using:

```r
install.packages("pak")
pak::pak("ezivot/introcompfinr").
```

How is this book different from other introductory book on financial econometrics?

There are now many good books written on the topic of computational/quantitative finance and financial econometrics with examples in R at various levels of sophistication. Books written for advanced U.S. undergraduates include Carmona (2014) and Tsay (2012). Books written for Masters students in computational finance or financial engineering courses include Fan and Yao (20xx), Lai and Xing (2008), Ruppert and Matteson (2015) and Tsay (2010). All of these books assume the readers have a broader mathematical and statistical background than what I assume here. Hence, this book is truly an “introduction” to the methods of computational finance and financial econometrics and is appropriate for undergraduate economics and finance majors at Universities worldwide. Another feature that distinguishes this book from others is the extensive use of R and illustrated R code throughout the book. In this regard, this book is a “hands on” book in which the student can readily execute the R examples on their own computer while reading the book. This allows the student to also learn R as well as the concepts in the book.

Eric Zivot
Seattle, Washington
August 2026

Acknowledgments

Many people have helped with the development of this book. First and foremost, I would like to thank the students at UW who have taken my course over the last fifteen years. I have received countless comments, corrections, and suggestions that have greatly improved the exposition and material in the book.

A lot of how I think about the statistical analysis of financial data has been shaped by many discussions and interactions with my long-term colleague and good friend R. Douglas Martin. Doug introduced me to the S language and I had the pleasure of working with him at Insightful where we developed the S+FinMetrics module for S-PLUS and I wrote the book *Modeling Financial Time Series with S-PLUS* with my former Ph.D. student Jiahui (Jeffrey) Wang.

My very patient wife Nina Sidneva carefully proof-read many early versions of the chapters as they were under development and caught many of my stupid errors. Xiao Yin proof-read most of the chapters after I converted my Latex notes to Lyx and incorporated the R examples with `knitr`. I could not have written the book without the expert technical help from Bethany Yollin and the wonderful book *Dynamics Documents with R and knitr* by Yihui Xie. Bethany helped with the conversion of my Latex notes to Lyx and the incorporation of the R examples in the text using `knitr`. She also helped to develop and maintain an early version the book's R package, `IntroCompfinR`. Any problem I found with Lyx and `knitr` was solved by referring to Yihui's book. In 2020 I decided to migrate the book from Lyx to Rmarkdown so that everything could be managed within Rstudio and the book could be generated using `bookdown`. My UW PhD RAs Yumming (Herbert) Liu, Pushpak Sarkar and Kovid Puria made this happen. In 2026, I moved from `bookdown` to Quarto.

My interactions with the R community, especially those involved with R/Finance, have been invaluable for learning and using R for the analysis of financial data. In particular, Brian Peterson and Peter Carl have been incredibly helpful and supportive of my book projects. I use their wonderful R package `PerformanceAnalytics` throughout the book. Working with time series data is made so much easier with Achiem Zieles' `zoo` package and Jeff Ryan's `xts` package. Many University of Washington (UW) econ PhD students (Brian Donhauser, Ming He, Wan-Jung Hsu, Kara Ng, Anthony Sanford, Galib Sultan, Cindy Huang) acted as TAs for my undergraduate course and gave constructive feedback. I also thank Coursera for helping me convert my undergraduate course into Coursera format.

1 Return Calculations

Updated: August 30, 2026

Copyright © Eric Zivot 2022, 2023, 2025, 2026

In this chapter I cover asset return calculations with an emphasis on equity returns. It is important to understand the different ways in which asset returns are calculated because most of the models presented in the book are for asset returns. Simple returns are most commonly used for money and portfolio calculations in practice. However, simple returns have some undesirable properties that make mathematical and statistical modeling difficult. Continuously compounded returns, in contrast, have nicer properties that make mathematical and statistical modeling easier.

This chapter is organized as follows. Section Section 1.1 covers basic time value of money calculations. Section Section 1.2 covers asset return calculations, including both simple and continuously compounded returns. Section Section 1.3 defines portfolios and details portfolio return calculations. Section Section 1.4 illustrates asset return calculations using R.

In this chapter I use the following R packages: **ggplot2**, **introcompfinr**, **methods**, **PerformanceAnalytics**, **quantmod**, **xts**, and **zoo**. Make sure these packages are installed and loaded before replicating the chapter examples.

1.1 The Time Value of Money

This section reviews basic time value of money calculations. The concepts of future value, present value and the compounding of interest are defined and discussed.

1.1.1 Future value, present value and simple interest

Consider an amount $\$V$ invested for n years at a *simple interest rate* of R per annum (where R is expressed as a decimal). If compounding takes place only at the end of the year, the *future value* after n years is:

$$FV_n = \$V(1+R) \times \cdots \times (1+R) = \$V(1+R)^n. \tag{1.1}$$

Over the first year, $\$V$ grows to $\$V(1+R) = \$V + \$V \times R$ which represents the initial principal $\$V$ plus the payment of simple interest $\$V \times R$ for the year. Over the second year, the new principal $\$V(1+R)$ grows to $\$V(1+R)(1+R) = \$V(1+R)^2$, and so on.

Example 1.1 (Future value with simple interest)**.** Consider putting \$1000 in an interest checking account that pays a simple annual percentage rate of 3%. The future value after $n = 1, 5$ and 10 years is, respectively,

$$\begin{aligned} FV_1 &= \$1000 \cdot (1.03)^1 = \$1030, \\ FV_5 &= \$1000 \cdot (1.03)^5 = \$1159.27, \\ FV_{10} &= \$1000 \cdot (1.03)^{10} = \$1343.92. \end{aligned}$$

Over the first year, \$30 in interest is accrued; over three years, \$159.27 in interest is accrued; over five years, \$343.92 in interest is accrued. These calculations are easy to perform in R.

```
V = 1000
R = 0.03
FV1 = V*(1 + R)
FV5 = V*(1 + R)^5
FV10 = V*(1 + R)^10
c(FV1, FV5, FV10)
#> [1] 1030 1159 1344
```

■

The future value formula Equation 1.1 defines a relationship between four variables: FV_n, V, R and n. Given three variables, the fourth variable can be determined. For example, given FV_n, R and n and solving for V gives the *present value* formula:

$$V = \frac{FV_n}{(1+R)^n}. \tag{1.2}$$

Here, V represents the present value (or current value) of the future value FV_n that is to be received n years from today. The future value is discounted by the factor $(1+R)^{-n}$ to reflect the fact that dollars to be received in n years are worth less than current dollars because current dollars can be invested today returning R per year. The discount factor $(1+R)^{-n}$ is less than 1, and represents the price today of \$1 to be received in n years when the interest rate is R per year.

Taking FV_n, n and V as given, the *annual interest rate* on the investment is defined as:

$$R = \left(\frac{FV_n}{V}\right)^{1/n} - 1. \tag{1.3}$$

Here, R represents a type of average rate of return on an investment known as the *compound annual return*.

Finally, given FV_n, V and R we can solve for n:

$$n = \frac{ln(FV_n/V)}{ln(1+R)}. \tag{1.4}$$

The expression Equation 1.4 is called the *horizon period* and can be used to determine the number of years it takes for an investment of \$$V$ to double. Setting $FV_n = 2V$ in Equation 1.4 gives:

$$n = \frac{\ln(2)}{\ln(1+R)} \approx \frac{0.7}{R},$$

which uses the approximations $\ln(2) = 0.6931 \approx 0.7$ and $\ln(1+R) \approx R$ for R close to zero. The approximation $n \approx 0.7/R$ is called the *rule of 70*.

Example 1.2 (Using the rule of 70)**.** The rule of 70 gives a good approximation as long as the interest rate is not too high.

```
R = seq(0.01, 0.10, by=0.01)
nExact = log(2)/log(1 + R)
nRule70 = 0.7/R
cbind(R, nExact, nRule70)
#>          R nExact nRule70
#>  [1,] 0.01  69.66   70.00
#>  [2,] 0.02  35.00   35.00
#>  [3,] 0.03  23.45   23.33
#>  [4,] 0.04  17.67   17.50
#>  [5,] 0.05  14.21   14.00
#>  [6,] 0.06  11.90   11.67
#>  [7,] 0.07  10.24   10.00
#>  [8,] 0.08   9.01    8.75
#>  [9,] 0.09   8.04    7.78
#> [10,] 0.10   7.27    7.00
```

For R values between 1% and 10% the approximation error in the rule of 70 is never more than half a year.

■

1.1.2 Multiple compounding periods

If interest is paid m times per year, then the future value after n years is:

$$FV_n^m = \$V \cdot \left(1 + \frac{R}{m}\right)^{m \cdot n}.$$

R/m is often referred to as the *periodic interest rate* . As m, the frequency of compounding, increases, the rate becomes continuously compounded and it can be shown that the future value becomes:

$$FV_n^c = \lim_{m \to \infty} \$V \cdot \left(1 + \frac{R}{m}\right)^{m \cdot n} = \$V \cdot e^{R \cdot n},$$

where $e^{(\cdot)}$ is the exponential function and $e^1 = 2.71828$.

Example 1.3 (Future value with different compounding frequencies)**.** For a simple annual percentage rate of 10%, the value of $1000 at the end of one year ($n = 1$) for different values of m is calculated below.

```
V = 1000
R = 0.1
m = c(1, 2, 4, 356, 10000)
FV = V*(1 + R/m)^(m)
print(cbind(m, FV), digits=7)
#>           m       FV
#> [1,]      1 1100.000
#> [2,]      2 1102.500
#> [3,]      4 1103.813
#> [4,]    356 1105.155
#> [5,] 10000 1105.170
```

The result with continuous compounding is

```
print(V*exp(R), digits=7)
#> [1] 1105.171
```

■

The continuously compounded return analogues to the present value, annual return and horizon period formulas Equation 1.2, Equation 1.3 and Equation 1.4 are:

$$V = e^{-Rn} FV_n,$$
$$R = \frac{1}{n} \ln\left(\frac{FV_n}{V}\right),$$
$$n = \frac{1}{R} \ln\left(\frac{FV_n}{V}\right).$$

1.1.3 Effective annual rate

We now consider the relationship between simple interest rates, periodic rates, effective annual rates and continuously compounded rates. Suppose an investment pays a periodic interest rate of 2% each quarter. This gives rise to a simple annual rate of 8% (2% $\times$ 4 quarters). At the end of the year, \$1000 invested accrues to:

$$\$1000 \cdot \left(1 + \frac{0.08}{4}\right)^{4 \cdot 1} = \$1082.40.$$

The *effective annual rate*, R_A, on the investment is determined by the relationship:

$$\$1000 \cdot (1 + R_A) = \$1082.40.$$

Solving for R_A gives:

$$R_A = \frac{\$1082.40}{\$1000} - 1 = 0.0824,$$

or $R_A = 8.24\%$. Here, the effective annual rate is the simple interest rate with annual compounding that gives the same future value that occurs with simple interest compounded four times per year. The effective annual rate is greater than the simple annual rate due to the payment of interest on interest.

The general relationship between the simple annual rate R with payments m times per year and the effective annual rate, R_A, is:

$$(1 + R_A) = \left(1 + \frac{R}{m}\right)^m.$$

Given the simple rate R, we can solve for the effective annual rate using:

$$R_A = \left(1 + \frac{R}{m}\right)^m - 1. \tag{1.5}$$

Given the effective annual rate R_A, we can solve for the simple rate using:

$$R = m\left[(1 + R_A)^{1/m} - 1\right].$$

The relationship between the effective annual rate and the simple rate that is compounded continuously is:

$$(1 + R_A) = e^R.$$

Hence,

$$\begin{aligned} R_A &= e^R - 1, \\ R &= \ln(1 + R_A). \end{aligned}$$

Example 1.4 (Determine effective annual rates)**.** The effective annual rates associated with the investments in example Example 1.2 are calculated below.

```r
RA = FV/V - 1
print(cbind(m, FV, RA), digits=7)
#>          m       FV        RA
#> [1,]     1 1100.000 0.1000000
#> [2,]     2 1102.500 0.1025000
#> [3,]     4 1103.813 0.1038129
#> [4,]   356 1105.155 0.1051554
#> [5,] 10000 1105.170 0.1051704
```

The effective annual rate with continuous compounding is

```r
print(exp(R) - 1, digits=7)
#> [1] 0.1051709
```

■

Example 1.5 (Determine continuously compounded rate from effective annual rate)**.** Suppose an investment pays a periodic interest rate of 5% every six months ($m = 2, R/2 = 0.05$). In the market this would be quoted as having an annual percentage rate, R_A, of 10%. An investment of \$100 yields $\$100 \cdot (1.05)^2 = \$110.25$ after one year. The effective annual rate, R_A, is then 10.25%. To find the continuously compounded simple rate that gives the same future value as investing at the effective annual rate we solve: $R = \ln(1.1025) = 0.09758$. That is, if interest is compounded continuously at a simple annual rate of 9.758% then \$100 invested today would grow to $\$100 \cdot e^{0.09758} = \$110.25$

■

1.2 Asset Return Calculations

In this section, we review asset return calculations given initial and future prices associated with an investment. We first summarize the main type of assets used for investment purposes. Next, we cover simple return calculations, which are typically reported in practice but are often not convenient for statistical modeling purposes. We then describe continuously compounded return calculations, which are more convenient for statistical modeling purposes. We finish the section with a discussion of portfolio return calculations.

1.2.1 Assets

When we talk about assets in this book, we mean the typical investment assets available for purchase to individuals through brokerage and retirement accounts. These include bonds, stocks, mutual funds, exchange traded funds and notes, and derivative securities. Investment assets can be purchased through online brokerage accounts at firms like Ally Invest, Charles Schwab, E*TRADE, Fidelity, Interactive Brokers, Merrill Lynch, Robinhood, TD Ameritrade, and Vanguard. Due to the growth and competition of online investment accounts, the cost of purchasing investment assets has dropped significantly over time.

Bonds

Bonds are assets that represent debt obligations from corporations or the government used to raise capital (money) to finance operations. A typical bond is characterized by (1) a principal amount, representing initial capital borrowed; (2) maturity date, representing when the principal must be repaid, (3) periodic coupon payments, representing interest paid on specific dates. Bonds issued by the U.S. government are called *Treasury bonds*, and are generally thought to be the safest bonds as it is very unlikely that the U.S. government will default on its debt obligations. Other government agencies as well as states and municipalities issue bonds to raise money. Corporate bonds are bonds issued by private and public corporations. Bonds other than Treasury bonds have the risk, called *default risk*, that the principal or coupon payments many not be made.

Stocks

A *stock* represents an ownership share in corporation. Public stock represents ownership in a publicly traded corporation, where the shares can be purchased or sold (traded) by the public in organized stock exchanges (markets). In the U.S. the main stock exchanges are the New York Stock Exchange (NYSE) and the National Association of Securities Dealers Automated Quotations (NASDAQ) exchange. Private stock represents ownership in a private corporation whose stock is not traded in an exchange. Private corporations become public companies when they decide to raise money (capital) by issuing new stock in a public exchange through an *initial public offering* (IPO). This new stock purchased by individuals can then be sold to others in the exchange. Ownership of stock often provides voting rights which allows stock holders to have a say in the management of the company. Owners of stock have a right to receive dividends, which are cash payments representing corporate profits. Corporations may or many not pay dividends. Typically, young growing companies reinvest their profits instead of paying dividends. The stocks of these companies are called *growth stocks*. More established companies more often pay dividends instead of reinvesting all profits.

Mutual Funds

A *mutual fund* is a professionally managed investment fund that pools money from many investors to purchase a collection of assets called a *portfolio*. Mutual funds typically invest in bonds, stocks, and money market instruments. Mutual funds are commonly used in *individual retirement accounts* (IRAs) and employee sponsored retirement accounts. Because mutual funds are managed portfolios they charge fees for this service. The amount of the fee depends on the activity of the portfolio manager. Actively managed funds, where the portfolio manager actively picks the assets in the portfolio, have the highest fees. Actively managed mutual funds are offered by many firms such as the Capital Group, Fidelity, Janice, Putnam, and T. Rowe

Price. Passively managed funds have lower fees because the managers follow a rule to match a particular asset index such as the S&P 500 index. Passive mutual funds were pioneered by Vandguard. Mutual funds are traded on exchanges but are only traded at the end of each day.

Exchange Traded Funds and Notes

An *exchange traded fund* (ETF) is similar in many ways to a mutual fund, except that an ETF can be bought and sold throughout the day on stock exchanges. Most ETFs are index funds, that is, they hold the same securities in the same proportions as a certain stock market index or bond market index. The most popular ETFs in the U.S. replicate the S&P 500 Index, the total market index, the NASDAQ-100 index, the price of gold, the "growth" stocks in the Russell 1000 Index, or the index of the largest technology companies. ETNs are similar to ETFs. The main companies issuing ETFs and ETNs are Barclays Capital Inc. (iPath), BlackRock (iShares), Credit Swisse, Direxion, StateStreet (SPDR), Invesco (PowerShares), and ProShares. Each of these companies provide online information on their ETFs and ETNs. You can easily find ETFs and ETNs for particular asset categories and investment strategies at www.etf.com.

Derivative Securities

A *derivative security* is an asset whose value is derived from the behavior of other underlying assets such as bonds, stocks, ETFs, etc. Common derivative securities are forwards, futures, options and swaps. We will not consider derivative securities is this book.

Information and historical price data on many publicly traded assets is freely available at https://finance.yahoo.com/.

1.2.2 Simple returns

Consider purchasing one share of an asset (e.g., stock, bond, ETF, mutual fund, option, etc.) at time t_0 for the price P_{t_0}, and then selling that share at time t_1 for the price P_{t_1}. If there are no intermediate cash flows (e.g., dividends) between t_0 and t_1, the rate of return over the period t_0 to t_1 is the percentage change in price:

$$R(t_0, t_1) = \frac{P_{t_1} - P_{t_0}}{P_{t_0}}. \tag{1.6}$$

The time between t_0 and t_1 is called the *holding period* and equation Equation 1.6 is called the *holding period return*. In principle, the holding period can be any amount of time: one second, five minutes, eight hours, two days six minutes and two seconds, fifteen years, etc.. To simplify matters, in this chapter we will assume that the holding period is some increment of calendar time; e.g., one day, one month or one year. In particular, we will assume a default holding period of one month in what follows.

Let P_t denote the price at the end of month t of an asset that pays no dividends and let P_{t-1} denote the price at the end of month $t-1$. Then the one-month *simple net return* on an investment in the asset between months $t-1$ and t is defined as:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \Delta P_t. \tag{1.7}$$

Writing $(P_t - P_{t-1})/P_{t-1} = P_t/P_{t-1} - 1$, we can define the *simple gross return* as:

$$1 + R_t = \frac{P_t}{P_{t-1}}. \tag{1.8}$$

The one-month gross return has the interpretation of the future value of \$1 invested in the asset for one-month with simple interest rate R_t. Solving Equation 1.8 for P_t gives

$$P_t = P_{t-1}(1 + R_t),$$

which shows P_t can be thought of as the future value of P_{t-1} invested for one month with simple return R_t. Unless otherwise stated, when we refer to returns we mean net returns. Since asset prices must always be non-negative (a long position in an asset is a limited liability investment), the smallest value for R_t is -1 or -100%.

Example 1.6 (Simple return calculation)**.** Consider a one-month investment in Microsoft stock. Suppose you buy the stock in month $t-1$ (e.g., end of February) at $P_{t-1} = \$85$ and sell the stock the next month (e.g., end of March) for $P_t = \$90$. Further assume that Microsoft does not pay a dividend between months $t-1$ and t. The one-month simple net and gross returns (over the month of March) are then:

$$\begin{aligned} R_t &= \frac{\$90 - \$85}{\$85} = \frac{\$90}{\$85} - 1 = 1.0588 - 1 = 0.0588, \\ 1 + R_t &= 1.0588. \end{aligned}$$

The one-month investment in Microsoft yielded a 5.88% per month return. Alternatively, \$1 invested in Microsoft stock in month $t-1$ grew to \$1.0588 in month t. The calculations using R are:

```
P0 = 90
Pm1 = 85
R0 = (P0 - Pm1)/Pm1
c(R0,1 + R0)
#> [1] 0.0588 1.0588
```

■

The return calculation works the same way if multiple shares of an asset are initially purchased. Let $s \geq 1$ denote the number of shares of an asset purchased at time $t-1$. The initial value of the investment is then $V_{t-1} = s \times P_{t-1}$. The value of the investment at time t is $V_t = s \times P_t$. The rate of return on the investment is the percentage change in value over the holding period:

$$R_t = \frac{V_t - V_{t-1}}{V_{t-1}} = \frac{s \times P_t - s \times P_{t-1}}{s \times P_{t-1}} \tag{1.9}$$

$$= \frac{s \times (P_t - P_{t-1})}{s \times P_{t-1}} = \frac{P_t - P_{t-1}}{P_{t-1}}. \tag{1.10}$$

As a result, the future value of the investment is equal to the current value times the gross return:

$$V_t = V_{t-1} \times (1 + R_t)$$

1.2.2.1 Multi-period returns

The simple two-month return on an investment in an asset between months $t-2$ and t is defined as:

$$R_t(2) = \frac{P_t - P_{t-2}}{P_{t-2}} = \frac{P_t}{P_{t-2}} - 1.$$

Writing $P_t/P_{t-2} = (P_t/P_{t-1}) \cdot (P_{t-1}/P_{t-2})$ the two-month return can be expressed as:

$$\begin{aligned} R_t(2) &= \frac{P_t}{P_{t-1}} \cdot \frac{P_{t-1}}{P_{t-2}} - 1 \\ &= (1 + R_t)(1 + R_{t-1}) - 1. \end{aligned}$$

Then the simple two-month gross return becomes:

$$1 + R_t(2) = (1 + R_t)(1 + R_{t-1}) = 1 + R_{t-1} + R_t + R_{t-1}R_t,$$

which is the product of the two simple one-month gross returns, called the *geometric average*, and not one plus the sum of the two one-month returns. Hence, the two-month net return is

$$R_t(2) = R_{t-1} + R_t + R_{t-1}R_t.$$

If, however, R_{t-1} and R_t are close to zero then $R_{t-1}R_t \approx 0$ and $1 + R_t(2) \approx 1 + R_{t-1} + R_t$ so that $R_t(2) \approx R_{t-1} + R_t$.

Example 1.7 (Computing two-period returns)**.** Continuing with the previous example, suppose that the price of Microsoft stock in month $t-2$ is \$80 and no dividend is paid between months $t-2$ and t. The two-month net return is:

$$R_t(2) = \frac{\$90 - \$80}{\$80} = \frac{\$90}{\$80} - 1 = 1.1250 - 1 = 0.1250,$$

or 12.50% per two months. The two one-month returns are:

$$R_{t-1} = \frac{\$85 - \$80}{\$80} = 1.0625 - 1 = 0.0625,$$
$$R_t = \frac{\$90 - 85}{\$85} = 1.0588 - 1 = 0.0588,$$

and the geometric average of the two one-month gross returns is: $1+R_t(2) = 1.0625 \times 1.0588 = 1.1250$.

The calculations can be *vectorized* in R as follows:

```
Pm2 = 80
# create vector of prices
P = c(Pm2, Pm1, P0)
# calculate vector of 1-month returns from vector of prices
R = (P[2:3] - P[1:2])/P[1:2]
R
#> [1] 0.0625 0.0588
# calculate 2-month return from 2 1-month returns
(1 + R[1])*(1 + R[2]) - 1
#> [1] 0.125
# same calculation vectorized using R function cumprod()
cumprod(1+R) - 1
#> [1] 0.0625 0.1250
```

In the above, `P[2:3]` represents the 2nd and 3rd elements of `P`, and `P[1:2]` represents the 1st and 2nd elements of `P`.

■

It is important to recognize that two-period simple returns are not symmetric in the following sense. Suppose $R_{t-1} = -0.5$ and $R_t = 0.5$ and initial wealth is \$1. After two-periods \$1 falls to

$$\$1 \times (1 + R_{t-1})(1 + R_t) = \$1 \times (1 - 0.5)(1 + 0.5) = \$0.75$$

That is, a 50% drop followed by a 50% gain does not return wealth to its initial level. In the second period, the return has to be 100% in order for wealth to return to its initial level.

In general, the k-month gross return is defined as the product of k one-month gross returns:

$$\begin{aligned} 1 + R_t(k) &= (1 + R_t)(1 + R_{t-1}) \cdots (1 + R_{t-k+1}) \\ &= \prod_{j=0}^{k-1} (1 + R_{t-j}). \end{aligned} \tag{1.11}$$

If all returns are close to zero then

$$R_t(k) \approx \sum_{j=0}^{k-1} R_{t-j}$$

As with the one-month return, the multi-period return can also be derived in the same way if multiple shares of an asset are initially purchased. Using $V_{t-j} = s \times P_{t-j}$ we have

$$\frac{V_t}{V_{t-k}} = \frac{V_t}{V_{t-1}} \times \frac{V_{t-1}}{V_{t-2}} \cdots \frac{V_{t-k+1}}{V_{t-k}} \tag{1.12}$$

$$= \frac{s \times P_t}{s \times P_{t-1}} \times \frac{s \times P_{t-1}}{s \times P_{t-2}} \cdots \frac{s \times P_{t-k+1}}{s \times P_{t-k}} \tag{1.13}$$

$$= (1 + R_t)(1 + R_{t-1}) \cdots (1 + R_{t-k+1}). \tag{1.14}$$

As a result, V_t can be interpreted as the future value of V_{t-k} based on the returns $R_{t-k+1}, \cdots, R_t$:

$$V_t = V_{t-k} \times (1 + R_{t-k+1})(1 + R_{t-k+2}) \cdots (1 + R_t).$$

1.2.2.2 Adjusting for dividends

If an asset pays a *dividend* (or any cash payment), D_t, sometime between months $t-1$ and t, the *total net return* calculation becomes:

$$R_t^{\text{total}} = \frac{P_t + D_t - P_{t-1}}{P_{t-1}} = \frac{P_t - P_{t-1}}{P_{t-1}} + \frac{D_t}{P_{t-1}}, \tag{1.15}$$

where $(P_t - P_{t-1})/P_{t-1}$ is referred as the *capital gain* and D_t/P_{t-1} is referred to as the *dividend yield*. The *total gross return* is:

$$1 + R_t^{\text{total}} = \frac{P_t + D_t}{P_{t-1}}. \tag{1.16}$$

The formula Equation 1.11 for computing multi-period return remains the same except that one-period gross returns are computed using Equation 1.16.

High dividend paying stocks are often called *income stocks* and have characteristics similar to bonds where the regular dividends play a similar role as coupon interest payments.

Example 1.8 (Compute total return when dividends are paid)**.** Consider a one-month investment in Microsoft stock. Suppose you buy the stock in month $t-1$ at $P_{t-1} = \$85$ and sell the stock the next month for $P_t = \$90$. Further assume that Microsoft pays a \$1 dividend between months $t-1$ and t. The capital gain, dividend yield and total return are then:

$$\begin{aligned} R_t &= \frac{\$90 + \$1 - \$85}{\$85} = \frac{\$90 - \$85}{\$85} + \frac{\$1}{\$85} \\ &= 0.0588 + 0.0118 \\ &= 0.0707. \end{aligned}$$

The one-month investment in Microsoft yields a 7.07% per month total return. The capital gain component is 5.88% and the dividend yield component is 1.18%

■

1.2.2.3 Adjusting for inflation

The return calculations considered so far are based on the *nominal* or current prices of assets. Returns computed from nominal prices are nominal returns. The *real return* on an asset over a particular horizon takes into account the growth rate of the general price level over the horizon. If the nominal price of the asset grows faster than the general price level, then the nominal return will be greater than the inflation rate and the real return will be positive. Conversely, if the nominal price of the asset increases less than the general price level, then the nominal return will be less than the inflation rate and the real return will be negative. Real returns reflect the increase or decrease in purchasing power from an investment.

The computation of real returns on an asset is a two-step process:

1. Deflate the nominal price of the asset by the general price level
2. Compute returns in the usual way using the deflated prices

To illustrate, consider computing the real simple one-period return on an asset. Let P_t denote the nominal price of the asset at time t and let CPI_t denote an index of the general price level (e.g. consumer price index) at time t[1]. The deflated or real price at time t is:

$$P_t^{\text{Real}} = \frac{P_t}{CPI_t},$$

and the real one-period return is:

$$\begin{aligned} R_t^{\text{Real}} &= \frac{P_t^{\text{Real}} - P_{t-1}^{\text{Real}}}{P_{t-1}^{\text{Real}}} = \frac{\frac{P_t}{CPI_t} - \frac{P_{t-1}}{CPI_{t-1}}}{\frac{P_{t-1}}{CPI_{t-1}}} \\ &= \frac{P_t}{P_{t-1}} \cdot \frac{CPI_{t-1}}{CPI_t} - 1. \end{aligned} \tag{1.17}$$

The one-period gross real return return is:

$$1 + R_t^{\text{Real}} = \frac{P_t}{P_{t-1}} \cdot \frac{CPI_{t-1}}{CPI_t}. \tag{1.18}$$

If we define inflation between periods $t-1$ and t as:

$$\pi_t = \frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} = \%\Delta CPI_t, \tag{1.19}$$

[1] The CPI is usually normalized to 1 or 100 in some base year. We assume that the CPI is normalized to 1 in the base year for simplicity.

then $1 + \pi_t = CPI_t/CPI_{t-1}$ and Equation 1.17 may be re-expressed as:

$$R_t^{\text{Real}} = \frac{1 + R_t}{1 + \pi_t} - 1. \tag{1.20}$$

Example 1.9 (Compute real return)**.** Consider, again, a one-month investment in Microsoft stock. Suppose the CPI in months $t-1$ and t is 1 and 1.01, respectively, representing a 1% monthly growth rate in the overall price level. The real prices of Microsoft stock are:

$$P_{t-1}^{\text{Real}} = \frac{\$85}{1} = \$85,$$
$$P_t^{\text{Real}} = \frac{\$90}{1.01} = \$89.1089,$$

and the real monthly return is:

$$R_t^{\text{Real}} = \frac{\$89.10891 - \$85}{\$85} = 0.0483.$$

The nominal return and inflation over the month are:

$$R_t = \frac{\$90 - \$85}{\$85} = 0.0588,$$
$$\pi_t = \frac{1.01 - 1}{1} = 0.01.$$

Then the real return computed using Equation 1.20 is:

$$R_t^{\text{Real}} = \frac{1.0588}{1.01} - 1 = 0.0483.$$

Notice that simple real return is almost, but not quite, equal to the simple nominal return minus the inflation rate: $R_t^{\text{Real}} \approx R_t - \pi_t = 0.0588 - 0.01 = 0.0488$.

■

1.2.2.4 Adjusting for taxes

Any investment that generates a capital gain is potentially subject to taxation by the government. This taxation reduces the investment return. To see how this works, let V_t denote an initial investment amount, let $R_t > 0$ denote the one-period rate of return on the investment, and let τ denote the tax rate on investment income (capital gain). In the U.S. the tax rate on long-term (holding period more than one year) capital gains is either 0%, 15%, or 20% (depending on income) and the tax rate on short-term capital gains (holding period less than one year) is the tax rate on ordinary income (which could be as high as 35%). The future value without taxes is $V_t(1+R_t)$, and V_tR_t is the capital gain. The amount owed in taxes is $V_tR_t\tau$ and the amount remaining after taxes is $V_tR_t(1-\tau)$. The after tax return is

$$\begin{aligned}R_t^{Atax} &= \frac{V_t + V_tR_t(1-\tau) - V_t}{V_t} \\ &= \frac{V_t(1+R_t(1-\tau)-1)}{V_t} \\ &= R_t(1-\tau).\end{aligned} \tag{1.21}$$

Hence, the after tax return is the before tax return multiplied by one minus the tax rate.

Example 1.10 (Compute after tax return)**.** Let $V_t = \$100$, $R_t = 0.05$, and $\tau = 0.20$. Then the after tax return is

$$R_t^{Atax} = (0.05)(1-0.20) = (0.05)(0.80) = 0.04.$$

The amount paid in taxes is $\$1000 \times (0.05) \times (0.2) = \$10$, and the after tax profit is $\$50 - \$10 = \$40$.

1.2.2.5 Return on foreign investment

It is common for an investor in one country to buy an asset in another country as an investment. If the two countries have different currencies then the exchange rate between the two currencies influences the return on the asset. To illustrate, consider an investment in a foreign stock (e.g. a stock trading on the London stock exchange) by a U.S. national (domestic investor). The domestic investor takes U.S. dollars, converts them to the foreign currency (e.g. British Pound) via the exchange rate (price of foreign currency in U.S. dollars) and then purchases the foreign stock using the foreign currency. When the stock is sold, the proceeds in the foreign currency must then be converted back to the domestic currency.

To be more precise, consider the information in the table below:

Month	U.S. \$ Cost of 1 Pound £, $E_t^{US/UK}$	Value of British Shares, P_t^{UK}	Value in U.S. \$, P_t^{US}
t - 1	\$ 1.50	£40	\$1.50 × £40 = \$60
t	\$ 1.30	£45	\$1.30 × £45 = \$58.5

Let P_t^{US}, P_t^{UK}, and $E_t^{US/UK}$ denote the U.S. dollar price of the asset, British Pound price of the asset, and currency exchange rate between U.S. dollars and British Pounds, at the end of month t, respectively. The U.S. dollar cost of one share of British stock in month $t-1$ is

$$P_{t-1}^{US} = E_{t-1}^{US/UK} \times P_{t-1}^{UK} = \$1.50 \times\ 40 = \$60,$$

and the U.S. dollar cost at time t is

$$P_t^{US} = E_t^{US/UK} \times P_t^{UK} = \$1.30 \times £45 = \$58.5.$$

The U.S. dollar rate of return is then

$$R_t^{US} = \frac{P_t^{US} - P_{t-1}^{US}}{P_{t-1}^{US}} = \frac{\$58.5 - \$60}{\$60} = -0.025.$$

Notice that

$$\begin{aligned} R_t^{US} = & \frac{E_t^{US/UK} \times P_t^{UK} - E_{t-1}^{US/UK} \times P_{t-1}^{UK}}{E_{t-1}^{US/UK} \times P_{t-1}^{UK}} \\ = & \frac{E_t^{US/UK} \times P_t^{UK}}{E_{t-1}^{US/UK} \times P_{t-1}^{UK}} - 1 \\ = & (1 + R_t^{US/UK})(1 + R_t^{UK}) - 1, \end{aligned} \tag{1.22}$$

where

$$\frac{E_t^{US/UK}}{E_{t-1}^{US/UK}} = 1 + R_t^{US/UK}, \frac{P_t^{UK}}{P_{t-1}^{UK}} = 1 + R_t^{UK}.$$

Hence,

$$1 + R_t^{US} = (1 + R_t^{US/UK})(1 + R_t^{UK}).$$

Here, $R_t^{US/UK}$ is the simple rate of return on foreign currency and R_t^{UK} is the simple rate of return on the British stock in British Pounds:

$$\begin{aligned} R_t^{US/UK} &= \frac{E_t^{US/UK} - E_{t-1}^{US/UK}}{E_{t-1}^{US/UK}} = \frac{\$1.30 - \$1.50}{\$1.50} = -0.133, \\ R_t^{UK} &= \frac{P_t^{UK} - P_{t-1}^{UK}}{P_{t-1}^{UK}} = \frac{£45 - £40}{£40} = 0.125. \end{aligned}$$

Hence, the gross return, $1 + R_t^{US}$, for the U.S. investor has two parts: (1) the gross return from an investment in foreign currency, $1 + R_t^{US/UK}$; and (2) the gross return of the foreign asset in the foreign currency, $1 + R_t^{UK}$:

$$\begin{aligned} 1 + R_t^{US} &= (1 + R_t^{US/UK})(1 + R_t^{UK}) \\ &= (0.867)(1.125) \\ &= 0.975. \end{aligned}$$

In this example, the U.S. investor loses money even though the British investment has a positive rate of return because of the large drop in the exchange rate.

1.2.2.6 Annualizing returns

Very often returns over different horizons are annualized, i.e., converted to an annual return, to facilitate comparisons with other investments. The annualization process depends on the holding period of the investment and an implicit assumption about compounding. We illustrate with several examples using monthly returns. The same techniques work with different holding periods.

To start, if our investment horizon is one year then the annual gross and net returns are computed directly from the 12 one-month returns:

$$\begin{aligned} 1 + R_A = 1 + R_t(12) &= \frac{P_t}{P_{t-12}} = (1 + R_t)(1 + R_{t-1}) \cdots (1 + R_{t-11}), \\ R_A &= R_t(12). \end{aligned}$$

In this case, no compounding is required to create an annual return.

Next, consider a one-month investment in an asset with return R_t. What is the annualized return on this investment? If we assume that we receive the same return $R = R_t$ every month for the year, then the gross annual return is:

$$1 + R_A = 1 + R_t(12) = (1 + R)^{12}.$$

That is, the *annual gross return* is defined as the assumed common monthly return compounded for 12 months. The net annual return is then:

$$R_A = (1 + R)^{12} - 1.$$

It is important to recognize that this annualization calculation assumes the same monthly return R for each of the 12 months. This can be highly inaccurate and misleading if the actual future monthly returns are much different than R. The following examples illustrate.

Example 1.11 (Compute annualized return from one-month return)**.** In the first example, the one-month return, R_t, on Microsoft stock was 5.88%. If we assume that we can get this return for 12 months then the annualized return is: $R_A = (1.0588)^{12} - 1 = 1.9850 - 1 = 0.9850$, or 98.50% per year. Pretty good! However, this is likely to be highly inaccurate since getting 5% every month for a year is highly unlikely for a risky stock like Microsoft.

■

Now, consider a two-month investment with return $R_t(2)$. If we assume that we receive the same two-month return $R(2) = R_t(2)$ for the next six two-month periods, then the gross and net annual returns are:

$$\begin{aligned} 1 + R_A &= (1 + R(2))^6, \\ R_A &= (1 + R(2))^6 - 1. \end{aligned}$$

Here the annual gross return is defined as the two-month return compounded for 6 months.

Example 1.12 (Compute annualized return from two-month return)**.** Suppose the two-month return, $R_t(2)$, on Microsoft stock is 12.5%. If we assume that we can get this two-month return for the next 6 two-month periods then the annualized return is: $R_A = (1.1250)^6 - 1 = 2.0273 - 1 = 1.0273$, or 102.73% per year. As with the previous example, this result is likely to be inaccurate as it is highly unlikely to get 12.5% every two months for the rest of the year.

■

As the last example, suppose that our investment horizon is two years. That is, we start our investment at time $t - 24$ and cash out at time t. The two-year gross return is then $1 + R_t(24) = P_t/P_{t-24}$. What is the annual return on this two-year investment? The process is the same as computing the effective annual rate. To determine the annual return we solve the following relationship for R_A:

$$
\begin{aligned}
(1 + R_A)^2 &= 1 + R_t(24) \rightarrow \\
R_A &= (1 + R_t(24))^{1/2} - 1.
\end{aligned}
$$

In this case, the annual return is compounded twice to get the two-year return and the relationship is then solved for the annual return.

Example 1.13 (Compute annualized return from two-year return)**.** Suppose that the price of Microsoft stock 24 months ago is $P_{t-24} = \$50$ and the price today is $P_t = \$90$. The two-year gross return is $1 + R_t(24) = \$90/\$50 = 1.8000$ which yields a two-year net return of $R_t(24) = 0.80 = 80\%$. The annual return for this investment is defined as: $R_A = (1.800)^{1/2} - 1 = 1.3416 - 1 = 0.3416$ or 34.16% per year.

■

1.2.2.7 Average returns

For investments over a given horizon, it is often of interest to compute a measure of the average rate of return over the horizon. To illustrate, consider a sequence of monthly investments over n months with monthly returns $R_1, R_2, \dots, R_n$. The $n-$month return is

$$
R(n) = (1 + R_1)(1 + R_2) \cdots (1 + R_n) - 1.
$$

What is the average monthly return? There are two possibilities. The first is the *arithmetic average return*:

$$
\bar{R}^{Arith} = \frac{1}{n}\left(R_1 + R_2 + \cdots + R_n\right).
$$

The second is the *geometric average return*:

$$
\bar{R}^{Geo} = \left[(1 + R_1)(1 + R_2) \cdots (1 + R_n)\right]^{1/n} - 1.
$$

Notice that the geometric average return is the monthly return which compounded monthly for n months gives the gross $n-$month return

$$(1+\bar{R}^{Geo})^n = 1+R(n).$$

As a measure of average investment performance, the geometric average return is preferred to the arithmetic average return. To see why, consider a two-month investment horizon with monthly returns $R_1 = 0.5$ and $R_2 = -0.5$. One dollar invested over these two months grows to

$$(1+R_1)(1+R_2) = (1.5)(0.5) = 0.75,$$

for a two-period return of $R(2) = 0.75 - 1 = -0.25$. Hence, the two-month investment loses 25%. The arithmetic average return is

$$\bar{R}^{Arith} = \frac{1}{2}\left(0.5 + (-0.5)\right) = 0.$$

As a measure of investment performance, this is misleading because the investment loses money over the two-month horizon. The average return indicates the investment breaks even. The geometric average return is

$$\bar{R}^{Geo} = \left[(1.5)(0.5)\right]^{1/2} - 1 = -0.134.$$

This is a more accurate measure of investment performance because it indicates that the investment eventually loses money. In addition, the compound two-month return using $\bar{R}^{Geo} = -0.134$ is equal to $R(2) = -0.25$.

While the arithmetic average return may not be the most appropriate measure of actual average investment performance, we shall see in the following chapters that it is a useful estimate of expected performance.

1.2.3 Continuously compounded returns

In this section we define continuously compounded returns from simple returns, and describe their properties.

1.2.3.1 One-period returns

Let R_t denote the simple monthly return on an investment. The *continuously compounded monthly return*, r_t, is defined as:

$$r_t = \ln(1+R_t) = \ln\left(\frac{P_t}{P_{t-1}}\right), \tag{1.23}$$

where $\ln(\cdot)$ is the natural log function[2]. To see why r_t is called the continuously compounded return, take the exponential of both sides of Equation 1.23 to give:

$$e^{r_t} = 1 + R_t = \frac{P_t}{P_{t-1}}.$$

Rearranging we get $P_t = P_{t-1}e^{r_t} = P_{t-t}(1 + R_t)$, so that r_t is the continuously compounded growth rate in prices between months $t-1$ and t. This is to be contrasted with R_t, which is the simple growth rate in prices between months $t-1$ and t without any compounding. As a result, r_t is always smaller than R_t. Furthermore, since $\ln(x/y) = \ln(x) - \ln(y)$ it follows that:

$$\begin{aligned} r_t &= \ln\left(\frac{P_t}{P_{t-1}}\right) \\ &= \ln(P_t) - \ln(P_{t-1}) \\ &= p_t - p_{t-1}, \end{aligned}$$

where $p_t = \ln(P_t)$. Hence, the continuously compounded monthly return, r_t, can be computed simply by taking the first difference of the natural logarithms of monthly prices.

If the simple return R_t is close to zero ($R_t \approx 0$) then $r_t = \ln(1 + R_t) \approx R_t$. This result can be shown using a first order Taylor series approximation to the function $f(x) = \ln(1+x)$. Recall, for a continuous and differentiable function $f(x)$ the first order Taylor series approximation of $f(x)$ about the point $x = x_0$ is

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \text{remainder},$$

where $f'(x_0)$ denotes the derivative of $f(x)$ evaluated at x_0. Let $f(x) = \ln(1+x)$ and set $x_0 = 0$. Note that

$$f(0) = \ln(1) = 0,\ f'(x) = \frac{1}{1+x},\ f'(0) = 1.$$

Then, using the first order Taylor series approximation, $\ln(1+x) \approx x$ when $x \approx 0$.

Example 1.14 (Compute continuously compounded returns)**.** Using the price and return data from Example Example 1.6, the continuously compounded monthly return on Microsoft stock can be computed in two ways:

$$\begin{aligned} r_t &= \ln(1.0588) = 0.0571 \\ r_t &= \ln(90) - \ln(85) = 4.4998 - 4.4427 = 0.0571 \end{aligned}$$

Notice that r_t is slightly smaller than R_t. Using R, the calculations are

[2] The continuously compounded return is always defined since asset prices, P_t, are always non-negative. Properties of logarithms and exponentials are discussed in the appendix to this chapter.

```r
c(log(1 + R0), log(P0) - log(Pm1))
#> [1] 0.0572 0.0572
```

Notice that the R function `log()` computes the natural logarithm.

■

Given a monthly continuously compounded return r_t, it is straightforward to solve back for the corresponding simple net return R_t:

$$R_t = e^{r_t} - 1. \tag{1.24}$$

Hence, nothing is lost by considering continuously compounded returns instead of simple returns. Continuously compounded returns are very similar to simple returns as long as the return is relatively small, which it generally will be for monthly or daily returns. Since R_t is bounded from below by -1, the smallest value for r_t is $r_t = ln(1-1) = -\infty$. This, however, does not mean that you could lose an infinite amount of money on an investment. The actual amount of money lost is determined by the simple return Equation 1.24.

Example 1.15 (Determine simple returns from continuously compounded returns)**.** In the previous example, the continuously compounded monthly return on Microsoft stock is $r_t = 5.71\%$. The simple net return is then: $R_t = e^{.0571} - 1 = 0.0588$.

■

For asset return modeling and statistical analysis it is often more convenient to use continuously compounded returns than simple returns. One reason is that continuously compounded returns take values on the entire real line, $-\infty < r_t < \infty$, whereas simple returns are only defined for values larger than -1, $-1 \leq R_t < \infty$. Another reason is due to the additivity property of multi-period continuously compounded returns discussed in the next sub-section.

1.2.3.2 Multi-period returns

The relationship between multi-period continuously compounded returns and one-period continuously compounded returns is more simple than the relationship between multi-period simple returns and one-period simple returns. To illustrate, consider the two-month continuously compounded return defined as:

$$r_t(2) = \ln(1 + R_t(2)) = \ln\left(\frac{P_t}{P_{t-2}}\right) = p_t - p_{t-2}.$$

Taking exponentials of both sides and rearranging gives:

$$P_t = P_{t-2}e^{r_t(2)}$$

so that $r_t(2)$ is the continuously compounded growth rate of prices between months $t-2$ and t. Using $P_t/P_{t-2} = (P_t/P_{t-1}) \cdot (P_{t-1}/P_{t-2})$ and the fact that $\ln(x \cdot y) = \ln(x) + \ln(y)$ it follows that:

$$\begin{aligned} r_t(2) &= \ln\left(\frac{P_t}{P_{t-1}} \cdot \frac{P_{t-1}}{P_{t-2}}\right) \\ &= \ln\left(\frac{P_t}{P_{t-1}}\right) + \ln\left(\frac{P_{t-1}}{P_{t-2}}\right) \\ &= r_t + r_{t-1}. \end{aligned}$$

Hence the continuously compounded two-month return is just the sum of the two continuously compounded one-month returns. Recall, with simple returns the two-month return is a multiplicative (geometric) sum of two one-month returns.

Example 1.16 (Compute two-month continuously compounded returns)**.** Using the data from Example 1.7, the continuously compounded two-month return on Microsoft stock can be computed in two equivalent ways. The first way uses the difference in the logs of P_t and P_{t-2}:

$$r_t(2) = \ln(90) - \ln(80) = 4.4998 - 4.3820 = 0.1178.$$

The second way uses the sum of the two continuously compounded one-month returns. Here $r_t = \ln(90) - \ln(85) = 0.0571$ and $r_{t-1} = \ln(85) - \ln(80) = 0.0607$ so that:

$$r_t(2) = 0.0571 + 0.0607 = 0.1178.$$

Notice that $r_t(2) = 0.1178 < R_t(2) = 0.1250$. The vectorized calculations in R are:

```r
p = log(P)
r = log(1 + R)
c(r[1] + r[2], sum(r), p[3] - p[1], diff(p, 2))
#> [1] 0.118 0.118 0.118 0.118
```

■

Previously, we saw that simple two-period returns were not symmetric. Continuously compounded two-period returns, however, are symmetric. To see this, suppose $r_{t-1} = -0.5$ and $r_t = 0.5$. Notice that the implied simple returns are not symmetric: $R_{t-1} = e^{-0.5} - 1 = -0.3935$ and $R_t = e^{0.5} - 1 = 0.6487$. Then one dollar invested over two periods becomes:

$$\$1 \times (1 - 0.3935)(1 + 0.6487) = \$1$$

The continuously compounded k-month return is defined by:

$$r_t(k) = \ln(1 + R_t(k)) = \ln\left(\frac{P_t}{P_{t-k}}\right) = p_t - p_{t-k}.$$

Using similar manipulations to the ones used for the continuously compounded two-month return, we can express the continuously compounded k-month return as the sum of k continuously compounded monthly returns:

$$r_t(k) = \sum_{j=0}^{k-1} r_{t-j}. \tag{1.25}$$

The additivity of continuously compounded returns to form multiperiod returns is an important property for statistical modeling purposes.

1.2.3.3 Adjusting for dividends

The continuously compounded one-period return adjusted for dividends is defined by Equation 1.23, where R_t is computed using Equation 1.15[3].

Example 1.17 (Compute continuously compounded total return)**.** From example Example 1.8, the total simple return using Equation 1.15 is $R_t = 0.0707$. The continuously compounded total return is then:

$$r_t = \ln(1 + R_t) = \ln(1.0707) = 0.0683.$$

■

1.2.3.4 Adjusting for inflation

Adjusting continuously compounded nominal returns for inflation is particularly simple. The continuously compounded one-period real return is defined as:

$$r_t^{\text{Real}} = \ln(1 + R_t^{\text{Real}}). \tag{1.26}$$

Using Equation 1.18, it follows that:

$$\begin{aligned}
r_t^{\text{Real}} &= \ln\left(\frac{P_t}{P_{t-1}} \cdot \frac{CPI_{t-1}}{CPI_t}\right) \\
&= \ln\left(\frac{P_t}{P_{t-1}}\right) + \ln\left(\frac{CPI_{t-1}}{CPI_t}\right) \\
&= \ln(P_t) - \ln(P_{t-1}) + \ln(CPI_{t-1}) - \ln(CPI_t) \\
&= \ln(P_t) - \ln(P_{t-1}) - (\ln(CPI_t) - \ln(CPI_{t-1})) \\
&= r_t - \pi_t^c,
\end{aligned} \tag{1.27}$$

[3] Show formula from Campbell, Lo and MacKinlay.

where $r_t = \ln(P_t) - \ln(P_{t-1}) = \ln(1 + R_t)$ is the nominal continuously compounded one-period return and $\pi_t^c = \ln(CPI_t) - \ln(CPI_{t-1}) = \ln(1 + \pi_t)$ is the one-period continuously compounded growth rate in the general price level (continuously compounded one-period inflation rate). Hence, the real continuously compounded return is simply the nominal continuously compounded return minus the the continuously compounded inflation rate.

Example 1.18 (Compute continuously compounded real returns)**.** From Example Example 1.9, the nominal simple return is $R_t = 0.0588$, the monthly inflation rate is $\pi_t = 0.01$, and the real simple return is $R_t^{\text{Real}} = 0.0483$. Using Equation 1.26, the real continuously compounded return is:

$$r\_t^{\text{Real}} = \ln(1 + R\_t\text{Real}) = \ln(1.0483) = 0.047.$$

Equivalently, using (Equation 1.27) the real return may also be computed as:

$$r_t^{\text{Real}} = r_t - \pi_t c = \ln(1.0588) - \ln(1.01) = 0.047.$$

■

1.2.3.5 Adjusting for taxes

Using Equation 1.21, the continuously compounded return adjusted for taxes is

$$r_t^{Atax} = ln\left(R_t(1-\tau)\right) = ln(R_t) + ln(1-\tau) = r_t + ln(1-\tau) \tag{1.28}$$

1.2.3.6 Return on a foreign investment

Consider again, the US/UK investment example from sub-section Section 1.2.2.5. Using Equation 1.22, the continuously compounded US return is

$$\begin{aligned} r_t^{US} &= ln\left((1+R_t^{US/UK})(1+R_t^{UK})\right) \\ &= ln(1+R_t^{US/UK}) + ln(1+R_t^{UK}) \\ &= r_t^{US/UK} + r_t^{UK} \end{aligned} \tag{1.29}$$

where $r_t^{US/UK} = ln(1 + R_t^{US/UK})$ is the continuously compounded currency return and $r_t^{UK} = ln(1 + R_t^{UK})$ is the continuously compounded UK return.

1.2.3.7 Annualizing continuously compounded returns

Just as we annualized simple monthly returns, we can also annualize continuously compounded monthly returns. We illustrate using monthly continuously compounded returns. First, if our investment horizon is one year and we observe 12 monthly continuously compounded returns then the annual continuously compounded return is just the sum of the twelve monthly continuously compounded returns:

$$\begin{aligned} r_A = r_t(12) &= r_t + r_{t-1} + \cdots + r_{t-11} \\ &= \sum_{j=0}^{11} r_{t-j}. \end{aligned}$$

The average continuously compounded monthly return is defined as:

$$\overline{r}_m = \frac{1}{12} \sum_{j=0}^{11} r_{t-j}.$$

Notice that:

$$12 \cdot \overline{r}_m = \sum_{j=0}^{11} r_{t-j}$$

so that we may alternatively express r_A as:

$$r_A = 12 \cdot \overline{r}_m.$$

That is, the continuously compounded annual return is twelve times the average of the continuously compounded monthly returns.

Next, consider a one-month investment in an asset with continuously compounded return $r_t = r$. If we assume that we receive the same return r every month for the year, then the annual continuously compounded return is just 12 times the monthly continuously compounded return:

$$r_A = r_t(12) = r + r + ... + r = 12 \cdot r.$$

If we have a two-month continuously compounded return of $r(2)$, and we assume that we get this same return every two months for the next year, then the annualized continuously compounded return is

$$r_A = 6 \cdot r(2).$$

Finally, suppose we have a 2-year continuously compounded return $r(24)$. Then the continuously compounded annual return can be defined using

$$2 \cdot r_A = r(24) \Longrightarrow r_A = \frac{r(24)}{2}.$$

1.3 Portfolios and Portfolio Returns

Consider an investment in two assets named asset A and asset B (e.g. Amazon and Boeing stock). Let V_A and V_B denote the dollar amounts invested in assets A and B, respectively. We call such an investment in multiple assets a *portfolio*. The total dollar amount invested in the portfolio is $V_p = V_A + V_B$. The shares of dollars invested in assets A and B are:

$$x_A = \frac{V_A}{V_p}, x_B = \frac{V_B}{V_p}.$$

Notice that $x_A + x_B = 1$ by construction (100% of all wealth is invested in the two assets).

In this book, the collection of investment shares (x_A, x_B) and initial wealth invested V_p defines a portfolio. If V_p is not specified then assume that $V_p = \$1$. For example, one portfolio may be $(x_A = 0.5, x_B = 0.5)$ (equally weighted portfolio) and another may be $(x_A = 0, x_B = 1)$ (everything invested in asset B). Negative values for x_A or x_B are possible and represent short sales (to be discussed in later chapters). Given x_A, x_B, and V_p, the dollar amounts invested in assets A and B are:

$$V_A = x_A \times V_p, V_B = x_B \times V_p.$$

The rate of return on a portfolio can be represented as a share weighted average of the rates of returns on the assets in the portfolio. To see this, let R_A and R_B denote the simple one-period returns on assets A and B. We wish to determine the simple one-period return on the portfolio defined by (x_A, x_B) and initial wealth V_p. To do this, note that at the end of the holding period, the investments in assets A and B are worth $V_p \times x_A(1+R_A)$ and $V_p \times x_B(1+R_B)$, respectively. Hence, at the end of the holding period the portfolio is worth:

$$V_p \times [x_A(1 + R_A) + x_B(1 + R_B)] = V_p \times (1 + R_p).$$

Hence, $x_A(1 + R_A) + x_B(1 + R_B) = 1 + R_p$ defines the *portfolio gross return*. The portfolio gross return is equal to a weighted average of the gross returns on assets A and B, where the weights are the portfolio shares x_A and x_B.

To determine the portfolio rate of return, re-write the portfolio gross return as:

$$1 + R_p = x_A + x_B + x_A R_A + x_B R_B = 1 + x_A R_A + x_B R_B,$$

since $x_A + x_B = 1$ by construction. Then the *portfolio rate of return* is:

$$R_p = x_A R_A + x_B R_B,$$

which is equal to a weighted average of the simple returns on assets A and B, where the weights are the portfolio shares x_A and x_B.

In the portfolio rate of return, the components $x_A R_A$ and $x_B R_B$ are the *contributions* of assets A and B to the portfolio return, respectively. These contributions add up to the total portfolio return R_p.

Example 1.19 (Compute portfolio return)**.** Consider a portfolio of Microsoft and Starbucks stock in which you initially purchase ten shares of each stock at the end of month $t-1$ at the prices $P_{\text{msft},t-1} = \$85$ and $P_{\text{sbux},t-1} = \$30$, respectively. The initial value of the portfolio is $V_{t-1} = 10 \times \$85 + 10 \times 30 = \$1,150$. The portfolio shares are $x_{\text{msft}} = 850/1150 = 0.7391$ and $x_{\text{sbux}} = 30/1150 = 0.2609$. Suppose at the end of month t, $P_{\text{msft},t} = \$90$ and $P_{\text{sbux},t} = \$28$. Assuming that Microsoft and Starbucks do not pay a dividend between periods $t-1$ and t, the one-period returns on the two stocks are:

$$
\begin{aligned}
R_{\text{msft},t} &= \frac{\$90 - \$85}{\$85} = 0.0588, \\
R_{\text{sbux},t} &= \frac{\$28 - \$30}{\$30} = -0.0667.
\end{aligned}
$$

The one-month rate of return on the portfolio is then:

$$
R_{p,t} = (0.7391)(0.0588) + (0.2609)(-0.0667) = 0.02609,
$$

The contributions of assets A and B to the portfolio return are:

$$
x_A R_A = (0.7391)(0.0588) = 0.0435, \; x_B R_B = (0.2609)(-0.0667) = -0.0174.
$$

Asset A contributes positively and asset B contributes negatively to the portfolio return. The portfolio value at the end of month t is:

$$
V_t = V_{t-1}(1 + R_{p,t}) = \$1,150 \times (1.02609) = \$1,180
$$

In R, the portfolio calculations are:

```r
P.msft = c(85, 90)
P.sbux = c(30, 28)
V = P.msft[1]*10 + P.sbux[1]*10
x.msft = 10*P.msft[1]/V
x.sbux = 10*P.sbux[1]/V
R.msft = (P.msft[2] - P.msft[1])/P.msft[1]
R.sbux = (P.sbux[2] - P.sbux[1])/P.sbux[1]
R.p = x.msft*R.msft + x.sbux*R.sbux
V1 = V*(1 + R.p)
c(R.p, x.msft*R.msft, x.sbux*R.sbux, V1)
#> [1]    0.0261    0.0435   -0.0174 1180.0000
```

■

In general, for a portfolio of n assets with investment shares x_i such that $x_1 + \cdots + x_n = 1$, the one-period portfolio gross and simple returns are defined as:

$$1 + R_{p,t} = \sum_{i=1}^{n} x_i(1 + R_{i,t}) \tag{1.30}$$

$$R_{p,t} = \sum_{i=1}^{n} x_i R_{i,t}. \tag{1.31}$$

where $R_{i,t}$ is the one-period simple return on asset i. Asset i's contribution to the portfolio return is

$$\text{ctr}_{i,p} = x_i R_{i,t}. \tag{1.32}$$

and

$$\sum_{i=i}^{n} \text{ctr}_{i,p} = \sum_{i=i}^{n} x_i R_{i,t} = R_{p,t}.$$

1.3.1 Multiperiod portfolio returns and rebalancing

How to compute multiperiod portfolio returns depends on the assumptions made about the portfolio weights each period. Several scenarios exist:

1. Portfolio weights from the initial portfolio are allowed to change over time as prices of the underlying assets change over time. In this case no rebalancing of the portfolio is done. This is called a *buy-and-hold* portfolio.
2. Portfolio weights remain constant over time, which implies that the portfolio is rebalanced at every time period (associated with holding period) to maintain constant weights.
3. Portfolio weights from the initial portfolio are rebalanced at specific time intervals that are different than the holding period (e.g. rebalance quarterly for monthly holding periods). This is a hybrid of scenarios 1 and 2.
4. Portfolio weights are actively changed at each time period associated with the holding period. This is called an *actively managed portfolio*.

How to deal with the above scenarios is best illustrated through examples.

1.3.1.1 No rebalancing of portolio weights

Prices and shares framework

We illustrate the buy-and-hold scenario 1 using a simple portfolio consisting of two assets A and B invested for three months starting at time $t-3$ in a framework where we know the asset prices each period and the initial amounts invested in each asset[4]. In particular, we assume that the initial portfolio consists of $s_A = 100$ shares of asset A and $s_B = 50$ shares of asset

[4]This example is inspired by the `return.portfolio` vignette in **PerformanceAnalytics**).

B. The end-of-month prices and values ($V_{i,t} = s_i \times P_{i,t}, i = A, B$) of the assets as well as the end-of-month values of the portfolio are given by

```
s.A = 100
s.B = 50
P.A = c(5,7,6,7)
V.A = s.A*P.A
P.B = c(10, 11, 12, 8)
V.B = s.B*P.B
V.p = V.A + V.B
table.vals = cbind(P.A, V.A, P.B, V.B, V.p)
rownames(table.vals) = c("t-3", "t-2", "t-1", "t")
table.vals
#>     P.A V.A P.B V.B  V.p
#> t-3   5 500  10 500 1000
#> t-2   7 700  11 550 1250
#> t-1   6 600  12 600 1200
#> t     7 700   8 400 1100
```

In the above table the time index t represents the end-of-month t. Slightly abusing notation, we can also say that the time index $t-1$ represents the beginning of month t. This is important because for the portfolio calculations the end-of-month values $V_{A,t-1}$, $V_{B,t-1}$ and $V_{p,t-1}$ are used to calculate the portfolio weights to be applied in month t. That is,

$$x_{A,t-1} = \frac{V_{A,t-1}}{V_{p,t-1}},\ x_{B,t-1} = \frac{V_{B,t-1}}{V_{p,t-1}}$$

represent the portfolio weights for assets A and B at the beginning of month t. Then, the portfolio return over month t is computed as

$$R_{p,t} = \%\Delta V_{p,t} = \frac{V_{p,t} - V_{p,t-1}}{V_{p,t-1}} \tag{1.33}$$

$$= x_{A,t-1}R_{A,t} + x_{B,t-1}R_{B,t}, \tag{1.34}$$

$$\tag{1.35}$$

where

$$R_{A,t} = \%\Delta V_{A,t} = \frac{V_{A,t} - V_{A,t-1}}{V_{A,t-1}} \tag{1.36}$$

$$= \%\Delta P_{A,t} = \frac{P_{A,t} - P_{A,t-1}}{P_{A,t-1}} \tag{1.37}$$

$$R_{B,t} = \%\Delta V_{B,t} = \frac{V_{B,t} - V_{B,t-1}}{V_{B,t-1}} \tag{1.38}$$

$$= \%\Delta P_{B,t} = \frac{P_{B,t} - P_{B,t-1}}{P_{B,t-1}}. \tag{1.39}$$

The portfolio return, $R_{p,t}$, is still a share weighted average of the individual asset returns but with time varying portfolio weights $x_{A,t-1}$ and $x_{B,t-1}$.

The R calculations for returns are:

```
R.A = (V.A[2:4] - V.A[1:3])/V.A[1:3]
R.B = (V.B[2:4] - V.B[1:3])/V.B[1:3]
R.p = (V.p[2:4] - V.p[1:3])/V.p[1:3]
table.returns = cbind(R.A, R.B, R.p)
rownames(table.returns) =  c("t-2", "t-1", "t")
table.returns
#>        R.A     R.B     R.p
#> t-2  0.400  0.1000  0.2500
#> t-1 -0.143  0.0909 -0.0400
#> t    0.167 -0.3333 -0.0833
```

The R calculations for the weights are:

```
x.A = V.A/V.p
x.B = V.B/V.p
table.weights = cbind(x.A, x.B)
rownames(table.weights) = c("t-3", "t-2", "t-1", "t")
table.weights
#>       x.A   x.B
#> t-3 0.500 0.500
#> t-2 0.560 0.440
#> t-1 0.500 0.500
#> t   0.636 0.364
```

Here, the initial portfolio formed at the end of month $t-3$ is an equally weighted portfolio of assets A and B. However, the portfolio weights at the end of month $t-2$ are not 0.5 and 0.5 as the prices of assets A and B have changed over the month. Asset A's price has gone up more in percentage terms ($R_{A,t-2} = 40\%$) than asset B's price ($R_{B,t-2} = 10\%$) so the portfolio weight in asset A at the beginning of $t-1$ ($x_{A,t-2} = 0.56$), has become bigger than the portfolio weight in asset B ($x_{B,t-2} = 0.44$).

For the calculation of the portfolio returns, the asset weights at time $t-3$ apply to the asset returns at time $t-2$, the weights at time $t-2$ apply to the returns at time $t-1$, and the weights at time $t-1$ apply to the returns at time t:

```
R.p = x.A[1:3]*R.A + x.B[1:3]*R.B
R.p
#> [1]  0.2500 -0.0400 -0.0833
```

The asset contributions to the portfolio return at time t are

$$\mathrm{ctr}_{A,t} = \frac{\Delta V_{A,t}}{V_{p,t-1}} = x_{A,t-1} \times R_{A,t} \tag{1.40}$$

$$\mathrm{ctr}_{B,t} = \frac{\Delta V_{B,t}}{V_{p,t-1}} = x_{B,t-1} \times R_{B,t} \tag{1.41}$$

The R calculations for the contributions derived from values are:

```
ctr.A = (V.A[2:4] - V.A[1:3])/V.p[1:3]
ctr.B = (V.B[2:4] - V.B[1:3])/V.p[1:3]
table.ctr = cbind(ctr.A, ctr.B, ctr.A+ctr.B)
colnames(table.ctr)[3] = "R.p"
rownames(table.ctr) = c("t-2", "t-1", "t")
table.ctr
#>       ctr.A  ctr.B     R.p
#> t-2  0.2000  0.050  0.2500
#> t-1 -0.0800  0.040 -0.0400
#> t    0.0833 -0.167 -0.0833
```

The R calculations for the contributions derived from returns are:

```
ctr.A = x.A[1:3]*R.A
ctr.B = x.B[1:3]*R.B
table.ctr = cbind(ctr.A, ctr.B, R.p)
rownames(table.ctr) = c("t-2", "t-1", "t")
table.ctr
#>       ctr.A  ctr.B     R.p
#> t-2  0.2000  0.050  0.2500
#> t-1 -0.0800  0.040 -0.0400
#> t    0.0833 -0.167 -0.0833
```

Returns and weights framework

The above calculations are based on knowing asset prices and initial shares purchased for each asset. In many situations, we only have return information and initial portfolio weights on each asset and the initial value of the portfolio. Using the above example, the initial portfolio weights are $x_{A,t-3} = x_{B,t-3} = 0.5$ and the initial value of the portfolio is $V_{p,t-3} = 1000$. We also know the returns, $R_{i,t}$, each period for the two assets:

```
table.returns[, c("R.A", "R.B")]
#>         R.A     R.B
#> t-2  0.400  0.1000
#> t-1 -0.143  0.0909
#> t    0.167 -0.3333
```

We deduce the initial asset values using

$$V_{A,t-3} = x_{A,t-3} \times V_{p,t-3} = 0.5 \times 1000 = 500, \tag{1.42}$$
$$V_{B,t-3} = x_{B,t-3} \times V_{p,t-3} = 0.5 \times 1000 = 500. \tag{1.43}$$

The end-of-month asset values for $i = A, B$ are computed using

$$\begin{aligned} V_{i,t-2} &= V_{i,t-3} \times (1 + R_{i,t-2}), \\ V_{i,t-1} &= V_{i,t-2} \times (1 + R_{i,t-1}) = V_{i,t-3} \times (1 + R_{i,t-2})(1 + R_{i,t-1}), \\ V_{i,t} &= V_{i,t-1} \times (1 + R_{i,t}) = V_{i,t-3} \times (1 + R_{i,t-2})(1 + R_{i,t-1})(1 + R_{i,t}). \end{aligned}$$

The end-of-month portfolio values are the sum of the end-of-month asset values. The R calculations are:

```
V.A = V.B = V.p = rep(0, 4)
V.p[1] = 1000
V.A[1] = V.B[1] = 0.5*V.p[1]
V.A[2:4] = cumprod(1+R.A)*V.A[1]
V.B[2:4] = cumprod(1+R.B)*V.B[1]
V.p[2:4] = V.A[2:4] + V.B[2:4]
table.vals = cbind(V.A, V.B, V.p)
rownames(table.vals) = c("t-3", "t-2", "t-1", "t")
table.vals
#>     V.A V.B  V.p
#> t-3 500 500 1000
#> t-2 700 550 1250
#> t-1 600 600 1200
#> t   700 400 1100
```

The portfolio weights at the end-of-month t are computed as

$$x_{i,t} = \frac{V_{i,t}}{V_{p,t}},$$

and the R calculations are:

```
x.A = V.A/V.p
x.B = V.B/V.p
table.weights = cbind(x.A, x.B)
rownames(table.weights) = c("t-3", "t-2", "t-1", "t")
table.weights
#>       x.A   x.B
#> t-3 0.500 0.500
#> t-2 0.560 0.440
#> t-1 0.500 0.500
#> t   0.636 0.364
```

The portfolio return over month t can be computed using

$$R_{p,t} = \%\Delta V_{p,t} = \frac{V_{p,t} - V_{p,t-1}}{V_{p,t-1}} \tag{1.44}$$

$$= x_{A,t-1} \times R_{A,t} + x_{B,t-1} \times R_{B,t} \tag{1.45}$$

The R calculations for returns based on values are:

```
diff(table.vals[, "V.p"])/table.vals[1:3, "V.p"]
#>     t-2     t-1       t
#>  0.2500 -0.0400 -0.0833
```

The calculation for returns based on weights is

```
x.A[1:3]*R.A + x.B[1:3]*R.B
#> [1]  0.2500 -0.0400 -0.0833
```

1.3.1.2 Constant portfolio weights

If the initial portfolio weights are to be held constant over time then the portfolio typically needs to be *rebalanced* at each time period to adjust the portfolio weights back to the initial weights. Given the constant portfolio weights x_A and x_B the return on the portfolio at every time period is given by

$$R_{p,t} = x_A R_{A,t} + x_B R_{B,t}.$$

The rebalancing of portfolio weights can be illustrated with the example from the previous sub-section. The initial portfolio is an equally weighted portfolio, $x_{A,t-3} = x_{B,t-3} = 0.5$. To have constant portfolio weights the portfolio will need to be rebalanced each period so that $x_{A,t-2} = x_{A,t-1} = x_{A,t} = 0.5$ and $x_{B,t-2} = x_{B,t-1} = x_{B,t} = 0.5$. Consider the portfolio at the end-of-month $t-2$ where

$$V_{A,t-2} = V_{A,t-3} \times (1 + R_{A,t-2}) = 500 \times (1 + 0.40) = 700 \tag{1.46}$$

$$V_{B,t-2} = V_{B,t-3} \times (1 + R_{B,t-2}) = 500 \times (1 + 0.10) = 550 \tag{1.47}$$

$$V_{p,t-2} = V_{A,t-2} + V_{B,t-2} = 700 + 550 = 1250 \tag{1.48}$$

$$x_{A,t-2} = \frac{V_{A,t-2}}{V_{p,t-2}} = \frac{700}{1250} = 0.56 \tag{1.49}$$

$$x_{B,t-2} = \frac{V_{B,t-2}}{V_{p,t-2}} = \frac{550}{1250} = 0.44 \tag{1.50}$$

The end-of-month portfolio weights are $x_{A,t-2} = 0.56$ and $x_{B,t-2} = 0.44$, which are not equal to 0.5. Since $V_{p,t-2} = 1250$, to rebalance the portfolio so that $x_{A,t-2} = x_{B,t-2} = 0.5$ we need $V_{A,t-2} = V_{B,t-2} = V_{p,t-2}/2 = 625$. This requires selling $700 - 625 = 75$ of asset A and purchasing 75 of asset B at the end-of-month $t-2$. Then, at the beginning of month $t-1$ we have $V_{A,t-2} = V_{B,t-2} = 625$ and $x_{A,t-2} = x_{B,t-2} = 0.5$.

At the end-of-month $t-1$, for the rebalanced portfolio we have

$$V_{A,t-1} = V_{A,t-2} \times (1 + R_{A,t-1}) = 625 \times (1 - 0.1429) = 535.69 \tag{1.51}$$

$$V_{B,t-1} = V_{B,t-2} \times (1 + R_{B,t-1}) = 625 \times (1 + 0.0909) = 681.81 \tag{1.52}$$

$$V_{p,t-1} = V_{A,t-2} + V_{B,t-2} = 535.69 + 681.81 = 1217.50 \tag{1.53}$$

$$x_{A,t-1} = \frac{V_{A,t-1}}{V_{p,t-1}} = \frac{535.69}{1217.50} = 0.44 \tag{1.54}$$

$$x_{B,t-1} = \frac{V_{B,t-1}}{V_{p,t-1}} = \frac{681.81}{1217.50} = 0.56 \tag{1.55}$$

The end-of-month portfolio weights are $x_{A,t-1} = 0.44$ and $x_{B,t-1} = 0.56$. Since $V_{p,t-1} = 1217.50$, to rebalance the portfolio we need $V_{A,t-1} = V_{B,t-1} = 608.75$. This requires selling 73.06 of asset B and purchasing 73.06 of asset A and the end-of-month t-1. Then, at the beginning of month t we have $V_{A,t-1} = V_{B,t-1} = 608.75$ and $x_{A,t-1} = x_{B,t-1} = 0.5$.

Finally, at the end-of-month t, for the rebalanced portfolio we have

$$V_{A,t} = V_{A,t-1} \times (1 + R_{A,t}) = 608.75 \times (1 + 0.1667) = 710.23 \tag{1.56}$$
$$V_{B,t} = V_{B,t-1} \times (1 + R_{B,t}) = 608.75 \times (1 - 0.3333) = 405.85 \tag{1.57}$$
$$V_{p,t} = V_{A,t-1} + V_{B,t-1} = 710.23 + 405.85 = 1116.08 \tag{1.58}$$
$$x_{A,t} = \frac{V_{A,t}}{V_{p,t}} = \frac{710.23}{1116.08} = 0.64 \tag{1.59}$$
$$x_{B,t} = \frac{V_{B,t}}{V_{p,t}} = \frac{405.85}{1116.08} = 0.36 \tag{1.60}$$

The end-of-month portfolio weights are $x_{A,t} = 0.64$ and $x_{B,t} = 0.36$. Since $V_{p,t} = 1116.08$, to rebalance the portfolio we need $V_{A,t} = V_{B,t} = 558.04$. This requires selling 152.19 of asset A and purchasing 152.19 of asset B and the end-of-month t. Then, at the beginning of month $t + 1$ we have $V_{A,t} = V_{B,t} = 558.04$ and $x_{A,t} = x_{B,t} = 0.5$.

The above calculations in R are:

```
value.table = matrix(0, 4, 3)
rownames(value.table) = c("t-3", "t-2", "t-1", "t")
colnames(value.table) = c("V.A", "V.B", "V.p")
value.table["t-3", ] = c(V.A[1], V.B[1], V.p[1])
for (t in 2:4) {
  value.table[t, "V.A"] = value.table[t-1, "V.A"]*(1 + R.A[t-1])
  value.table[t, "V.B"] = value.table[t-1, "V.B"]*(1 + R.B[t-1])
  value.table[t, "V.p"] = value.table[t, "V.A"] + value.table[t, "V.B"]
  RB = max(value.table[t, "V.A"], value.table[t, "V.B"]) - value.table[t, "V.p"]/2
  if (value.table[t, "V.A"] > value.table[t, "V.B"]) {
    value.table[t, "V.A"] = value.table[t, "V.A"] - RB
    value.table[t, "V.B"] = value.table[t, "V.B"] + RB
    }
  else{
    value.table[t, "V.A"] = value.table[t, "V.A"] + RB
    value.table[t, "V.B"] = value.table[t, "V.B"] - RB
  }
  value.table[t, "V.p"] = value.table[t, "V.A"] + value.table[t, "V.B"]
}
value.table
#>     V.A V.B  V.p
#> t-3 500 500 1000
#> t-2 625 625 1250
#> t-1 609 609 1218
#> t   558 558 1116
```

Notice that the rebalanced portfolio slightly outperforms the buy-and-hold portfolio.

The portfolio return each period can be computed using $R_{p,t} = \%\Delta V_{p,t}$:

```
diff(value.table[, "V.p"])/value.table[1:3, "V.p"]
#>     t-2     t-1       t
#>  0.2500 -0.0260 -0.0833
```

It can also be computed using $R_{p,t} = 0.5 \times R_{A,t} + 0.5 \times R_{B,t}$:

```
0.5*R.A + 0.5*R.B
#> [1]  0.2500 -0.0260 -0.0833
```

Here, we see that the rebalanced portfolio loses less money over month $t-1$ than the buy-and-hold portfolio.

In the above example, at each rebalancing date we sell the asset that has the best performance (sell high) and we buy the asset that has the worst performance (buy low). Hence, rebalancing forces the sage investing advice: "buy low and sell high".

1.3.1.3 Rebalance portfolio at specified dates

When the portfolio is rebalanced to a fixed initial portfolio, such as an equally weighted portfolio, on specific dates the portfolio return calculations become a hybrid of those used for scenarios 1 and 2. The portfolio return is computed as

$$R_{p,t} = x_{A,t-1} \times R_{A,t} + x_{B,t-1} \times R_{B,t},$$

where $x_{i,t-1}(i = A, B)$ are equal to the initial portfolio weights on the rebalancing dates and is computed like the buy-and-hold weights in between the rebalancing dates.

1.3.1.4 Actively managed portfolio

In an actively managed portfolio, the portfolio weights are chosen by the portfolio manager at each time period. Many mutual funds are managed in this way. The return on the actively managed portfolio is computed as

$$R_{p,t} = x_{A,t-1} \times R_{A,t} + x_{B,t-1} \times R_{B,t}$$

where $x_{A,t-1}$ and $x_{B,t-1}$ are the portolio weights chosen by the portfolio manager at the beginning of time t.

1.3.2 Continuously Compounded Portfolio Returns

The continuously compounded portfolio return is defined by Equation 1.23, where R_t is computed using the portfolio return Equation 1.31. However, notice that:

$$r_{p,t} = \ln(1 + R_{p,t}) = \ln\left(1 + \sum_{i=1}^{n} x_i R_{i,t}\right) \neq \sum_{i=1}^{n} x_i r_{i,t}, \tag{1.61}$$

where $r_{i,t}$ denotes the continuously compounded one-period return on asset i. Hence, the continuously compounded portfolio return is not a share weighted average of the individual asset continuously compounded returns. However, if the portfolio return $R_{p,t} = \sum_{i=1}^{n} x_i R_{i,t}$ is not too large then $r_{p,t} \approx R_{p,t}$ otherwise, $R_{p,t} > r_{p,t}$.

Example 1.20 (Compute continuously compounded portfolio returns)**.** Consider a portfolio of Microsoft and Starbucks stock with $x_{\text{msft}} = 0.25$, $x_{\text{sbux}} = 0.75$, $R_{\text{msft},t} = 0.0588$, $R_{\text{sbux},t} = -0.0503$ and $R_{p,t} = -0.02302$. Using (Equation 1.61), the continuously compounded portfolio return is: $r_{p,t} = \ln(1-0.02302) = \ln(0.977) = -0.02329$. Using $r_{\text{msft},t} = \ln(1+0.0588) = 0.0572$ and $r_{\text{sbux},t} = \ln(1-0.0503) = -0.05161$, notice that: $x_{\text{msft}}r_{\text{msft}} + x_{\text{sbux}}r_{\text{sbux}} = -0.02442 \neq r_{p,t}$

■

Because the continuously compounded portfolio return is not a share weighted average of the individual asset continuously compounded returns, the analysis of portfolios is typically performed using simple returns and not continuously compounded returns.

1.4 Return Calculations with Data in R

This section discusses representing time series data in R using `xts` objects, the calculation of returns from historical prices in R, as well as the graphical display of prices and returns.

1.4.1 Representing time series data using `xts` objects

The examples in this section are based on the daily adjusted closing price data for Microsoft and Starbucks stock over the period January 4, 1993 through December 31, 2014[5]. These data are available as the `xts` objects `msftDailyPrices` and `sbuxDailyPrices` in the R package **introcompfinr** [6]

[5]These are prices that are adjusted for dividends and stock splits. That is, any dividend payments have been included in the prices and historical prices have been divided by the split ratio associated with any stock splits.

[6]See the Appendix Working with Time Series Data in R for an overview of `xts` objects and Zivot (2016) for a comprehensive coverage.

```r
head(cbind(msftDailyPrices,sbuxDailyPrices), 3)
#>            MSFT SBUX
#> 1993-01-04 1.89 1.08
#> 1993-01-05 1.92 1.10
#> 1993-01-06 1.98 1.12
```

```r
class(msftDailyPrices)
#> [1] "xts" "zoo"
```

There are many different ways of representing a time series of data in R. For financial time series `xts` (extensible time series) objects from the **xts** package are especially convenient and useful. An `xts` object consists of two pieces of information: (1) a `matrix` of numeric data with different time series in the columns, (2) an R object representing the common time indexes associated with the rows of the data. `xts` objects extend and enhance the `zoo` class of time series objects from the **zoo** package written by Achim Zeileis (zoo stands for Z's ordered observations).

The `matrix` of time series data can be extracted from the `xts` object using the **xts** function `coredata()`:

```r
matrix.data = coredata(msftDailyPrices)
head(matrix.data, 3)
#>      MSFT
#> [1,] 1.89
#> [2,] 1.92
#> [3,] 1.98
```

The resulting `matrix` of data does not have any date information. The date index can be extracted using the **xts** function `index()`:

```r
date.index = index(msftDailyPrices)
head(date.index, 3)
#> [1] "1993-01-04" "1993-01-05" "1993-01-06"
```

The object `date.index` is of class `Date`:

```r
class(date.index)
#> [1] "Date"
```

When a `Date` object is printed, the date is displayed in the format `yyyy-mm-dd`. However, internally each date represents the number of days since January 1, 1970:

```r
as.numeric(date.index[1])
#> [1] 8404
```

Here, `1993-01-04` (January 4, 1993) is 8,404 days after January 1, 1970. This allows for simple *date arithmetic* (e.g. adding and subtracting dates, etc). More sophisticated date arithmetic is available using the functions in the **lubridate** package.

There are several advantages of using `xts` objects to represent financial time series data. Most financial time series do not follow an equally spaced regular calendar. The time index of a `xts` object can be any strictly increasing date sequence which can match the times and dates for which assets trade on an exchange. For example, assets traded on the main US stock exchanges (e.g. NYSE and NASDAQ) trade only on weekdays between 9:30AM and 4:00PM Eastern Standard Time and do not trade on a number of holidays.

Another advantage of `xts` objects is that you can easily extract observations at or between specific dates. For example, to extract the daily price on January 3, 2014 use

```r
msftDailyPrices["2014-01-03"]
#>            MSFT
#> 2014-01-03 35.7
```

To extract prices between January 3, 2014 and January 7, 2014, use

```r
msftDailyPrices["2014-01-03::2014-01-07"]
#>            MSFT
#> 2014-01-03 35.7
#> 2014-01-06 34.9
#> 2014-01-07 35.2
```

Or to extract all of the prices for January, 2014 use:

```r
msftDailyPrices["2014-01"]
#>            MSFT
#> 2014-01-02 35.9
#> 2014-01-03 35.7
#> 2014-01-06 34.9
#> 2014-01-07 35.2
#> 2014-01-08 34.6
#> 2014-01-09 34.3
#> 2014-01-10 34.8
#> 2014-01-13 33.8
#> 2014-01-14 34.6
#> 2014-01-15 35.5
```

```
#> 2014-01-16 35.6
#> 2014-01-17 35.2
#> 2014-01-21 35.0
#> 2014-01-22 34.7
#> 2014-01-23 34.9
#> 2014-01-24 35.6
#> 2014-01-27 34.8
#> 2014-01-28 35.0
#> 2014-01-29 35.4
#> 2014-01-30 35.6
#> 2014-01-31 36.6
```

Two or more `xts` objects can be merged together and aligned to a common date index using the **xts** function `merge()`:

```
msftSbuxDailyPrices = merge(msftDailyPrices, sbuxDailyPrices)
head(msftSbuxDailyPrices, n=3)
#>            MSFT SBUX
#> 1993-01-04 1.89 1.08
#> 1993-01-05 1.92 1.10
#> 1993-01-06 1.98 1.12
```

If two or more `xts` objects have a common date index then they can be combined using `cbind()`.

1.4.1.1 Basic calculations on `xts` objects.

Because `xts` objects typically contain a matrix of numeric data, many R functions that operate on matrices also operate on `xts` objects. For example, to transform `msftDailyPrices` to log prices use

```
msftDailyLogPrices = log(msftDailyPrices)
head(msftDailyLogPrices, 3)
#>             MSFT
#> 1993-01-04 0.637
#> 1993-01-05 0.652
#> 1993-01-06 0.683
```

If two or more `xts` objects have the same time index then they can be added, subtracted, multiplied, and divided. For example,

```r
msftPlusSbuxDailyPrices = msftDailyPrices + sbuxDailyPrices
colnames(msftPlusSbuxDailyPrices) = "MSFT+SBUX"
head(msftPlusSbuxDailyPrices,3)
#>            MSFT+SBUX
#> 1993-01-04      2.97
#> 1993-01-05      3.02
#> 1993-01-06      3.10
```

To compute the average of the Microsoft daily prices use

```r
mean(msftDailyPrices)
#> [1] 19.9
```

If an R function, for some unknown reason, is not operating correctly on an `xts` object first extract the data using `coredata()` and then call the R function. For example,

```r
mean(coredata(msftDailyPrices))
#> [1] 19.9
```

1.4.1.2 Changing the frequency of an `xts` object

Data at the daily frequency is the highest frequency of data considered in this book. However, in many of the examples we want to use data at the weekly or monthly frequency. The conversion of data at a daily frequency to data at a monthly frequency is easy to do with `xts` objects. For example, end-of-month prices can be extracted from the daily prices using the `xts` function `to.monthly()`:

```r
msftMonthlyPrices = to.monthly(msftDailyPrices, OHLC=FALSE)
sbuxMonthlyPrices = to.monthly(sbuxDailyPrices, OHLC=FALSE)
msftSbuxMonthlyPrices = to.monthly(msftSbuxDailyPrices, OHLC=FALSE)
head(msftMonthlyPrices, 3)
#>          MSFT
#> Jan 1993 1.92
#> Feb 1993 1.85
#> Mar 1993 2.06
```

By default, `to.monthly()` extracts the data for the last day of the month and creates a **zoo** `yearmon` date index. For monthly data, the `yearmon` date index is convenient for printing and plotting as the month and year are nicely printed. To preserve the `Date` class of the time index and show the end-of-month date, use the optional argument `indexAt = "lastof"` in the call to `to.monthly()`:

```r
head(to.monthly(msftDailyPrices, OHLC=FALSE, indexAt="lastof"), 3)
#>            MSFT
#> 1993-01-31 1.92
#> 1993-02-28 1.85
#> 1993-03-31 2.06
```

In the above calls to `to.monthly()`, the optional argument `OHLC=FALSE` prevents the creation of open, high, low, and closing prices for the month.

In a similar fashion, you can extract end-of-week prices using the `xts` function `to.weekly()`:

```r
msftWeeklyPrices = to.weekly(msftDailyPrices, OHLC=FALSE)
head(msftWeeklyPrices, 3)
#>            MSFT
#> 1993-01-08 1.94
#> 1993-01-15 2.00
#> 1993-01-22 1.99
```

Here, the weekly data are the closing prices on each Friday of the week:

```r
head(weekdays(index(msftWeeklyPrices)),3)
#> [1] "Friday" "Friday" "Friday"
```

1.4.1.3 Plotting `xts` objects with `plot.xts()` and `plot.zoo`

Time plots of `xts` objects can be created with the generic `plot()` function as there is a method function in the **xts** package for objects of class `xts`. For example, Figure Figure 1.1 shows a basic time plot of the monthly closing prices of Microsoft and Starbucks created with:

```r
plot(msftSbuxMonthlyPrices, main="Monthly Closing Prices",
     legend.loc="topleft")
```

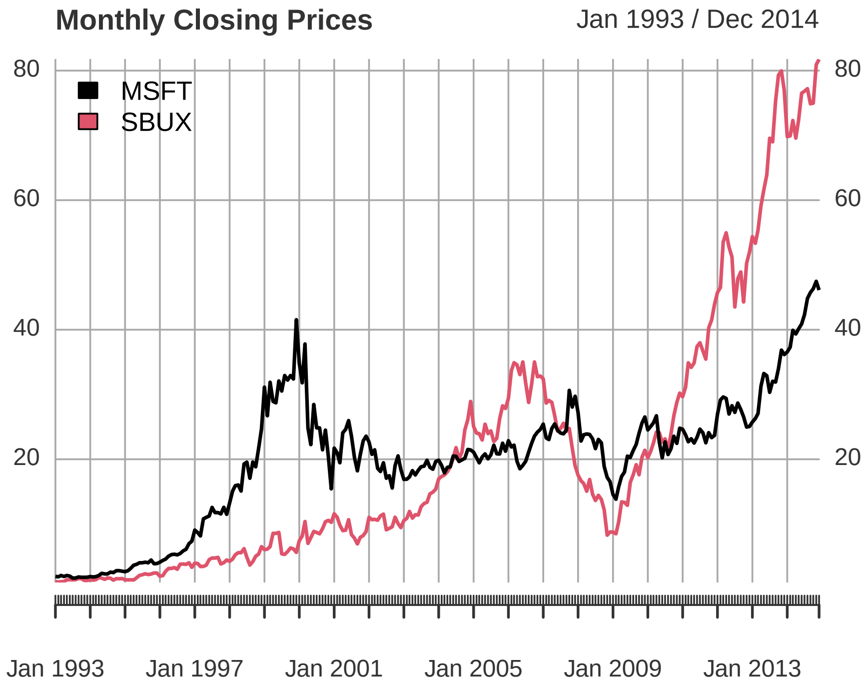


Figure 1.1: Single panel plot (default) with `plot.xts()`

The default plot style in `plot.xts()` is a single-panel plot with multiple series. You can also create multi-panel plots, as in Figure Figure 1.2, by setting the optional argument `multi.panel = TRUE` in the call to `plot.xts()`

```
plot(msftSbuxMonthlyPrices, main="Monthly Closing Prices",
     multi.panel=TRUE)
```

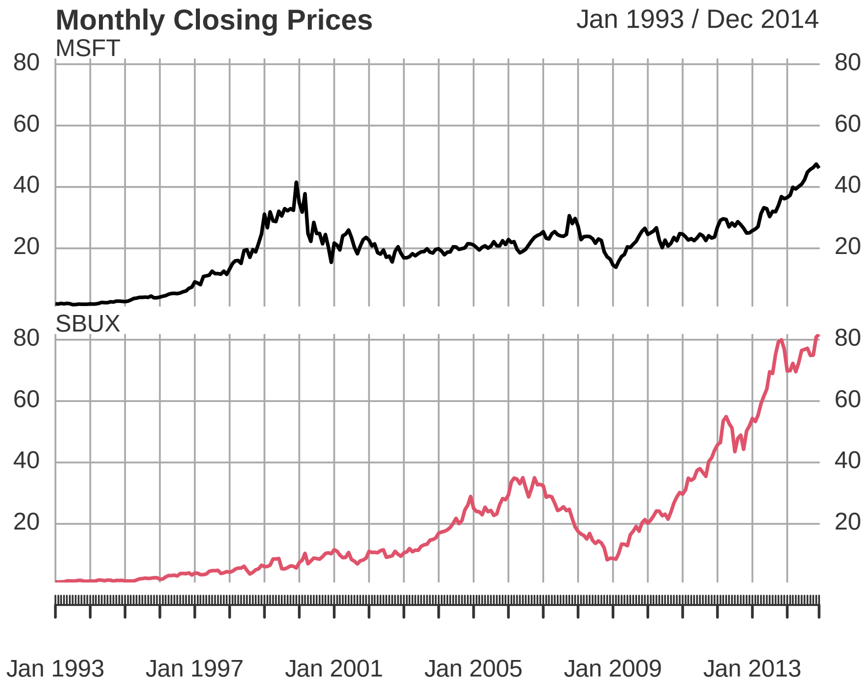


Figure 1.2: Multi-panel time series plot with `plot.xts()`

See the help file for `plot.xts()` for more examples of plotting `xts` objects.

Because the `xts` class inherits from the `zoo` class, the **zoo** method function `plot.zoo()` can also be used to plot `xts` objects. Figure Figure 1.3 is created with

```
plot.zoo(msftSbuxMonthlyPrices, plot.type="single",
         main="Monthly Closing Prices",
         lwd = 2, col=c("black", "red"), ylab="Price")
grid()
legend(x="topleft", legend=colnames(msftSbuxMonthlyPrices),
      lwd = 2, col=c("black", "red"))
```

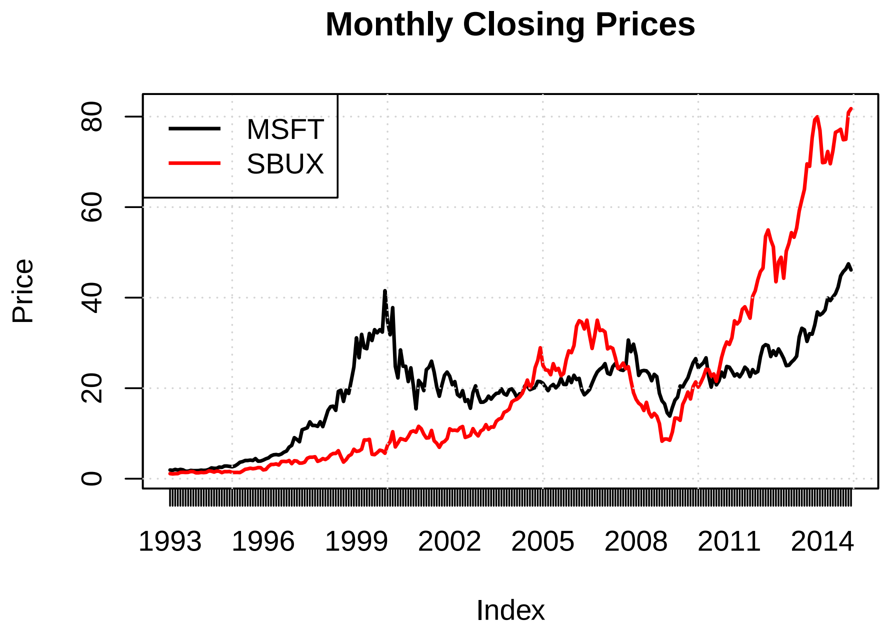


Figure 1.3: Single panel time series plot (default) with `plot.zoo()`

A muti-panel plot, shown in Figure Figure 1.4, can be created using

```
plot.zoo(msftSbuxMonthlyPrices, plot.type="multiple", main="",
         lwd = 2, col=c("black", "red"), cex.axis=0.8)
```

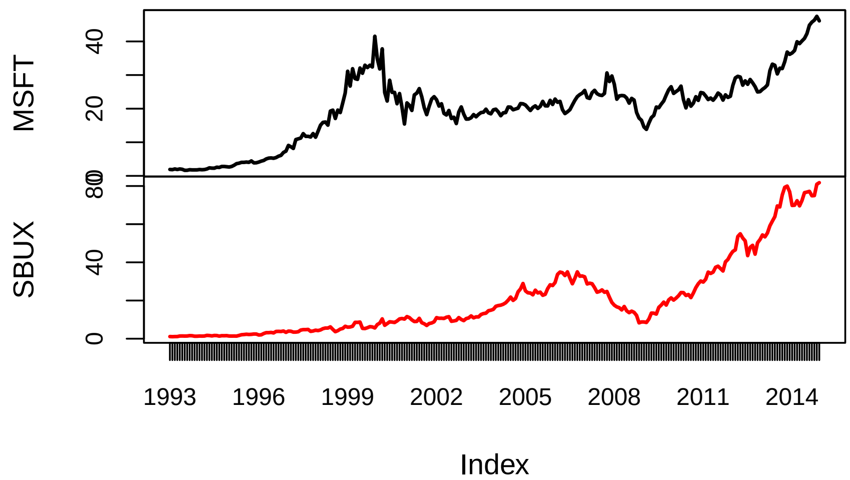


Figure 1.4: Multi-panel time series plot with `plot.zoo()`

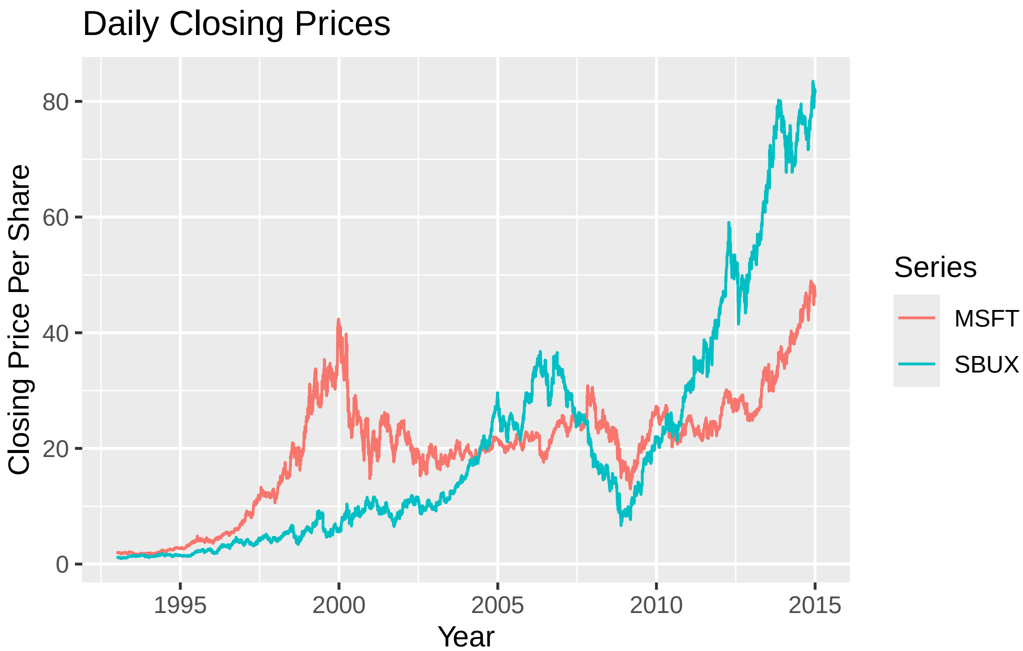
Creating plots with `plot.zoo()` is a bit more involved than with `plot.xts()`, as `plot.xts()` is newer. See the help file for `plot.zoo()` for more examples.

1.4.1.4 Ploting `xts` objects using `autoplot()` from ggplot2

The plots created using `plot.xts()` and `plot.zoo()` are created using the base graphics system in R. This allows for a lot of flexibility but often the resulting plots do not look very exciting or modern. In addition, these functions are not well suited for plotting many time series together in a single plot.

Another very popular graphics system in R is provided by the **ggplot2** package by Hadley Wickham of RStudio. For plotting `xts` objects, especially with multiple columns (data series), the **ggplot2** function `autoplot()` is especially convenient and easy:

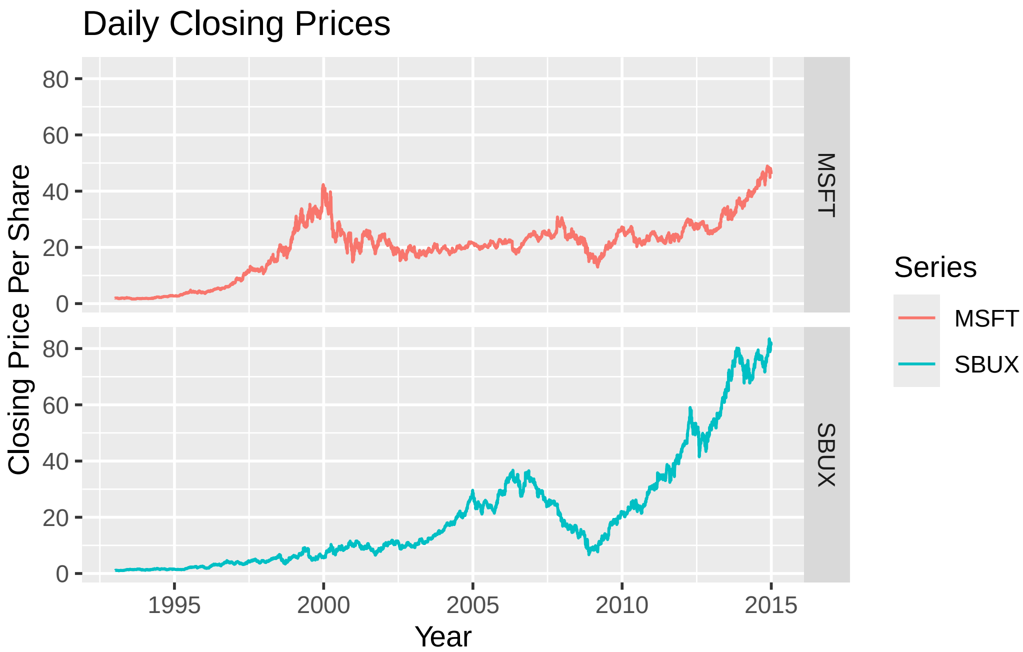
```
autoplot(msftSbuxDailyPrices, facets = NULL) +
  ggtitle("Daily Closing Prices") +
  ylab("Closing Price Per Share") +
  xlab("Year")
```



The syntax for creating graphs with `autoplot()` uses the *grammar of graphics* syntax from the **ggplot2** package. See the R Graphics Cookbook for more details and examples on using the **ggplot2** package. The call to `autoplot()` first invokes the method function `autoplot.zoo()` from the **zoo** package and creates a basic time series plot. Additional layers showing a main title and axes labels are added using `+`. The optional argument `facets = NULL` specifies that all series are plotted together on a single plot with different colors.

To produce a multi-panel plot call `autoplot()` with `facets = Series ~ .`:

```
autoplot(msftSbuxDailyPrices, facets = Series ~ .) +
  ggtitle("Daily Closing Prices") +
  ylab("Closing Price Per Share") +
  xlab("Year")
```



See the help file for `autoplot.zoo()` for more examples.

1.4.2 Calculating returns

In this sub-section, we illustrate how to calculate a time series of returns from a time series of prices.

1.4.2.1 Brute force return calculations

Consider computing simple monthly returns, $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$, from historical prices using the `xts` object `sbuxMonthlyPrices`. The R code for a brute force calculation is:

```
sbuxMonthlyReturns = diff(sbuxMonthlyPrices)/stats::lag(sbuxMonthlyPrices)
head(sbuxMonthlyReturns, n=3)
#>             SBUX
#> Jan 1993      NA
#> Feb 1993 -0.0625
#> Mar 1993  0.0476
```

Here, the `diff()` function computes the first difference in the prices, $P_t - P_{t-1}$, and the `lag()` function computes the lagged price, P_{t-1}.[7] An equivalent calculation is $R_t = \frac{P_t}{P_{t-1}} - 1$:

```
head(sbuxMonthlyPrices/stats::lag(sbuxMonthlyPrices) - 1, n=3)
#>              SBUX
#> Jan 1993       NA
#> Feb 1993 -0.0625
#> Mar 1993  0.0476
```

Notice that the return for January, 1993 is `NA` (missing value). To automatically remove this missing value, use the R function `na.omit()` or explicitly exclude the first observation:

```
head(na.omit(sbuxMonthlyReturns), n=3)
#>              SBUX
#> Feb 1993 -0.0625
#> Mar 1993  0.0476
#> Apr 1993  0.0182
```

```
head(sbuxMonthlyReturns[-1], n=3)
#>              SBUX
#> Feb 1993 -0.0625
#> Mar 1993  0.0476
#> Apr 1993  0.0182
```

To compute continuously compounded returns from simple returns, use:

```
sbuxMonthlyReturnsC = na.omit(log(1 + sbuxMonthlyReturns))
head(sbuxMonthlyReturnsC, n=3)
#>              SBUX
#> Feb 1993 -0.0645
#> Mar 1993  0.0465
#> Apr 1993  0.0180
```

Or, equivalently, to compute continuously compounded returns directly from prices, use:

```
sbuxMonthlyReturnsC = na.omit(diff(log(sbuxMonthlyPrices)))
head(sbuxMonthlyReturnsC, n=3)
#>              SBUX
```

[7] The method function `lag.xts()` works differently than generic `lag()` function and the method function `lag.zoo()`. In particular, `lag.xts()` with optional argument `k=1` gives the same result as `lag()` and `lag.zoo()` with `k=-1`.

```
#> Feb 1993 -0.0645
#> Mar 1993  0.0465
#> Apr 1993  0.0180
```

The above calculations used an `xts` object with a single column. If an `xts` object has multiple columns the same calculations work for each column. For example,

```
msftSbuxMonthlyReturns = diff(msftSbuxMonthlyPrices)/stats::lag(msftSbuxMonthlyPrices)
head(msftSbuxMonthlyReturns, n=3)
#>             MSFT    SBUX
#> Jan 1993      NA      NA
#> Feb 1993 -0.0365 -0.0625
#> Mar 1993  0.1135  0.0476
```

Also,

```
msftSbuxMonthlyReturnsC = diff(log(msftSbuxMonthlyPrices))
head(msftSbuxMonthlyReturnsC, n=3)
#>             MSFT    SBUX
#> Jan 1993      NA      NA
#> Feb 1993 -0.0371 -0.0645
#> Mar 1993  0.1075  0.0465
```

These are examples of *vectorized* calculations. That is, the calculations are performed on all columns simultaneously. This is computationally more efficient than looping the same calculation for each column. In R, you should try to vectorize calculations whenever possible.

1.4.2.2 Calculating returns using `Return.calculate()`

Simple and continuously compounded returns can be computed using the **PerformanceAnalytics** function `Return.calculate()`:

```
# Simple returns
sbuxMonthlyReturns = Return.calculate(sbuxMonthlyPrices)
head(sbuxMonthlyReturns, n=3)
#>             SBUX
#> Jan 1993      NA
#> Feb 1993 -0.0625
#> Mar 1993  0.0476
```

```
# CC returns
sbuxMonthlyReturnsC = Return.calculate(sbuxMonthlyPrices, method="log")
head(sbuxMonthlyReturnsC, n=3)
#>             SBUX
#> Jan 1993      NA
#> Feb 1993 -0.0645
#> Mar 1993  0.0465
```

`Return.calculate()` also works with `xts` objects with multiple columns:

```
msftSbuxMonthlyReturns = Return.calculate(msftSbuxMonthlyPrices)
head(msftSbuxMonthlyReturns, 3)
#>             MSFT    SBUX
#> Jan 1993      NA      NA
#> Feb 1993 -0.0365 -0.0625
#> Mar 1993  0.1135  0.0476
```

The advantages of using `Return.calculate()` instead of brute force calculations are: (1) clarity of code (you know what is being computed), (2) the same function is used for both simple and continuously compounded returns, (3) built-in error checking.

1.4.2.3 Equity curves

To directly compare the investment performance of two or more assets, plot the simple multi-period cumulative returns of each asset on the same graph. This type of graph, sometimes called an *equity curve*, shows how a one dollar investment amount in each asset grows over time. Better performing assets have higher equity curves. For simple returns, the k-period returns are $R_t(k) = \prod_{j=0}^{k-1}(1+R_{t-j})$ and represent the growth of one dollar invested for k periods. For continuously compounded returns, the k-period returns are $r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$. However, this continuously compounded k-period return must be converted to a simple k-period return, using $R_t(k) = \exp\left(r_t(k)\right) - 1$, to properly represent the growth of one dollar invested for k periods.

Example 1.21 (Equity curves for Microsoft and Starbucks monthly returns)**.** To create the equity curves for Microsoft and Starbucks based on simple returns use (You can also use the **PerformanceAnalytics** function `chart.CumReturns()`)

```
msftMonthlyReturns = na.omit(Return.calculate(msftMonthlyPrices))
sbuxMonthlyReturns = na.omit(Return.calculate(sbuxMonthlyPrices))
equityCurveMsft = cumprod(1 + msftMonthlyReturns)
```

```r
equityCurveSbux = cumprod(1 + sbuxMonthlyReturns)
dataToPlot = merge(equityCurveMsft, equityCurveSbux)
plot(dataToPlot, main="Monthly Equity Curves", legend.loc="topleft")
```

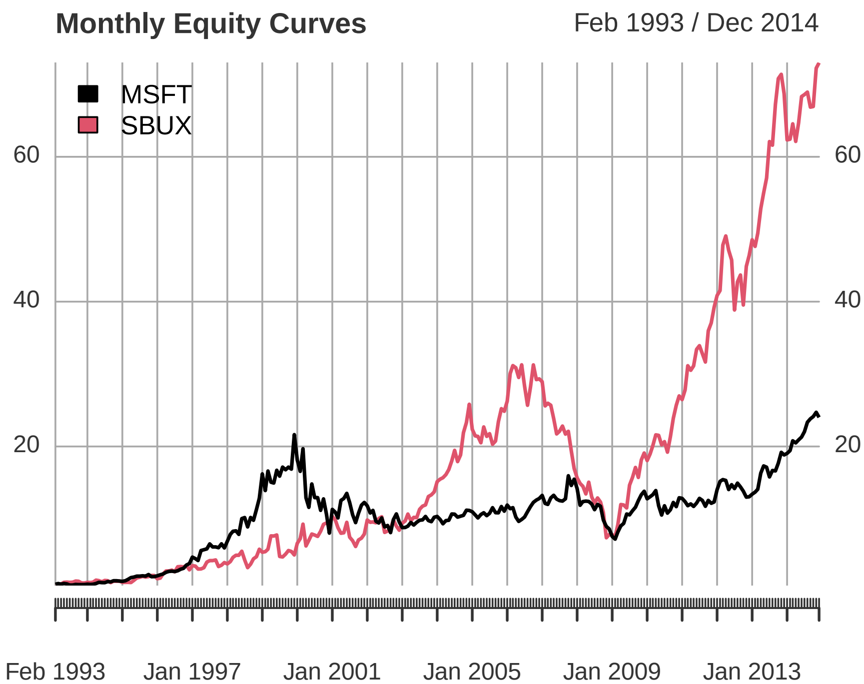


Figure 1.5: Monthly equity curve for Microsoft and Starbucks.

The R function `cumprod()` creates the cumulative products needed for the equity curves. Figure Figure 1.5 shows that a one dollar investment in Starbucks dominated a one dollar investment in Microsoft over the given period. In particular, \$1 invested in Microsoft grew to about only \$22 (over about 20 years) whereas \$1 invested in Starbucks grew over \$70. Notice the huge increases and decreases in value of Microsoft during the dot-com bubble and bust over the period 1998 - 2001, and the enormous increase in value of Starbucks from 2009 - 2014.

■

1.4.3 Calculating portfolio returns from time series data

As discussed in sub-section Section 1.3, the calculation of portfolio returns depends on the assumptions made about the portfolio weights over time.

1.4.3.1 Constant portfolio weights

The easiest case assumes that portfolio weights are constant over time. To illustrate, consider an equally weighted portfolio of Microsoft and Starbucks stock. A time series a portfolio monthly returns can be created directly from the monthly returns on Microsoft and Starbucks stock:

```
x.msft = x.sbux = 0.5
msftMonthlyReturns = na.omit(Return.calculate(msftMonthlyPrices))
sbuxMonthlyReturns = na.omit(Return.calculate(sbuxMonthlyPrices))
equalWeightPortMonthlyReturns = x.msft*msftMonthlyReturns +
  x.sbux*sbuxMonthlyReturns
head(equalWeightPortMonthlyReturns, 3)
#>              MSFT
#> Feb 1993 -0.0495
#> Mar 1993  0.0806
#> Apr 1993 -0.0297
```

The same calculation can be performed using the **PerformanceAnalytics** function `Return.portfolio()`:

```
equalWeightPortMonthlyReturns =
  Return.portfolio(na.omit(msftSbuxMonthlyReturns),
                   weights = c(x.msft, x.sbux),
                   rebalance_on = "months")
head(equalWeightPortMonthlyReturns, 3)
#>          portfolio.returns
#> Feb 1993           -0.0495
#> Mar 1993            0.0806
#> Apr 1993           -0.0297
```

Here, the optional argument `rebalance_on = "months"` specifies that the portfolio is to be rebalanced to the specified weights at the beginning of every month.

If you also want the asset contributions to the portfolio return set the optional argument `contribution = TRUE` (recall, asset contributions sum to the portfolio return):

```
equalWeightPortMonthlyReturns =
  Return.portfolio(na.omit(msftSbuxMonthlyReturns),
                   weights = c(x.msft, x.sbux),
                   rebalance_on = "months",
                   contribution = TRUE)
```

```r
head(equalWeightPortMonthlyReturns, 3)
#>          portfolio.returns    MSFT     SBUX
#> Feb 1993           -0.0495 -0.0182 -0.03125
#> Mar 1993            0.0806  0.0568  0.02381
#> Apr 1993           -0.0297 -0.0388  0.00909
```

You can see the rebalancing calculations by setting the optional argument `verbose = TRUE`:

```r
equalWeightPortMonthlyReturns =
  Return.portfolio(na.omit(msftSbuxMonthlyReturns),
                   weights = c(x.msft, x.sbux),
                   rebalance_on = "months",
                   verbose = TRUE)
names(equalWeightPortMonthlyReturns)
#> [1] "returns"      "contribution" "BOP.Weight"   "EOP.Weight"   "BOP.Value"
#> [6] "EOP.Value"
```

In this case `Return.portfolio()` returns a list with the above components. The component `BOP.Weight` shows the beginning-of-period portfolio weights, and the component `EOP.Weight` shows the end-of-period portfolio weights (before any rebalancing).

```r
head(equalWeightPortMonthlyReturns$BOP.Weight, 3)
#>          MSFT SBUX
#> Feb 1993  0.5  0.5
#> Mar 1993  0.5  0.5
#> Apr 1993  0.5  0.5
```

With monthly returns and with `rebalance_on = "months"`, the beginning-of-period weights are always rebalanced to the initially supplied weights.

```r
head(equalWeightPortMonthlyReturns$EOP.Weight, 3)
#>           MSFT  SBUX
#> Feb 1993 0.507 0.493
#> Mar 1993 0.515 0.485
#> Apr 1993 0.475 0.525
```

Because the returns for Microsoft and Starbux are different every month the end-of-period weights drift away from the beginning-of-period weights.

1.4.3.2 Buy-and-hold portfolio

In the buy-and-hold portfolio, no rebalancing of the initial portfolio is done and the portfolio weights are allowed to evolve over time based on the returns of the assets. Hence, the computation of the portfolio return each month requires knowing the weights on each asset each month. You can use `Return.portfolio()` to compute the buy-and-hold portfolio returns as follows:

```
buyHoldPortMonthlyReturns =
  Return.portfolio(na.omit(msftSbuxMonthlyReturns),
                   weights = c(x.msft, x.sbux))
head(buyHoldPortMonthlyReturns, 3)
#>          portfolio.returns
#> Feb 1993           -0.0495
#> Mar 1993            0.0810
#> Apr 1993           -0.0319
```

Omitting the optional argument `rebalance_on` tells `Return.portfolio()` to not rebalance the portfolio at all (default behavior of the function).

1.4.3.3 Periodic rebalancing

In practice, a portfolio may be rebalanced at a specific frequency that is different than the frequency of the returns. For example, with monthly returns the portfolio might be rebalanced quarterly or annually. For example, to rebalance the equally weighted portfolio of Microsoft and Starbucks stock each quarter (every three months) use

```
equalWeightQtrPortMonthlyReturns =
  Return.portfolio(na.omit(msftSbuxMonthlyReturns),
                   weights = c(x.msft, x.sbux),
                   rebalance_on = "quarters")
head(equalWeightQtrPortMonthlyReturns, 3)
#>          portfolio.returns
#> Feb 1993           -0.0495
#> Mar 1993            0.0810
#> Apr 1993           -0.0297
```

In between quarters, the portfolio evolves as a buy-and-hold portfolio. To see the details of the rebalancing set `verbose = TRUE` in the call to `Return.Portfolio()`.

1.4.3.4 Active portfolio weights

If the weights are actively chosen at specific time periods, these weights can be supplied to the function `Return.portfolio()` to compute the portfolio return.

1.4.4 Downloading financial data from the internet

The data for the examples in this book, available in the **introcompFinr** package, were downloaded from finance.yahoo.com using the `getSymbols()` function from the **quantmod** package.

For example, to download daily data on Amazon and Google stock (ticker symbols AMZN and GOOG) from January 3, 2007 through January 3, 2020 use `getSymbols()` as follows:

```
options("getSymbols.warning4.0"=FALSE)
suppressPackageStartupMessages(library(quantmod))
getSymbols(Symbols = c("AMZN", "GOOG"), from="2007-01-03", to="2020-01-03",
           auto.assign=TRUE, warnings=FALSE)
#> [1] "AMZN" "GOOG"
class(AMZN)
#> [1] "xts" "zoo"
```

The `xts` objects `AMZN` and `GOOG` each have six columns containing the daily open price for the day, high price for the day, low price for the day, close price for the day, volume for the day, and (dividend and split) adjusted closing price for the day:

```
head(AMZN, 3)
#>            AMZN.Open AMZN.High AMZN.Low AMZN.Close AMZN.Volume AMZN.Adjusted
#> 2007-01-03      1.93      1.95     1.90       1.93    2.48e+08          1.93
#> 2007-01-04      1.93      1.96     1.91       1.95    1.26e+08          1.95
#> 2007-01-05      1.94      1.94     1.88       1.92    1.32e+08          1.92
```

```
head(GOOG, 3)
#>            GOOG.Open GOOG.High GOOG.Low GOOG.Close GOOG.Volume GOOG.Adjusted
#> 2007-01-03      11.6      11.9     11.5       11.6    3.09e+08          11.5
#> 2007-01-04      11.7      12.1     11.7       12.0    3.17e+08          11.9
#> 2007-01-05      12.0      12.1     11.9       12.1    2.76e+08          12.0
```

1.5 Further Reading: Return Calculations

This chapter describes basic asset return calculations with an emphasis on equity calculations. Similar material is covered in Chapter 2 of (Ruppert and Matteson 2015). Comprehensive coverage of general return calculations is given in (Bacon 2008) and (Christopherson, Carino, and Ferson 2009). The virtues of portfolio rebalancing are discussed in (Ang 2014) and (Qian 2019).

This chapter introduced you to the R packages **PerformanceAnalytics**, **quantmod**, and **xts** for working with financial time series. It is recommended to read the vignettes included in each of these packages. Jeff Ryan's original quantmod page, while a bit outdated, has many useful examples of working with **quantmod** and **xts**. A good online tutorial for the **xts** package is provided by Joshua Ulrich, one of the creators of **xts**.

There are many useful R packages for the analysis of financial data. An up-to-date list of R packages relevant for finance is given on the Empirical Finance Task View on the comprehensive R archive network (CRAN).

1.6 Appendix: Properties of Exponentials and Logarithms

The computation of continuously compounded returns requires the use of natural logarithms. The natural logarithm function, $\ln(\cdot)$, is the inverse of the exponential function, $e^{(\cdot)} = \exp(\cdot)$, where $e^1 = 2.718$. That is, $\ln(x)$ is defined such that $x = \ln(e^x)$. Figure Figure 1.6 plots e^x and $\ln(x)$. Notice that e^x is always positive and increasing in x. $\ln(x)$ is monotonically increasing in x and is only defined for $x > 0$. Also note that $\ln(1) = 0$ and $\ln(0) = -\infty$.

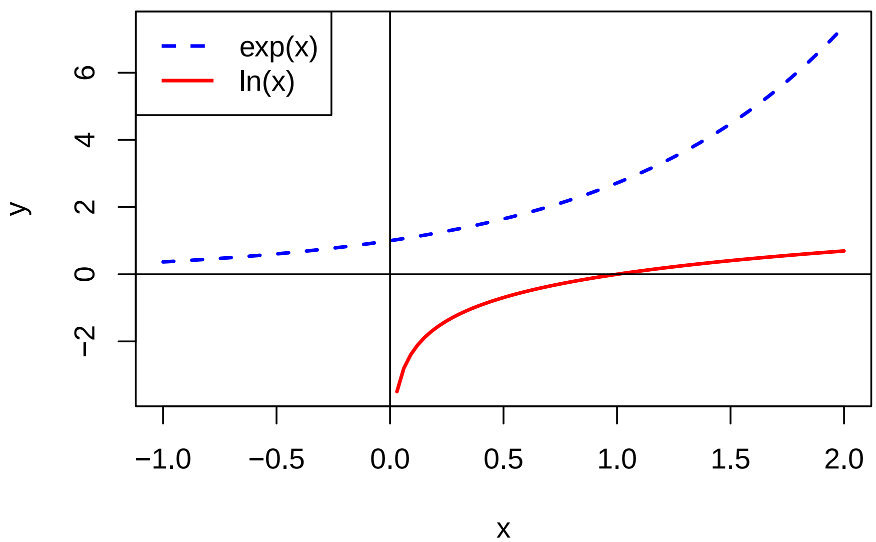


Figure 1.6: Exponential and natural logarithm functions.

The exponential and natural logarithm functions have the following properties:

1. $\ln(x \cdot y) = \ln(x) + \ln(y)$, $x, y > 0$
2. $\ln(x/y) = \ln(x) - \ln(y)$, $x, y > 0$
3. $\ln(x^y) = y \ln(x)$, $x > 0$
4. $\frac{d \ln(x)}{dx} = \frac{1}{x}$, $x > 0$
5. $\frac{d}{dx} \ln(f(x)) = \frac{1}{f(x)} \frac{d}{dx} f(x)$ (chain-rule)
6. $e^x e^y = e^{x+y}$
7. $e^x e^{-y} = e^{x-y}$
8. $(e^x)^y = e^{xy}$
9. $e^{\ln(x)} = x$
10. $\frac{d}{dx} e^x = e^x$

11. $\frac{d}{dx}e^{f(x)} = e^{f(x)}\frac{d}{dx}f(x)$ (chain-rule)

1.7 Exercises

Analytical Problems

Exercise 1.1. In this question you will see how the future value of an investment, FV_n varies with the interest rate, R, and the investment horizon, n. Let the initial amount invested be $V = \$1,000$. For a grid of values $R = 0.01, 0.02, \ldots, 0.10$ and $n = 1, 2, \ldots 10$, compute and plot FV_n.

Exercise 1.2. Consider the following credit card offer from a hypothetical bank. The card advertises an annual percentage rate (APR) of 19.34%. However, the bank actually compounds interest charges on a daily basis.

1. Assuming 365 days in the year, what is the daily periodic interest rate being charged?
2. If you carried a $1,000 balance throughout the year, how much would you owe at the end of the year?

Exercise 1.3. Consider "payday loans". Payday loans are short-term loans made to consumers, often for less than two weeks, and are offered by companies such as AmeriCash Advance and National Payday. The loan works like this: You write a check today that is postdated and the company gives you an amount of money less than the check. When the check date arrives, you go to the store and pay cash for the check, or the company cashes the check. For example, suppose AmeriCash Advance allows you to write a postdated check for $125 for 15 days later in return for a $100 loan today.

1. What is the annual percentage rate (APR) on this loan?
2. What is the effective annual rate (EAR)?

Exercise 1.4. Consider the following (actual) monthly adjusted closing price data for Starbucks stock over the period December 2004 through December 2005

Table 1.2: End of Month Price Data for Starbucks Stock

Date	Price
December, 2004	31.2
January, 2005	27.0
February, 2005	25.9
March, 2005	25.8
April, 2005	24.8
May, 2005	27.4
June, 2005	25.8
July, 2005	26.3

Date	Price
August, 2005	24.5
September, 2005	25.1
October, 2005	28.3
November, 2005	30.4
December, 2005	30.5

For the following questions, first do the return calculations by hand. That is, give the appropriate return calculation formula and evaluate the result. Then, check your hand calculations using the corresponding R calculations.

1. Using the data in the table, what is the simple monthly return between the end of December 2004 and the end of January 2005? If you invested $10,000 in Starbucks at the end of December 2004, how much would the investment be worth at the end of January 2005?
2. Using the data in the table, what is the continuously compounded monthly return between December 2004 and January 2005? Convert this continuously compounded return to a simple return (you should get the same answer as in part 1).
3. Assuming that the simple monthly return you computed in part 1 is the same for 12 months, what is the annual return with monthly compounding?
4. Assuming that the continuously compounded monthly return you computed in part 2 is the same for 12 months, what is the continuously compounded annual return?
5. Using the data in the table, compute the actual simple annual return between December 2004 and December 2005. If you invested $10,000 in Starbucks at the end of December 2004, how much would the investment be worth at the end of December 2005? Compare with your result in part 3.
6. Using the data in the table, compute the actual annual continuously compounded return between December 2004 and December 2005. Compare with your result in part 4. Convert this continuously compounded return to a simple return (you should get the same answer as in part 5).

Exercise 1.5. Consider a one month investment in two Northwest stocks: Amazon and Costco. Suppose you buy Amazon and Costco at the end of September at $P_{A,t-1} = \$38.23$, $P_{C,t-1} = \$41.11$ and then sell at the end of the October for $P_{A,t} = \$41.29$ and $P_{C,t} = \$41.74$. (Note: these are actual closing prices for 2004 taken from Yahoo!)

For the following questions, first do the return calculations by hand. That is, give the appropriate return calculation formula and evaluate the result. Then, check your hand calculations using the corresponding R calculations.

1. What are the simple monthly returns for the two stocks?
2. What are the continuously compounded returns for the two stocks?

3. Suppose Costco paid a \$0.10 per share cash dividend at the end of October. What is the monthly simple total return on Costco? What is the monthly dividend yield?
4. Suppose the monthly returns on Amazon and Costco from part 1 above are the same every month for 1 year. Compute the simple annual returns as well as the continuously compounded annual returns for the two stocks.
5. At the end of September 2004, suppose you have \$10,000 to invest in Amazon and Costco over the next month. If you invest \$8000 in Amazon and \$2000 in Costco, what are your portfolio shares, x_A and x_C?
6. Continuing with part 5, compute the monthly simple return and the monthly continuously compounded return on the portfolio. Assume that Costco does not pay a dividend.

Exercise 1.6. Consider a 48-month (4 year) investment in two assets: the Vanguard S&P 500 index (VFINX) and Amazon stock (AMZN). Suppose you buy one share of the S&P 500 fund and one share of Amazon stock at the end of January, 2015 for $P_{VFINX,t-48} = 165$, $P_{AMZN,t-48} = 355$, and then sell these shares at the end of January, 2019 for $P_{VFINX,t} = 241$,$P_{AMZN,t} = 1719$ (Note: these are actual adjusted closing prices taken from Yahoo!). In this question, you will see how much money you could have made if you invested in these assets right before the COVID-19 pandemic.

1. What are the simple 48-month (4-year) returns for the two investments?

2. What are the continuously compounded (cc) 48-month (4-year) returns for the two investments? Why are the cc returns smaller?

3. Suppose you invested \$10,000 in each asset at the end of January, 2015. How much would each investment be worth at the end of January, 2019?

4. What are the compound annual returns on the two 4-year investments?

5. At the end of January, 2015, suppose you plan to invest in a portfolio of VFINX and AMZN over the next 48 months (4 years). Suppose you purchase 5 shares of the VFINX mutual fund (at \$165/share) and 10 shares of AMZN stock (at \$355/share). What is the value of your initial investment? What are the portfolio weights in the two assets as of the end of January, 2015?

6. Using the results from part 1, compute the 4-year simple and cc portfolio returns.

7. What is the value of your portfolio at the end of January, 2019? What is the value of your investments in VFINX and AMZN? What is share of wealth invested in VFINX and AMZN (i.e., your portfolio weights)?

Exercise 1.7. Consider a 60-month (5 year) investment in two assets: the Vanguard S&P 500 index (VFINX) and Apple stock (AAPL). Suppose you buy one share of the S&P 500 fund and one share of Apple stock at the end of January, 2010 for $P_{VFINX,t-60} = \$89.91$, $P_{AAPL,t-60} = \$25.88$, and then sell these shares at the end of January, 2015 for $P_{VFINX,t} = \$184.2$, $P_{AAPL,t} = \$116.7$. (Note: these are actual adjusted closing prices taken from Yahoo!).

In this question, you will see how much money you could have made if you invested in these assets right after the 2008 financial crisis.

1. What are the simple 60-month (5-year) returns for the two investments?
2. What are the continuously compounded 60-month (5-year) returns for the two investments?
3. Are the simple and continuously compounded returns close to each other?
4. Suppose you invested \$1,000 in each asset at the end of January,
5. How much would each investment be worth at the end of January, 2015?
6. What is the compound annual return on the two 5 year investments?
7. At the end of January, 2010, suppose you have $\$1,000$ to invest in VFINX and AAPL over the next 60 months (5 years). Suppose you purchase \$400 worth of VFINX and the remainder in AAPL. What are the portfolio weights in the two assets? Using the results from parts 1 and 2, compute the 5-year simple and continuously compounded portfolio returns.

Exercise 1.8. Consider a one month investment in two Northwest stocks: Amazon and Costco. Suppose you buy Amazon and Costco at the end of September at $P_{A,t-1} = \$38.23, P_{C,t-1} = \$41.11$ and then sell at the end of the October for $P_{A,t} = \$41.29, P_{C,t} = \$41.74$. (Note: these are actual closing prices for 2004 taken from Yahoo!)

1. What are the simple monthly returns for the two stocks?
2. What are the continuously compounded returns for the two stocks?
3. Suppose Costco paid a \$0.10 per share cash dividend at the end of October. What is the monthly simple total return on Costco? What is the monthly dividend yield?
4. Suppose the monthly returns on Amazon and Costco from question (a) above are the same every month for 1 year. Compute the simple annual returns as well as the continuously compounded annual returns for the two stocks.
5. At the end of September, 2004, suppose you have $\$10,000$ to invest in Amazon and Costco over the next month. If you invest $\$8,000$ in Amazon and $\$2,000$ in Costco, what are your portfolio shares, x_A and x_C. (Assume partial share purchases are possible)
6. Continuing with the previous question, compute the monthly simple return and the monthly continuously compounded return on the portfolio. Assume that Costco does not pay a dividend.

Exercise 1.9. Consider an investment in a foreign stock (e.g., a stock trading on the London stock exchange) by a U.S. national (domestic investor). The domestic investor takes U.S. dollars, converts them to the foreign currency (e.g. British Pound) via the exchange rate (price of foreign currency in U.S. dollars) and then purchases the foreign stock using the foreign currency. When the stock is sold, the proceeds in the foreign currency must then be converted back to the domestic currency. To be more precise, consider the information in the table below:

Time	Cost of 1 Pound	Value of UK Shares	Value in U.S. $
0	\$1.50	£40	$1.5 \times 40 = 60$
1	\$1.30	£45	$1.3 \times 45 = 58.5$

1. Compute the simple rate of return, R_e, from the prices of foreign currency. This is the return to the domestic investor of investing in the foreign currency.
2. Compute the simple rate of return, R_{UK}, from the UK stock prices.
3. Compute the simple rate of return, R_{US}, from the prices in US dollars.
4. What is the mathematical relationship between R_{US}, R_{UK} and R_e?

Exercise 1.10. Consider an investing in a long-only portfolio of two assets A and B for one period between months $t-1$ and t. The beginning of period investment values in A and B are $V_{A,t-1} > 0$ and $V_{B,t-1} > 0$, and the one-month returns are $R_{A,t-1}$ and $R_{B,t-1}$. The initial value of the portfolio is $V_{p,t-1} = V_{A,t-1} + V_{B,t-1}$. The initial investment weights are $x_{A,t-1} = V_{A,t-1}/V_{p,t-1} > 0$ and $x_{B,t-1} = V_{B,t-1}/V_{p,t-1} > 0$ such that $x_{A,t-1} + x_{B,t-1} = 1$.

1. Show that
$$x_{A,t} = x_{A,t-1} \times \frac{1+R_{A,t-1}}{1+R_{p,t-1}},\ x_{B,t} = x_{B,t-1} \times \frac{1+R_{B,t-1}}{1+R_{p,t-1}},$$
and that $x_{A,t} + x_{B,t} = 1$.
2. Show that the change in portfolio weights are
$$\Delta x_{A,t} = x_{A,t} - x_{A,t-1} = \frac{x_{A,t-1}(R_{A,t-1} - R_{p,t-1})}{1+R_{p,t-1}},$$
$$\Delta x_{B,t} = x_{B,t} - x_{B,t-1} = \frac{x_{B,t-1}(R_{B,t-1} - R_{p,t-1})}{1+R_{p,t-1}}.$$
3. Under what conditions will $\Delta x_{A,t} > 0$, $\Delta x_{A,t} < 0$ and $\Delta x_{A,t} = 0$?
4. Using $R_{p,t-1} = x_{A,t-1}R_{A,t-1} + x_{B,t-1}R_{B,t-1}$ show that
$$\Delta x_{A,t} = \frac{x_{A,t-1}x_{B,t-1}(R_{A,t-1} - R_{B,t-1})}{1 + x_{A,t-1}R_{A,t-1} + x_{B,t-1}R_{B,t-1}}$$
5. Under what conditions will $\Delta x_{A,t} > 0$, $\Delta x_{A,t} < 0$ and $\Delta x_{A,t} = 0$?
6. To rebalance the portfolio at the end-of-month, you need to adjust the portfolio weights such that $x_{A,t} = x_{A,t-1}$ and $x_{B,t} = x_{B,t-1}$. If $R_{A,t-1} > R_{B,t-1}$ do you buy or sell asset A to rebalance the portfolio?

Return Calculations using R Packages

- Give examples of using the **PerformanceAnalytics package**
- Download different asset data from Yahoo!. Extract adjusted closing price data. Compute returns and plot returns and equity curves.

2 Review of Random Variables

Updated: August 30, 2026

Copyright © Eric Zivot 2026

This chapter reviews the probability concepts that are necessary for the modeling and statistical analysis of financial data presented in this book. This material is typically covered in an introductory probability and statistics course using calculus. In the course of the review, many examples related to finance will be presented and some important risk concepts, such as value-at-risk, will be introduced. The examples will also show how to use R for probability calculations and working with probability distributions.

The chapter is outlined as follows. Section Section 2.1 reviews univariate random variables and properties of univariate probability distributions. Important univariate distributions, such as the normal distribution, that are used throughout the book for modeling financial data, are described in detail. Section Section 2.2 covers bivariate probability distributions for two random variables. The concepts of dependence and independence between two random variables are discussed. The measures of linear dependence between two random variables, covariance and correlation, are defined and the bivariate normal distribution is introduced. Properties of linear combinations of two random variables are given and illustrated using an example to describe the return and risk properties of a portfolio of two assets. More than two random variables and Multivariate distributions are briefly discussed in section Section 2.3.

The R packages used in this chapter are **mvtnorm** and **sn**. Make sure these packages are downloaded and installed prior to replicating the R examples in this chapter.

2.1 Random Variables

We start with the basic definition of a *random variable*:

Definition 2.1 (Random variable)**.** A *random variable*, X, is a variable that can take on a given set of values, called the *sample space* and denoted S_X, where the likelihood of the values in S_X is determined by X's *probability distribution function* (pdf).

Example 2.1 (Future price of Microsoft stock)**.** Consider the price of Microsoft stock next month, P_{t+1}. Since P_{t+1} is not known with certainty today (t), we can consider it as a random variable. P_{t+1} must be positive, and realistically it can't get too large. Therefore, the sample space is the set of positive real numbers bounded above by some large number: $S_{P_{t+1}} = \{P_{t+1} : P_{t+1} \in [0, M],\ M > 0\}$. It is an open question as to what is the best characterization of the probability distribution of stock prices. The log-normal distribution is one possibility.[1] ■

Example 2.2 (Return on Microsoft stock)**.** Consider a one-month investment in Microsoft stock. That is, we buy one share of Microsoft stock at the end of month $t-1$ (e.g., end of February) and plan to sell it at the end of month t (e.g., end of March). The simple return over month t on this investment, $R_t = (P_t - P_{t-1})/P_{t-1}$, is a random variable because we do not know what the price will be at the end of the month. In contrast to prices, returns can be positive or negative and are bounded from below by -100%. We can express the sample space as $S_{R_t} = \{R_t : R_t \in [-1, M],\ M > 0\}$. The normal distribution is often a good approximation to the distribution of simple monthly returns, and is a better approximation to the distribution of continuously compounded monthly returns. ■

Example 2.3 (Up-down indicator variable)**.** As a final example, consider a variable X_{t+1} defined to be equal to one if the monthly price change on Microsoft stock, $P_{t+1} - P_t$, is positive, and is equal to zero if the price change is zero or negative. Here, the sample space is the set $S_{X_{t+1}} = \{0, 1\}$. If it is equally likely that the monthly price change is positive or negative (including zero) then the probability that $X_{t+1} = 1$ or $X_{t+1} = 0$ is 0.5. This is an example of a Bernoulli random variable. ■

The next sub-sections define discrete and continuous random variables.

2.1.1 Discrete random variables

Consider a random variable generically denoted X and its set of possible values or sample space denoted S_X.

Definition 2.2 (Discrete random variable)**.** A *discrete random variable* X is one that can take on a finite number of n different values $S_X = \{x_1, x_2, \ldots, x_n\}$ or, at most, a countably infinite number of different values $S_X = \{x_1, x_2, \ldots.\}$.

Definition 2.3 (Probability density function)**.** The pdf of a discrete random variable, denoted $p(x)$, is a function such that $p(x) = \Pr(X = x)$. The pdf must satisfy (i) $p(x) \geq 0$ for all $x \in S_X$; (ii) $p(x) = 0$ for all $x \notin S_X$; and (iii) $\sum_{x \in S_X} p(x) = 1$.

[1] If P_{t+1} is a positive random variable such that $\ln P_{t+1}$ is normally distributed then P_{t+1} has a log-normal distribution.

Example 2.4 (Annual return on Microsoft stock)**.**

Table 2.1: Probability distribution for the annual return on Microsoft.

State of Economy	Sample Space	$\Pr(X = x)$
Depression	-0.30	0.05
Recession	0.0	0.20
Normal	0.10	0.50
Mild Boom	0.20	0.20
Major Boom	0.50	0.05

Let X denote the annual return on Microsoft stock over the next year. We might hypothesize that the annual return will be influenced by the general state of the economy. Consider five possible states of the economy: depression, recession, normal, mild boom and major boom. A stock analyst might forecast different values of the return for each possible state. Hence, X is a discrete random variable that can take on five different values. Table Table 2.1 describes such a probability distribution of the return and a graphical representation of the probability distribution is presented in Figure **?@fig-ProbRevMsftDiscreteDistn** created with

```
r.msft = c(-0.3, 0, 0.1, 0.2, 0.5)
prob.vals = c(0.05, 0.20, 0.50, 0.20, 0.05)
plot(r.msft, prob.vals, type="h", lwd=4, xlab="annual return",
     ylab="probability", xaxt="n")
points(r.msft, prob.vals, pch=16, cex=2, col="blue")
axis(1, at=r.msft)
```

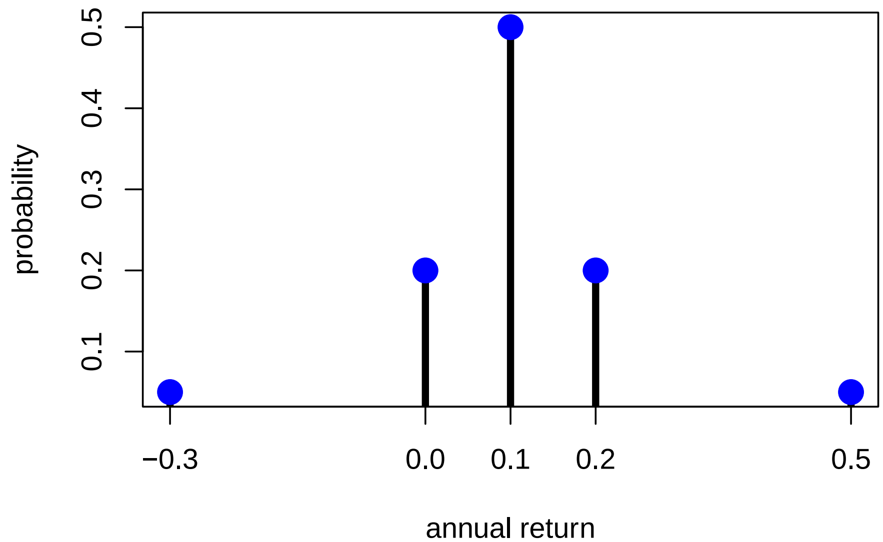


Figure 2.1: Discrete distribution for Microsoft stock.

The shape of the distribution is revealing. The center and most likely value is at 0.1 and the distribution is symmetric about this value (has the same shape to the left and to the right). Most of the probability is concentrated between 0 and 0.2 and there is very little probability associated with very large or small returns. ■

2.1.1.1 The Bernoulli distribution

Let $X = 1$ if the price next month of Microsoft stock goes up and $X = 0$ if the price goes down (assuming it cannot stay the same). Then X is clearly a discrete random variable with sample space $S_X = \{0, 1\}$. If the probability of the stock price going up or down is the same then $p(0) = p(1) = 1/2$ and $p(0) + p(1) = 1$.

The probability distribution described above can be given an exact mathematical representation known as the *Bernoulli* distribution. Consider two mutually exclusive events generically called "success" and "failure". For example, a success could be a stock price going up or a coin landing heads and a failure could be a stock price going down or a coin landing tails. The process creating the success or failure is called a *Bernoulli trial*. In general, let $X = 1$ if success occurs and let $X = 0$ if failure occurs. Let $\Pr(X = 1) = \pi$, where $0 < \pi < 1$, denote the probability of success. Then $\Pr(X = 0) = 1 - \pi$ is the probability of failure. A mathematical model describing this distribution is:

$$p(x) = \Pr(X = x) = \pi^x (1 - \pi)^{1-x}, \ x = 0, 1. \tag{2.1}$$

When $x = 0$, $p(0) = \pi^0 (1 - \pi)^{1-0} = 1 - \pi$ and when $x = 1, p(1) = \pi^1 (1 - \pi)^{1-1} = \pi$.

2.1.1.2 The binomial distribution

Consider a sequence of independent Bernoulli trials with success probability π generating a sequence of 0-1 variables indicating failures and successes. A *binomial* random variable X counts the number of successes in n Bernoulli trials, and is denoted $X \sim B(n, \pi)$. The sample space is $S_X = \{0, 1, \ldots, n\}$ and:

$$\Pr(X = k) = \binom{n}{k} \pi^k (1 - \pi)^{n-k}.$$

The term $\binom{n}{k}$ is the binomial coefficient, and counts the number of ways k objects can be chosen from n distinct objects. It is defined by:

$$\binom{n}{k} = \frac{n!}{(n-k)!k!},$$

where $n!$ is the factorial of n, or $n(n-1)\cdots 2 \cdot 1$.

2.1.2 Continuous random variables

Definition 2.4 (Continuous random variable)**.** A *continuous random variable* X is one that can take on any real value. That is, $S_X = \{x : x \in \mathbb{R}\}$.

Definition 2.5 (Probability density function for continuous random variable)**.** The probability density function (pdf) of a continuous random variable X is a nonnegative function f, defined on the real line, such that for any interval A:

$$\Pr(X \in A) = \int_A f(x)dx.$$

That is, $\Pr(X \in A)$ is the area under the probability curve over the interval A. The pdf $f(x)$ must satisfy (i) $f(x) \geq 0$; and (ii) $\int_{-\infty}^{\infty} f(x)dx = 1$.

A typical "bell-shaped" pdf is displayed in Figure Figure 2.2 and the shaded area under the curve between -2 and 1 represents $\Pr(-2 \leq X \leq 1)$. For a continuous random variable, $f(x) \neq \Pr(X = x)$ but rather gives the height of the probability curve at x. In fact, $\Pr(X = x) = 0$ for all values of x. That is, probabilities are not defined over single points. They are only defined over intervals. As a result, for a continuous random variable X we have:

$$\Pr(a \leq X \leq b) = \Pr(a < X \leq b) = \Pr(a < X < b) = \Pr(a \leq X < b).$$

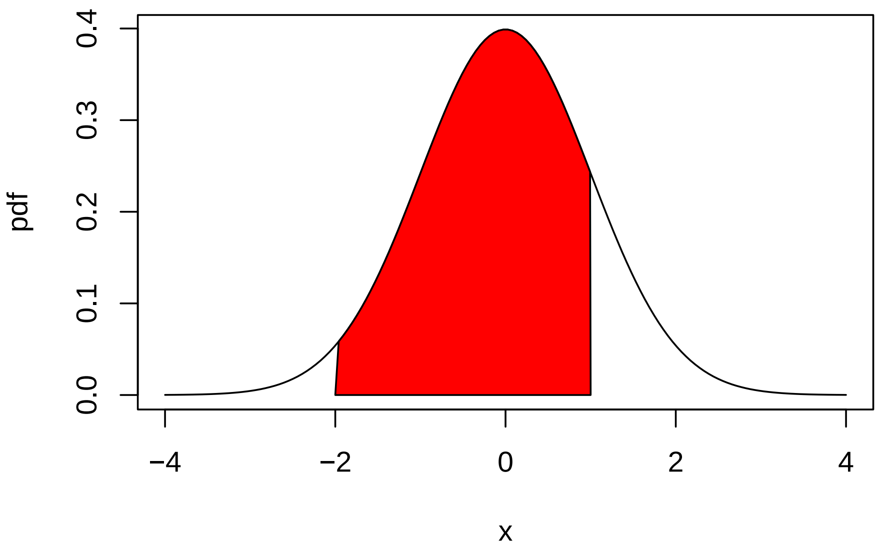


Figure 2.2: $Pr(-2 \leq X \leq 1)$ is represented by the area under the probability curve.

2.1.2.1 The uniform distribution on an interval

Let X denote the annual return on Microsoft stock and let a and b be two real numbers such that $a < b$. Suppose that the annual return on Microsoft stock can take on any value between a and b. That is, the sample space is restricted to the interval $S_X = \{x \in \mathcal{R} : a \leq x \leq b\}$. Further suppose that the probability that X will belong to any subinterval of S_X is proportional to the length of the interval. In this case, we say that X is *uniformly distributed* on the interval $[a, b]$. The pdf of X has a very simple mathematical form:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{otherwise,} \end{cases}$$

and is presented graphically as a rectangle. It is easy to see that the area under the curve over the interval $[a, b]$ (area of rectangle) integrates to one:

$$\int_a^b \frac{1}{b-a}\,dx = \frac{1}{b-a}\int_a^b dx = \frac{1}{b-a}\,[x]_a^b = \frac{1}{b-a}[b-a] = 1.$$

Example 2.5 (Uniform distribution on [-1 1])**.** Let $a = -1$ and $b = 1$, so that $b - a = 2$. Consider computing the probability that X will be between -0.5 and 0.5. We solve:

$$\Pr(-0.5 \leq X \leq 0.5) = \int_{-0.5}^{0.5} \frac{1}{2}\,dx = \frac{1}{2}\,[x]_{-0.5}^{0.5} = \frac{1}{2}\,[0.5 - (-0.5)] = \frac{1}{2}.$$

Next, consider computing the probability that the return will fall in the interval $[0, \delta]$ where δ is some small number less than $b = 1$:

$$\Pr(0 \leq X \leq \delta) = \frac{1}{2}\int_0^\delta dx = \frac{1}{2}[x]_0^\delta = \frac{1}{2}\delta.$$

As $\delta \to 0$, $\Pr(0 \leq X \leq \delta) \to \Pr(X = 0)$.

Using the above result we see that

$$\lim_{\delta \to 0} \Pr(0 \leq X \leq \delta) = \Pr(X = 0) = \lim_{\delta \to 0} \frac{1}{2}\delta = 0.$$

Hence, for continuous random variables probabilities are defined on intervals but not at distinct points. ■

2.1.2.2 The standard normal distribution

The normal or Gaussian distribution is perhaps the most famous and most useful continuous distribution in all of statistics. The shape of the normal distribution is the familiar "bell curve". As we shall see, it can be used to describe the probabilistic behavior of stock returns although other distributions may be more appropriate.

If a random variable X follows a *standard normal distribution* then we often write $X \sim N(0,1)$ as short-hand notation. This distribution is centered at zero and has inflection points at ± 1.[2] The pdf of a standard normal random variable is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}x^2}, \; -\infty \leq x \leq \infty. \tag{2.2}$$

It can be shown via the change of variables formula in calculus that the area under the standard normal curve is one:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}x^2} dx = 1.$$

The standard normal distribution is illustrated in Figure Figure 2.3. Notice that the distribution is *symmetric* about zero; i.e., the distribution has exactly the same form to the left and right of zero. Because the standard normal pdf formula is used so often it is given its own special symbol $\phi(x)$.

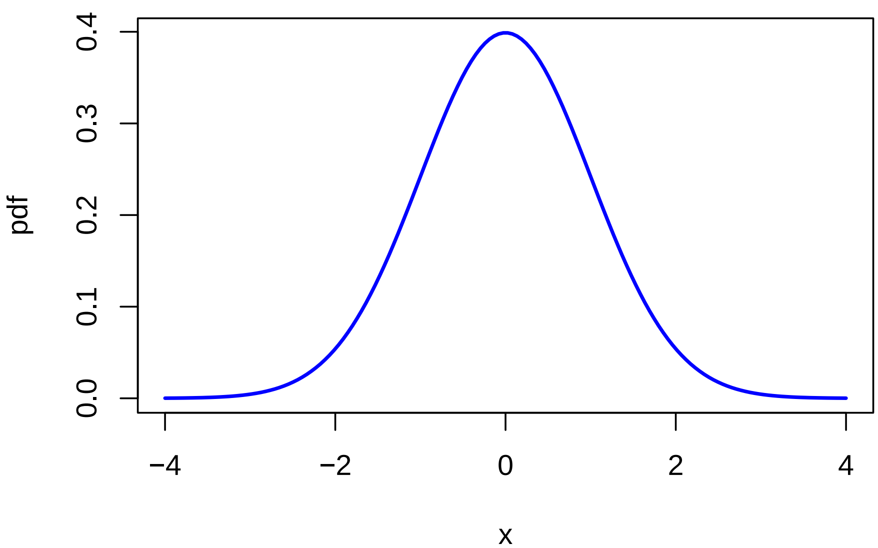


Figure 2.3: Standard normal density.

[2] An inflection point is a point where a curve goes from concave up to concave down, or vice versa.

The normal distribution has the annoying feature that the area under the normal curve cannot be evaluated analytically. That is:

$$\Pr(a \leq X \leq b) = \int_a^b \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}x^2}\ dx,$$

does not have a closed form solution. The above integral must be computed by numerical approximation. Areas under the normal curve, in one form or another, are given in tables in almost every introductory statistics book and standard statistical software can be used to find these areas. Some useful approximate results are:

$$\begin{aligned}
\Pr(-1 \leq X \leq 1) &\approx 0.67, \\
\Pr(-2 \leq X \leq 2) &\approx 0.95, \\
\Pr(-3 \leq X \leq 3) &\approx 0.99.
\end{aligned}$$

2.1.3 The cumulative distribution function

Definition 2.6 (Cumulative distribution function)**.** The *cumulative distribution function* (cdf) of a random variable X (discrete or continuous), denoted F_X, is the probability that $X \leq x$:

$$F_X(x) = \Pr(X \leq x),\ -\infty \leq x \leq \infty.$$

The cdf has the following properties:

(i) If $x_1 < x_2$ then $F_X(x_1) \leq F_X(x_2)$
(ii) $F_X(-\infty) = 0$ and $F_X(\infty) = 1$
(iii) $\Pr(X > x) = 1 - F_X(x)$
(iv) $\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$
(v) $F_X'(x) = \frac{d}{dx} F_X(x) = f(x)$ if X is a continuous random variable and $F_X(x)$ is continuous and differentiable.

Example 2.6 (CDF for a discrete random variable)**.** The cdf for the discrete distribution of Microsoft from Table Table 2.1 is given by:

$$F_X(x) = \begin{cases} 0, & x < -0.3 \\ 0.05, & -0.3 \leq x < 0 \\ 0.25, & 0 \leq x < 0.1 \\ 0.75 & 0.1 \leq x < 0.2 \\ 0.95 & 0.2 \leq x < 0.5 \\ 1 & x > 0.5 \end{cases}$$

and is illustrated in Figure **?@fig-figProbReviewDiscreteCDF**. ■

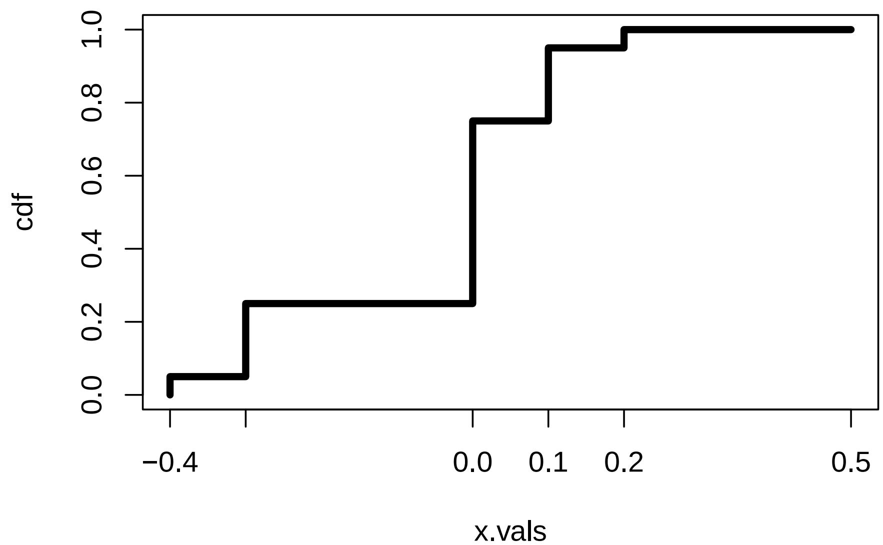


Figure 2.4: CDF of Discrete Distribution for Microsoft Stock Return.

Example 2.7 (CDF for a uniform random variable)**.** The cdf for the uniform distribution over $[a, b]$ can be determined analytically:

$$
\begin{aligned}
F_X(x) = \Pr(X < x) &= \int_{-\infty}^{x} f(t)\ dt \\
&= \frac{1}{b-a}\int_{a}^{x} dt = \frac{1}{b-a}[t]_a^x = \frac{x-a}{b-a}.
\end{aligned}
$$

We can determine the pdf of X directly from the cdf via,

$$
f(x) = F_X'(x) = \frac{d}{dx}F_X(x) = \frac{1}{b-a}.
$$

■

Example 2.8 (CDF for a standard normal random variable)**.** The cdf of standard normal random variable X is used so often in statistics that it is given its own special symbol:

$$
\Phi(x) = F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}\ dz. \tag{2.3}
$$

The cdf $\Phi(x)$, however, does not have an analytic representation like the cdf of the uniform distribution and so the integral in Equation 2.3 must be approximated using numerical techniques. A graphical representation of $\Phi(x)$ is given in Figure Figure 2.5. Because the standard normal pdf, $\phi(x)$, is symmetric about zero and bell-shaped the standard normal CDF, $\Phi(x)$, has an "S" shape where the middle of the "S" is at zero. ■

2.1.4 Quantiles of the distribution of a random variable

Definition 2.7 (Quantiles of the distribution of a random variable)**.** Consider a random variable X with continuous cdf $F_X(x)$. For $0 \leq \alpha \leq 1$, the $100 \cdot \alpha\%$ *quantile* of the distribution for X is the value q_α that satisfies:

$$
F_X(q_\alpha) = \Pr(X \leq q_\alpha) = \alpha.
$$

For example, the 5% quantile of X, $q_{0.05}$, satisfies,

$$
F_X(q_{0.05}) = \Pr(X \leq q_{0.05}) = 0.05.
$$

The *median* of the distribution is the 50% quantile. That is, the median, $q_{0.5}$, satisfies,

$$
F_X(q_{0.5}) = \Pr(X \leq q_{0.5}) = 0.5.
$$

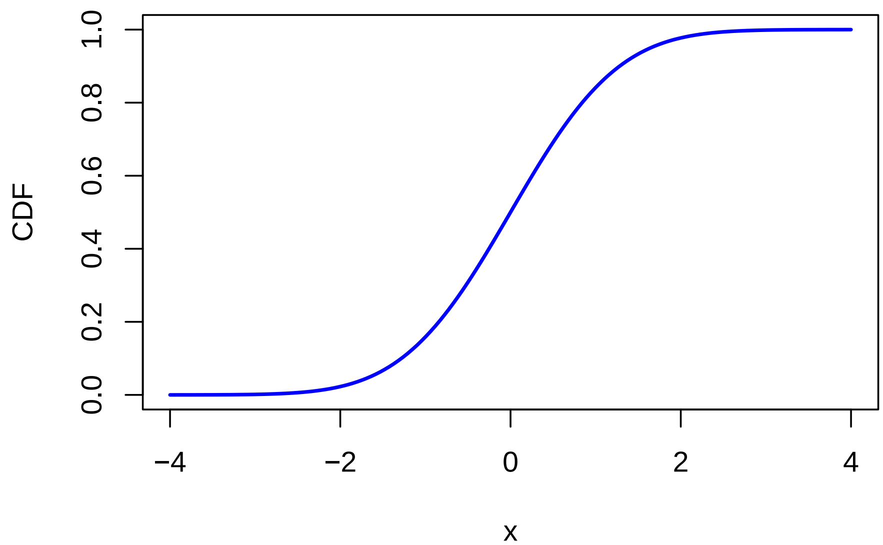


Figure 2.5: Standard normal cdf $\Phi(x)$.

If F_X is invertible then q_α may be determined analytically as:

$$q_\alpha = F_X^{-1}(\alpha) \tag{2.4}$$

where F_X^{-1} denotes the inverse function of F_X.[3] Hence, the 5% quantile and the median may be determined from

$$q_{0.05} = F_X^{-1}(.05), q_{0.5} = F_X^{-1}(.5).$$

The inverse cdf F_X^{-1} is sometimes called the *quantile function*.

Example 2.9 (Quantiles from a uniform distribution)**.** Let $X \sim U[a, b]$ where $b > a$. Recall, the cdf of X is given by:

$$F_X(x) = \frac{x-a}{b-a}, \; a \leq x \leq b,$$

which is continuous and strictly increasing. Given $\alpha \in [0, 1]$ such that $F_X(x) = \alpha$, solving for x gives the inverse cdf:

$$x = F_X^{-1}(\alpha) = \alpha(b-a) + a. \tag{2.5}$$

Using Equation 2.5, the 5% quantile and median, for example, are given by:

$$\begin{aligned} q_{0.05} &= F_X^{-1}(.05) = .05(b-a) + a = .05b + .95a, \\ q_{0.5} &= F_X^{-1}(.5) = .5(b-a) + a = .5(a+b). \end{aligned}$$

If $a = 0$ and $b = 1$, then $q_{0.05} = 0.05$ and $q_{0.5} = 0.5$. ∎

Example 2.10 (Quantiles from a standard normal distribution)**.** Let $X \sim N(0, 1)$. The quantiles of the standard normal distribution are determined by solving

$$q_\alpha = \Phi^{-1}(\alpha), \tag{2.6}$$

where Φ^{-1} denotes the inverse of the cdf Φ. This inverse function must be approximated numerically and is available in most spreadsheets and statistical software. Using the numerical approximation to the inverse function, the 1%, 2.5%, 5%, 10% quantiles and median are given by:

$$\begin{aligned} q_{0.01} &= \Phi^{-1}(.01) = -2.33, \; q_{0.025} = \Phi^{-1}(.025) = -1.96, \\ q_{0.05} &= \Phi^{-1}(.05) = -1.645, \; q_{0.10} = \Phi^{-1}(.10) = -1.28, \\ q_{0.5} &= \Phi^{-1}(.5) = 0. \end{aligned}$$

Often, the standard normal quantile is denoted z_α. ∎

[3] The inverse function of F_X , denoted F_X^{-1}, has the property that $F_X^{-1}(F_X(x)) = x$. F_X^{-1} will exist if F_X is strictly increasing and is continuous.

2.1.5 R functions for discrete and continuous distributions

R has built-in functions for a number of discrete and continuous distributions that are commonly used in probability modeling and statistical analysis. These are summarized in Table Table 2.2. For each distribution, there are four functions starting with `d`, `p`, `q`, and `r` that compute `d`ensity (pdf) values, cumulative `p`robabilities (cdf), `q`uantiles (inverse cdf) and `r`andom draws, respectively. Consider, for example, the functions associated with the normal distribution. The functions `dnorm()`, `pnorm()` and `qnorm()` evaluate the standard normal density Equation 2.2, the cdf Equation 2.3, and the inverse cdf or quantile function Equation 2.6, respectively, with the default values `mean = 0` and `sd = 1`. The function `rnorm()` returns a specified number of simulated values from the normal distribution.

Table 2.2: Probability distributions in base R.

Distribution	**Function (root)**	Parameters	Defaults
beta	`beta`	shape1, shape2	_, _
binomial	`binom`	size, prob	_, _
cauchy	`cauchy`	location, scale	0, 1
chi-squared	`chisq`	df, ncp	_, 1
F	`f`	df1, df2	_, _
gamma	`gamma`	shape, rate, scale	_, 1, 1/rate
geometric	`geom`	prob	_
hyper-geometric	`hyper`	m, n, k	_, _, _
log-normal	`lnorm`	meanlog, sdlog	0, 1
logistic	`logis`	location, scale	0, 1
negative binomial	`nbinom`	size, prob, mu	_,_,_
normal	`norm`	mean, sd	0, 1
Poisson	`pois`	Lambda	1
Student's t	`t`	df, ncp	_, 1
uniform	`unif`	min, max	0, 1
Weibull	`weibull`	shape, scale	_, 1
Wilcoxon	`wilcoxon`	m, n	_, _

Commonly used distributions used in finance application are the binomial, chi-squared, log normal, normal, and Student's t.

2.1.5.1 Finding areas under the standard normal curve

Let X denote a standard normal random variable. Given that the total area under the normal curve is one and the distribution is symmetric about zero, the following results hold:

- $\Pr(X \leq z) = 1 - \Pr(X \geq z)$ and $\Pr(X \geq z) = 1 - \Pr(X \leq z)$
- $\Pr(X \geq z) = \Pr(X \leq -z)$
- $\Pr(X \geq 0) = \Pr(X \leq 0) = 0.5$

The following examples show how to do probability calculations with standard normal random variables.

Example 2.11 (Finding areas under the normal curve using R)**.** First, consider finding $\Pr(X \geq 2)$. By the symmetry of the normal distribution, $\Pr(X \geq 2) = \Pr(X \leq -2) = \Phi(-2)$. In R use:

```
pnorm(-2)
#> [1] 0.0228
```

Next, consider finding $\Pr(-1 \leq X \leq 2)$. Using the cdf, we compute $\Pr(-1 \leq X \leq 2) = \Pr(X \leq 2) - \Pr(X \leq -1) = \Phi(2) - \Phi(-1)$. In R use:

```
pnorm(2) - pnorm(-1)
#> [1] 0.819
```

Finally, using R the exact values for $\Pr(-1 \leq X \leq 1)$, $\Pr(-2 \leq X \leq 2)$ and $\Pr(-3 \leq X \leq 3)$ are:

```
pnorm(1) - pnorm(-1)
#> [1] 0.683
pnorm(2) - pnorm(-2)
#> [1] 0.954
pnorm(3) - pnorm(-3)
#> [1] 0.997
```

■

2.1.5.2 Plotting distributions

When working with a probability distribution, it is a good idea to make plots of the pdf or cdf to reveal important characteristics. The following examples illustrate plotting distributions using R.

Example 2.12 (Plotting the standard normal curve)**.** The graphs of the standard normal pdf and cdf in Figures Figure 2.3 and Figure 2.5 were created using the following R code:

```r
# plot pdf
x.vals = seq(-4, 4, length=150)
plot(x.vals, dnorm(x.vals), type="l", lwd=2, col="blue",
     xlab="x", ylab="pdf")
# plot cdf
plot(x.vals, pnorm(x.vals), type="l", lwd=2, col="blue",
     xlab="x", ylab="CDF")
```

■

Example 2.13 (Shading a region under the standard normal curve)**.** Figure Figure 2.2 showing $\Pr(-2 \leq X \leq 1)$ as a red shaded area is created with the following code:

```r
lb = -2
ub = 1
x.vals = seq(-4, 4, length=150)
d.vals = dnorm(x.vals)
# plot normal density
plot(x.vals, d.vals, type="l", xlab="x", ylab="pdf")
i = x.vals >= lb & x.vals <= ub
# add shaded region between -2 and 1
polygon(c(lb, x.vals[i], ub), c(0, d.vals[i], 0), col="red")
```

■

2.1.6 Shape characteristics of probability distributions

Very often we would like to know certain shape characteristics of a probability distribution. We might want to know where the distribution is centered, and how spread out the distribution is about the central value. We might want to know if the distribution is symmetric about the center or if the distribution has a long left or right tail. For stock returns we might want to know about the likelihood of observing extreme values for returns representing market crashes. This means that we would like to know about the amount of probability in the extreme tails of the distribution. In this section we discuss four important shape characteristics of a probability distribution:

1. *expected value* (mean): measures the center of mass of a distribution
2. *variance* and *standard deviation* (volatility): measures the spread about the mean
3. *skewness*: measures symmetry about the mean
4. *kurtosis*: measures "tail thickness"

2.1.6.1 Expected Value

The *expected value* of a random variable X, denoted $E[X]$ or μ_X, measures the center of mass of the pdf.

Definition 2.8 (Expected value of discrete random variable)**.** For a discrete random variable X with sample space S_X, the expected value is defined as:

$$\mu_X = E[X] = \sum_{x \in S_X} x \cdot \Pr(X = x). \tag{2.7}$$

Equation Equation 2.7 shows that $E[X]$ is a probability weighted average of the possible values of X.

Example 2.14 (Expected value of discrete random variable)**.** Using the discrete distribution for the return on Microsoft stock in Table **?@tbl-Table-msftdiscretedistribution**, the expected return is computed as:

$$\begin{aligned} E[X] =& (-0.3) \cdot (0.05) + (0.0) \cdot (0.20) + (0.1) \cdot (0.5) \\ &+ (0.2) \cdot (0.2) + (0.5) \cdot (0.05) \\ =& 0.10. \end{aligned}$$

■

Example 2.15 (Expected value of Bernoulli and binomial random variables)**.** Let X be a Bernoulli random variable with success probability π. Then,

$$E[X] = 0 \cdot (1 - \pi) + 1 \cdot \pi = \pi.$$

That is, the expected value of a Bernoulli random variable is its probability of success. Now, let $Y \sim B(n, \pi)$. It can be shown that:

$$E[Y] = \sum_{k=0}^{n} k \binom{n}{k} \pi^k (1 - \pi)^{n-k} = n\pi. \tag{2.8}$$

■

Definition 2.9 (Expected value of a continuous random variable)**.** For a continuous random variable X with pdf $f(x)$, the *expected value* is defined as:

$$\mu_X = E[X] = \int_{-\infty}^{\infty} x \cdot f(x) \; dx. \tag{2.9}$$

Example 2.16 (Expected value of a uniform random variable)**.** Suppose X has a uniform distribution over the interval $[a, b]$. Then,

$$\begin{aligned}
E[X] &= \frac{1}{b-a}\int_a^b x\ dx = \frac{1}{b-a}\left[\frac{1}{2}x^2\right]_a^b \\
&= \frac{1}{2(b-a)}\left[b^2 - a^2\right] \\
&= \frac{(b-a)(b+a)}{2(b-a)} = \frac{b+a}{2}.
\end{aligned}$$

If $b = -1$ and $a = 1$, then $E[X] = 0$. ■

Example 2.17 (Expected value of a standard normal random variable)**.** Let $X \sim N(0, 1)$. Then it can be shown that:

$$E[X] = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}\ dx = 0.$$

Hence, the standard normal distribution is centered at zero. ■

2.1.6.2 Expectation of a function of a random variable

The other shape characteristics of the distribution of a random variable X are based on expectations of certain functions of X.

Definition 2.10 (Expected value of a function of a random variable)**.** Let $g(X)$ denote some function of the random variable X. If X is a discrete random variable with sample space S_X then:

$$E[g(X)] = \sum_{x \in S_X} g(x) \cdot \Pr(X = x), \tag{2.10}$$

and if X is a continuous random variable with pdf f then,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f(x)\ dx. \tag{2.11}$$

2.1.6.3 Variance and standard deviation

The variance of a random variable X, denoted $\text{var}(X)$ or σ_X^2, measures the spread of the distribution about the mean using the function $g(X) = (X - \mu_X)^2$. If most values of X are close to μ_X then on average $(X - \mu_X)^2$ will be small. In contrast, if many values of X are far below and/or far above μ_X then on average $(X - \mu_X)^2$ will be large. Squaring the deviations about μ_X guarantees a positive value.

Definition 2.11 (Variance and standard deviation)**.** For a discrete or continuous random variable X, the variance and standard deviation of X are defined as:

$$\sigma_X^2 = \text{var}(X) = E[(X - \mu_X)^2], \tag{2.12}$$

$$\sigma_X = \text{sd}(X) = \sqrt{\sigma_X^2} = \sigma_X. \tag{2.13}$$

Because σ_X^2 represents an average squared deviation, it is not in the same units as X. The standard deviation of X is the square root of the variance and is in the same units as X. For "bell-shaped" distributions, σ_X measures the typical size of a deviation from the mean value.

The computation of Equation 2.12 can often be simplified by using the result:

$$\text{var}(X) = E[(X - \mu_X)^2] = E[X^2] - \mu_X^2 \tag{2.14}$$

We will learn how to derive this result later in the chapter.

Example 2.18 (Variance and standard deviation for a discrete random variable)**.** Using the discrete distribution for the return on Microsoft stock in Table Table 2.1 and the result that $\mu_X = 0.1$, we have:

$$\begin{aligned}
\text{var}(X) = (-0.3 - 0.1)^2 \cdot (0.05) + (0.0 - 0.1)^2 \cdot (0.20) + (0.1 - 0.1)^2 \cdot (0.5) \\
+ (0.2 - 0.1)^2 \cdot (0.2) + (0.5 - 0.1)^2 \cdot (0.05) \\
= 0.020.
\end{aligned}$$

Alternatively, we can compute $\text{var}(X)$ using Equation 2.14:

$$\begin{aligned}
E[X^2] - \mu_X^2 &= (-0.3)^2 \cdot (0.05) + (0.0)^2 \cdot (0.20) + (0.1)^2 \cdot (0.5) \\
&+ (0.2)^2 \cdot (0.2) + (0.5)^2 \cdot (0.05) - (0.1)^2 \\
&= 0.020.
\end{aligned}$$

The standard deviation is $\sqrt{0.020} = 0.141$. Given that the distribution is fairly bell-shaped we can say that typical values deviate from the mean value of 10% by about ±14.1%. ■

Example 2.19 (Variance and standard deviation of a Bernoulli and binomial random variables)**.** Let X be a Bernoulli random variable with success probability π. Given that $\mu_X = \pi$ it follows that:

$$\begin{aligned}
\text{var}(X) &= (0 - \pi)^2 \cdot (1 - \pi) + (1 - \pi)^2 \cdot \pi \\
&= \pi^2(1 - \pi) + (1 - \pi)^2\pi \\
&= \pi(1 - \pi)\left[\pi + (1 - \pi)\right] \\
&= \pi(1 - \pi), \\
\text{sd}(X) &= \sqrt{\pi(1 - \pi)}.
\end{aligned}$$

Now, let $Y \sim B(n, \pi)$: It can be shown that:

$$\text{var}(Y) = n\pi(1-\pi)$$

and so $\text{sd}(Y) = \sqrt{n\pi(1-\pi)}$. ■

Example 2.20 (Variance and standard deviation of a uniform random variable)**.** Let $X \sim U[a, b]$. Using Equation 2.14 and $\mu_X = \frac{a+b}{2}$, after some algebra, it can be shown that:

$$\text{var}(X) = E[X^2] - \mu_X^2 = \frac{1}{b-a}\int_a^b x^2 \; dx - \left(\frac{a+b}{2}\right)^2 = \frac{1}{12}(b-a)^2,$$

and $\text{sd}(X) = (b-a)/\sqrt{12}$. ■

Example 2.21 (Variance and standard deviation of a standard normal random variable)**.** Let $X \sim N(0, 1)$. Here, $\mu_X = 0$ and it can be shown that:

$$\sigma_X^2 = \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \; dx = 1.$$

It follows that $\text{sd}(X) = 1$. ■

2.1.6.4 The general normal distribution

Recall, if X has a standard normal distribution then $E[X] = 0$, $\text{var}(X) = 1$. A *general normal* random variable X has $E[X] = \mu_X$ and $\text{var}(X) = \sigma_X^2$ and is denoted $X \sim N(\mu_X, \sigma_X^2)$. Its pdf is given by:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} \exp\left\{-\frac{1}{2\sigma_X^2}\left(x-\mu_X\right)^2\right\}, \; -\infty \leq x \leq \infty. \tag{2.15}$$

Showing that $E[X] = \mu_X$ and $\text{var}(X) = \sigma_X^2$ is a bit of work and is good calculus practice. As with the standard normal distribution, areas under the general normal curve cannot be computed analytically. Using numerical approximations, it can be shown that:

$$\begin{aligned} \Pr(\mu_X - \sigma_X < X < \mu_X + \sigma_X) &\approx 0.67, \\ \Pr(\mu_X - 2\sigma_X < X < \mu_X + 2\sigma_X) &\approx 0.95, \\ \Pr(\mu_X - 3\sigma_X < X < \mu_X + 3\sigma_X) &\approx 0.99. \end{aligned}$$

Hence, for a general normal random variable about 95% of the time we expect to see values within $\pm$ 2 standard deviations from its mean. Observations more than three standard deviations from the mean are very unlikely.

Example 2.22 (Normal distribution for monthly returns)**.** Let R denote the monthly return on an investment in Microsoft stock, and assume that it is normally distributed with mean $\mu_R = 0.01$ and standard deviation $\sigma_R = 0.10$. That is, $R \sim N(0.01, (0.10)^2)$. Notice that $\sigma_R^2 = 0.01$ and is not in units of return per month. Figure Figure 2.6 illustrates the distribution and is created using:

```r
mu.r = 0.01
sd.r = 0.1
x.vals = seq(-4, 4, length=150)*sd.r + mu.r
plot(x.vals, dnorm(x.vals, mean=mu.r, sd=sd.r), type="l", lwd=2,
     col="blue", xlab="x", ylab="pdf")
```

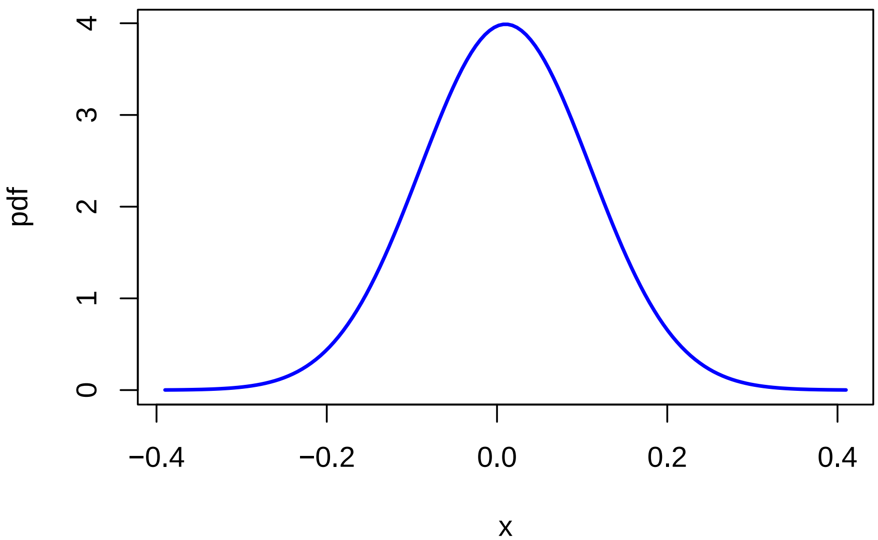


Figure 2.6: Normal distribution for the monthly returns on Microsoft: $R \sim N(0.01, (0.10)^2)$.

Notice that essentially all of the probability lies between -0.3 and 0.3. Using the R function `pnorm()`, we can easily compute the probabilities $\Pr(R < -0.5)$, $\Pr(R < 0)$, $\Pr(R > 0.5)$ and $\Pr(R > 1)$:

```r
pnorm(-0.5, mean=0.01, sd=0.1)
#> [1] 1.7e-07
pnorm(0, mean=0.01, sd=0.1)
#> [1] 0.46
1 - pnorm(0.5, mean=0.01, sd=0.1)
#> [1] 4.79e-07
1 - pnorm(1, mean=0.01, sd=0.1)
#> [1] 0
```

Using the R function `qnorm()`, we can find the quantiles $q_{0.01}$, $q_{0.05}$, $q_{0.95}$ and $q_{0.99}$:

```
a.vals = c(0.01, 0.05, 0.95, 0.99)
qnorm(a.vals, mean=0.01, sd=0.10)
#> [1] -0.223 -0.154  0.174  0.243
```

Hence, over the next month, there are 1% and 5% chances of losing more than 22.2% and 15.5%, respectively. In addition, there are 5% and 1% chances of gaining more than 17.5% and 24.3%, respectively. ■

Example 2.23 (Risk-return tradeoff)**.** Consider the following investment problem. We can invest in two non-dividend paying stocks, Amazon and Boeing, over the next month. Let R_A denote the monthly return on Amazon and R_B denote the monthly return on Boeing. Assume that $R_A \sim N(0.02, (0.10)^2)$ and $R_B \sim N(0.01, (0.05)^2)$. Figure Figure 2.7 shows the pdfs for the two returns. Notice that $\mu_A = 0.02 > \mu_B = 0.01$ but also that $\sigma_A = 0.10 > \sigma_B = 0.05$. The return we expect on Amazon is bigger than the return we expect on Boeing but the variability of Amazon's return is also greater than the variability of Boeing's return. The high return variability (volatility) of Amazon reflects the higher risk associated with investing in Amazon compared to investing in Boeing. If we invest in Boeing we get a lower expected return, but we also get less return variability or risk. For example, the probability of losing more than 10% in one month for Amazon is

```
pnorm(-0.1, 0.02, 0.1)
#> [1] 0.115
```

and the probability of losing more than 10% in one month for Boeing is

```
pnorm(-0.1, 0.01, 0.05)
#> [1] 0.0139
```

Of course, there is also a higher probability for upside return for Amazon than for Boeing. This example illustrates the fundamental "no free lunch" principle of economics and finance: you can't get something for nothing. In general, to get a higher expected return you must be prepared to take on higher risk. ■

Example 2.24 (Why the normal distribution may not be appropriate for simple returns)**.** Let R_t denote the simple annual return on an asset, and suppose that $R_t \sim N(0.05, (0.50)^2)$. Because asset prices must be non-negative, R_t must always be larger than -1. However, the normal distribution is defined for $-\infty \leq R_t \leq \infty$ and based on the assumed normal distribution $\Pr(R_t < -1) = 0.018$. That is, there is a 1.8% chance that R_t is smaller than -1. This implies that there is a 1.8% chance that the asset price at the end of the year will be negative! This is one reason why the normal distribution may not be appropriate for simple returns. ■

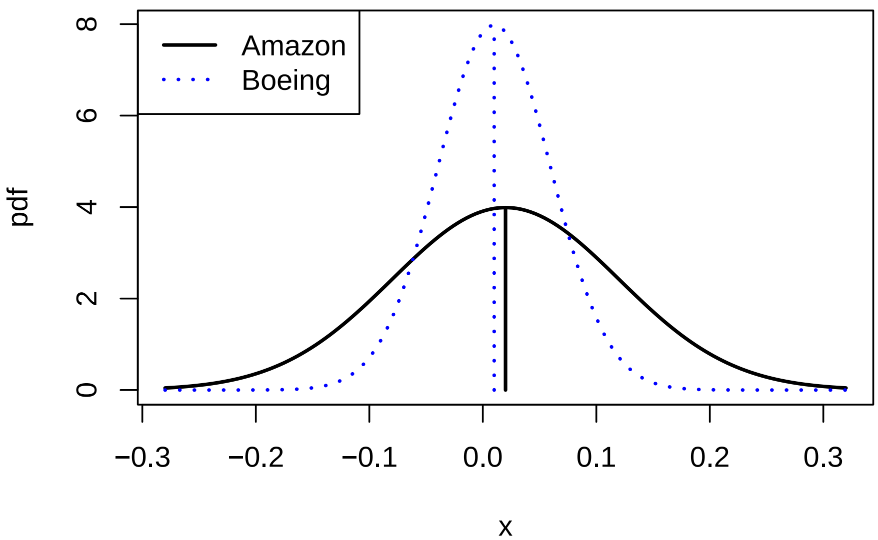


Figure 2.7: Risk-return tradeoff between one-month investments in Amazon and Boeing stock.

Example 2.25 (The normal distribution is more appropriate for continuously compounded returns)**.** Let $r_t = \ln(1 + R_t)$ denote the continuously compounded annual return on an asset, and suppose that $r_t \sim N(0.05, (0.50)^2)$. Unlike the simple return, the continuously compounded return can take on values less than -1. In fact, r_t is defined for $-\infty \leq r_t \leq \infty$. For example, suppose $r_t = -2$. This implies a simple return of $R_t = e^{-2} - 1 = -0.865$.[4] Then $\Pr(r_t \leq -2) = \Pr(R_t \leq -0.865) = 0.00002$. Although the normal distribution allows for values of r_t smaller than -1, the implied simple return R_t will always be greater than -1. ■

2.1.6.5 The Log-Normal Distribution

Let $X \sim N(\mu_X, \sigma_X^2)$, which is defined for $-\infty < X < \infty$. The *log-normally* distributed random variable Y is determined from the normally distributed random variable X using the transformation $Y = e^X$. In this case, we say that Y is log-normally distributed and write:

$$Y \sim \ln N(\mu_X, \sigma_X^2),\ 0 < Y < \infty. \tag{2.16}$$

The pdf of the log-normal distribution for Y can be derived from the normal distribution for X using the change-of-variables formula from calculus and is given by:

$$f(y) = \frac{1}{y\sigma_X\sqrt{2\pi}} e^{-\frac{(\ln y - \mu_X)^2}{2\sigma_X^2}}. \tag{2.17}$$

Due to the exponential transformation, Y is only defined for non-negative values. It can be shown that:

$$\begin{aligned}
\mu_Y &= E[Y] = e^{\mu_X + \sigma_X^2/2}, \\
\sigma_Y^2 &= \mathrm{var}(Y) = e^{2\mu_X + \sigma_X^2}(e^{\sigma_X^2} - 1).
\end{aligned}$$

Example 2.26 (Log-normal distribution for simple returns)**.** Let $r_t = \ln(P_t/P_{t-1})$ denote the continuously compounded monthly return on an asset and assume that $r_t \sim N(0.05, (0.50)^2)$. That is, $\mu_r = 0.05$ and $\sigma_r = 0.50$. Let $R_t = (P_t - P_{t-1})/P_t$ denote the simple monthly return. The relationship between r_t and R_t is given by $r_t = \ln(1 + R_t)$ and $1 + R_t = e^{r_t}$. Since r_t is normally distributed, $1 + R_t$ is log-normally distributed. Notice that the distribution of $1 + R_t$ is only defined for positive values of $1 + R_t$. This is appropriate since the smallest value that R_t can take on is -1. Using Equation 2.18, the mean and variance for $1 + R_t$ are given by:

$$\begin{aligned}
\mu_{1+R} &= e^{0.05 + (0.5)^2/2} = 1.191 \\
\sigma_{1+R}^2 &= e^{2(0.05) + (0.5)^2}(e^{(0.5)^2} - 1) = 0.563
\end{aligned}$$

The pdfs for r_t and R_t are shown in Figure **?@fig-figNormalLognormal**. ■

[4] If $r_t = -\infty$ then $R_t = e^{r_t} - 1 = e^{-\infty} - 1 = 0 - 1 = -1$.

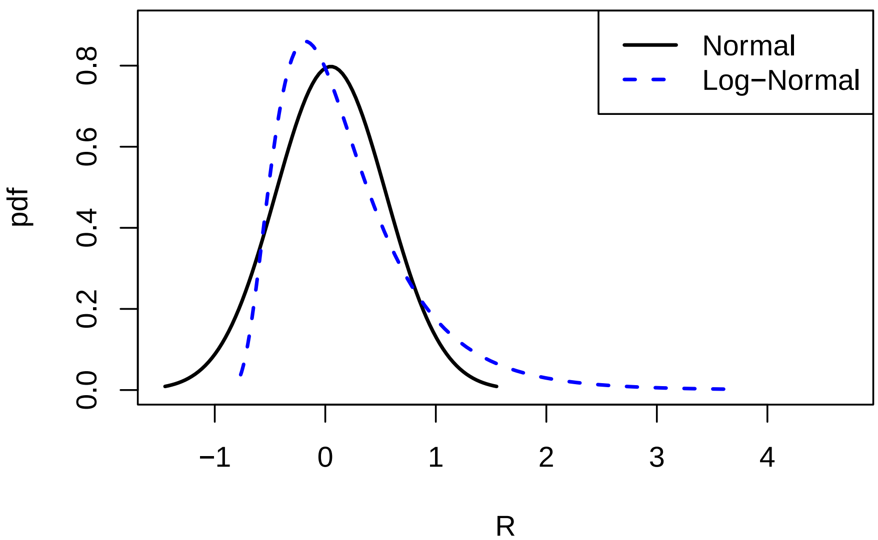


Figure 2.8: Normal distribution for r_t and log-normal distribution for $R_t = e^{r_t} - 1$.

2.1.6.6 Skewness

Definition 2.12 (Skewness)**.** The *skewness* of a random variable X, denoted $\text{skew}(X)$, measures the symmetry of a distribution about its mean value using the function $g(X) = (X - \mu_X)^3/\sigma_X^3$, where σ_X^3 is just $\text{sd}(X)$ raised to the third power:

$$\text{skew}(X) = \frac{E[(X - \mu_X)^3]}{\sigma_X^3} \tag{2.18}$$

When X is far below its mean, $(X - \mu_X)^3$ is a big negative number. When X is far above its mean, $(X - \mu_X)^3$ is a big positive number. Hence, if there are more big values of X below μ_X then $\text{skew}(X) < 0$. Conversely, if there are more big values of X above μ_X then $\text{skew}(X) > 0$. If X has a symmetric distribution, then $\text{skew}(X) = 0$ since positive and negative values in Equation 2.18 cancel out. If $\text{skew}(X) > 0$, then the distribution of X has a *long right tail* and if $\text{skew}(X) < 0$ the distribution of X has a *long left tail.* These cases are illustrated in Figure Figure 2.9.

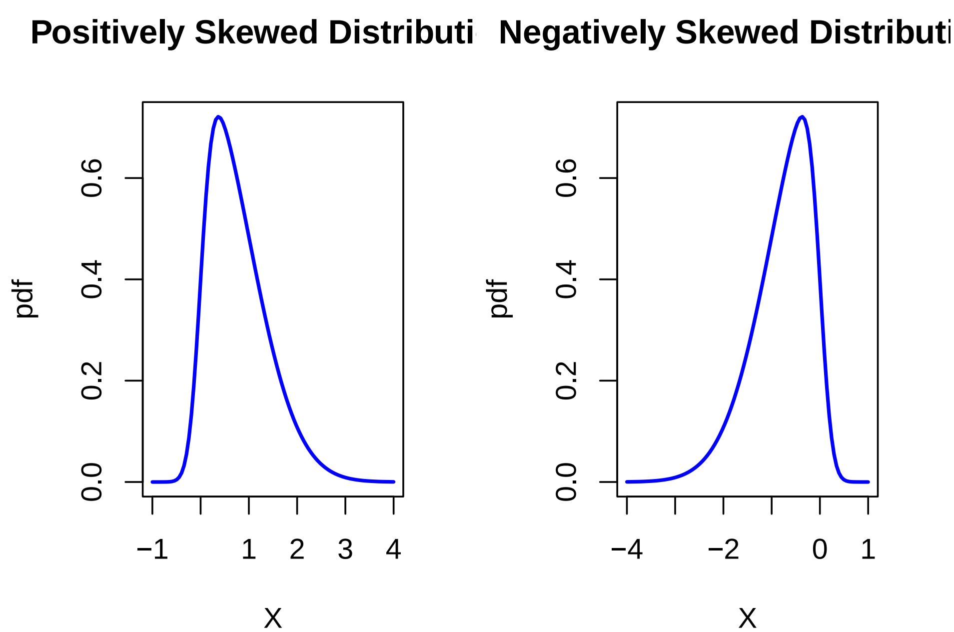


Figure 2.9: Skewed distributions.

Example 2.27 (Skewness for a discrete random variable)**.** Using the discrete distribution for the return on Microsoft stock in Table **?@tbl-Table-msftdiscretedistribution**, the results that $\mu_X = 0.1$ and $\sigma_X = 0.141$, we have:

$$\begin{aligned}\text{skew}(X) &= [(-0.3 - 0.1)^3 \cdot (0.05) + (0.0 - 0.1)^3 \cdot (0.20) + (0.1 - 0.1)^3 \cdot (0.5) \\ &\quad + (0.2 - 0.1)^3 \cdot (0.2) + (0.5 - 0.1)^3 \cdot (0.05)]/(0.141)^3 \\ &= 0.0.\end{aligned}$$

■

Example 2.28 (Skewness for a normal random variable)**.** Suppose X has a general normal distribution with mean μ_X and variance σ_X^2. Then it can be shown that:

$$\text{skew}(X) = \int_{-\infty}^{\infty} \frac{(x-\mu_X)^3}{\sigma_X^3} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma_X^2}(x-\mu_X)^2}\ dx = 0.$$

This result is expected since the normal distribution is symmetric about it's mean value μ_X. ■

Example 2.29 (Skewness for a log-Normal random variable)**.** Let $Y = e^X$, where $X \sim N(\mu_X, \sigma_X^2)$, be a log-normally distributed random variable with parameters μ_X and σ_X^2. Then it can be shown that:

$$\text{skew}(Y) = \left(e^{\sigma_X^2} + 2\right)\sqrt{e^{\sigma_X^2} - 1} > 0.$$

Notice that skew(Y) is always positive, indicating that the distribution of Y has a long right tail, and that it is an increasing function of σ_X^2. This positive skewness is illustrated in Figure **?@fig-figNormalLognormal**. ■

2.1.6.7 Kurtosis

Definition 2.13 (Kurtosis)**.** The *kurtosis* of a random variable X, denoted kurt(X), measures the thickness in the tails of a distribution and is based on $g(X) = (X - \mu_X)^4/\sigma_X^4$:

$$\text{kurt}(X) = \frac{E[(X-\mu_X)^4]}{\sigma_X^4}, \tag{2.19}$$

where σ_X^4 is just sd(X) raised to the fourth power.

Since kurtosis is based on deviations from the mean raised to the fourth power, large deviations get lots of weight. Hence, distributions with large kurtosis values are ones where there is the possibility of extreme values. In contrast, if the kurtosis is small then most of the observations are tightly clustered around the mean and there is very little probability of observing extreme values.

Example 2.30 (Kurtosis for a discrete random variable)**.** Using the discrete distribution for the return on Microsoft stock in Table Table 2.1, the results that $\mu_X = 0.1$ and $\sigma_X = 0.141$, we have:

$$\begin{aligned}\text{kurt}(X) &= [(-0.3-0.1)^4 \cdot (0.05) + (0.0-0.1)^4 \cdot (0.20) + (0.1-0.1)^4 \cdot (0.5) \\ &+ (0.2-0.1)^4 \cdot (0.2) + (0.5-0.1)^4 \cdot (0.05)]/(0.141)^4 \\ &= 6.5.\end{aligned}$$

■

Example 2.31 (Kurtosis for a normal random variable)**.** Suppose X has a general normal distribution with mean μ_X and variance σ_X^2. Then it can be shown that:

$$\text{kurt}(X) = \int_{-\infty}^{\infty} \frac{(x-\mu_X)^4}{\sigma_X^4} \cdot \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{1}{2}(\frac{x-\mu_X}{\sigma_X})^2}\, dx = 3.$$

Hence, a kurtosis of 3 is a benchmark value for tail thickness of bell-shaped distributions. If a distribution has a kurtosis greater than 3, then the distribution has thicker tails than the normal distribution. If a distribution has kurtosis less than 3, then the distribution has thinner tails than the normal. ■

Sometimes the kurtosis of a random variable is described relative to the kurtosis of a normal random variable. This relative value of kurtosis is referred to as *excess kurtosis* and is defined as:

$$\text{ekurt}(X) = \text{kurt}(X) - 3 \tag{2.20}$$

If the excess kurtosis of a random variable is equal to zero then the random variable has the same kurtosis as a normal random variable. If excess kurtosis is greater than zero, then kurtosis is larger than that for a normal; if excess kurtosis is less than zero, then kurtosis is less than that for a normal.

2.1.6.8 The Student's t Distribution

The kurtosis of a random variable gives information on the tail thickness of its distribution. The normal distribution, with kurtosis equal to three, gives a benchmark for the tail thickness of symmetric distributions. A distribution similar to the standard normal distribution but with fatter tails, and hence larger kurtosis, is the Student's t distribution. If X has a Student's t distribution with degrees of freedom parameter v, denoted $X \sim t_v$, then its pdf has the form:

$$f(x) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi}\Gamma\left(\frac{v}{2}\right)}\left(1+\frac{x^2}{v}\right)^{-\left(\frac{v+1}{2}\right)},\ -\infty < x < \infty,\ v > 0. \tag{2.21}$$

where $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}dt$ denotes the gamma function. It can be shown that:

$$
\begin{aligned}
E[X] &= 0,\ v > 1 \\
\mathrm{var}(X) &= \frac{v}{v-2},\ v > 2, \\
\mathrm{skew}(X) &= 0,\ v > 3, \\
\mathrm{kurt}(X) &= \frac{6}{v-4} + 3,\ v > 4.
\end{aligned}
$$

The parameter v controls the scale and tail thickness of the distribution. If v is close to four, then the kurtosis is large and the tails are thick. If $v < 4$, then $\mathrm{kurt}(X) = \infty$. As $v \to \infty$ the Student's t pdf approaches that of a standard normal random variable and $\mathrm{kurt}(X) = 3$. Figure **?@fig-ProbReviewStudentT** shows plots of the Student's t density for various values of v.

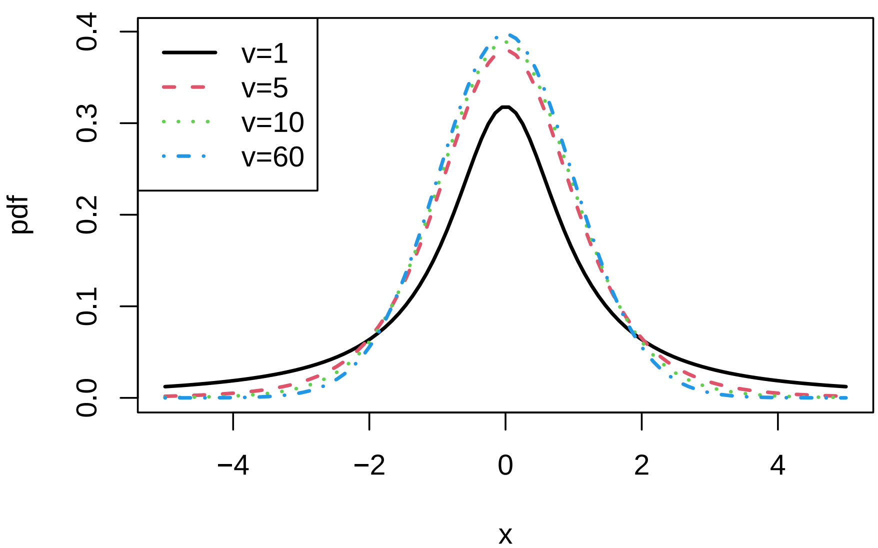


Figure 2.10: Student's t density with $v = 1, 5, 10$ and 60.

Example 2.32 (Computing tail probabilities and quantiles from the Student's t distribution)**.** The R functions `pt()` and `qt()` can be used to compute the cdf and quantiles of a Student's t random variable. For $v = 1, 2, 5, 10, 60, 100$ and ∞ the 1% quantiles can be computed using:

```r
v = c(1, 2, 5, 10, 60, 100, Inf)
qt(0.01, df=v)
#> [1] -31.82  -6.96  -3.36  -2.76  -2.39  -2.36  -2.33
```

For $v = 1, 2, 5, 10, 60, 100$ and ∞ the values of $\Pr(X < -3)$ are:

```r
pt(-3, df=v)
#> [1] 0.10242 0.04773 0.01505 0.00667 0.00196 0.00170 0.00135
```

■

2.1.7 Linear functions of a random variable

Let X be a random variable either discrete or continuous with $E[X] = \mu_X$, $\text{var}(X) = \sigma_X^2$, and let a and b be known constants. Define a new random variable Y via the linear function of X:

$$Y = g(X) = aX + b.$$

Then the following results hold:

$$\begin{aligned}
\mu_Y &= E[Y] = a \cdot E[X] + b = a \cdot \mu_X + b, \\
\sigma_Y^2 &= \text{var}(Y) = a^2 \text{var}(X) = a^2 \sigma_X^2, \\
\sigma_Y &= \text{sd}(Y) = a \cdot \text{sd}(X) = a \cdot \sigma_X.
\end{aligned}$$

The first result shows that expectation is a linear operation. That is,

$$E[aX + b] = aE[X] + b.$$

The second result shows that adding a constant to X does not affect its variance, and that the effect of multiplying X by the constant a increases the variance of X by the square of a. These results will be used often enough that it is instructive to go through the derivations.

Consider the first result. Let X be a discrete random variable. By the definition of $E[g(X)]$, with $g(X) = b + aX$, we have:

$$\begin{aligned}
E[Y] &= \sum_{x \in S_X} (ax + b) \cdot \Pr(X = x) \\
&= a \sum_{x \in S_X} x \cdot \Pr(X = x) + b \sum_{x \in S_X} \Pr(X = x) \\
&= a \cdot E[X] + b \cdot 1 \\
&= a \cdot \mu_X + b.
\end{aligned}$$

If X is a continuous random variable then by the linearity of integration:

$$\begin{aligned} E[Y] &= \int_{-\infty}^{\infty} (ax+b)f(x)\ dx = a\int_{-\infty}^{\infty} xf(x)\ dx + b\int_{-\infty}^{\infty} f(x)\ dx \\ &= aE[X] + b. \end{aligned}$$

Next consider the second result. Since $\mu_Y = a\mu_X + b$ we have:

$$\begin{aligned} \mathrm{var}(Y) &= E[(Y-\mu_Y)^2] \\ &= E[(aX+b-(a\mu_X+b))^2] \\ &= E[(a(X-\mu_X)+(b-b))^2] \\ &= E[a^2(X-\mu_X)^2] \\ &= a^2E[(X-\mu_X)^2] \text{ (by the linearity of } E[\cdot]) \\ &= a^2\mathrm{var}(X) \end{aligned}$$

Notice that the derivation of the second result works for discrete and continuous random variables.

A normal random variable has the special property that a linear function of it is also a normal random variable. The following proposition states the result.

Proposition 2.1 (Linear function of a normal random variable)**.** *Let $X \sim N(\mu_X, \sigma_X^2)$ and let a and b be constants. Let $Y = aX + b$. Then $Y \sim N(a\mu_X + b, a^2\sigma_X^2)$.*

The above property is special to the normal distribution and may or may not hold for a random variable with a distribution that is not normal.

Example 2.33 (Standardizing a random variable)**.** Let X be a random variable with $E[X] = \mu_X$ and $\mathrm{var}(X) = \sigma_X^2$. Define a new random variable Z as:

$$Z = \frac{X-\mu_X}{\sigma_X} = \frac{1}{\sigma_X}X - \frac{\mu_X}{\sigma_X},$$

which is a linear function $aX + b$ where $a = \frac{1}{\sigma_X}$ and $b = -\frac{\mu_X}{\sigma_X}$. This transformation is called *standardizing* the random variable X since,

$$\begin{aligned} E[Z] &= \frac{1}{\sigma_X}E[X] - \frac{\mu_X}{\sigma_X} = \frac{1}{\sigma_X}\mu_X - \frac{\mu_X}{\sigma_X} = 0, \\ \mathrm{var}(Z) &= \left(\frac{1}{\sigma_X}\right)^2 \mathrm{var}(X) = \frac{\sigma_X^2}{\sigma_X^2} = 1. \end{aligned}$$

Hence, standardization creates a new random variable with mean zero and variance 1. In addition, if X is normally distributed then $Z \sim N(0,1)$. ■

Example 2.34 (Computing probabilities using standardized random variables)**.** Let $X \sim N(2, 4)$ and suppose we want to find $\Pr(X > 5)$ but we only know probabilities associated with a standard normal random variable $Z \sim N(0, 1)$. We solve the problem by standardizing X as follows:

$$\Pr(X > 5) = \Pr\left(\frac{X - 2}{\sqrt{4}} > \frac{5 - 2}{\sqrt{4}}\right) = \Pr\left(Z > \frac{3}{2}\right) = 0.06681.$$

■

Standardizing a random variable is often done in the construction of test statistics. For example, the so-called *t-statistic* or *t-ratio* used for testing simple hypotheses on coefficients is constructed by the standardization process.

A non-standard random variable X with mean μ_X and variance σ_X^2 can be created from a standard random variable via the linear transformation:

$$X = \mu_X + \sigma_X \cdot Z. \tag{2.22}$$

This result is useful for modeling purposes as illustrated in the next examples.

Example 2.35 (Quantile of general normal random variable)**.** Let $X \sim N(\mu_X, \sigma_X^2)$. Quantiles of X can be conveniently computed using:

$$q_\alpha = \mu_X + \sigma_X z_\alpha, \tag{2.23}$$

where $\alpha \in (0, 1)$ and z_α is the $\alpha \times 100\%$ quantile of a standard normal random variable. This formula is derived as follows. By the definition z_α and Equation 2.22

$$\alpha = \Pr(Z \leq z_\alpha) = \Pr(\mu_X + \sigma_X \cdot Z \leq \mu_X + \sigma_X \cdot z_\alpha) = \Pr(X \leq \mu_X + \sigma_X \cdot z_\alpha),$$

which implies Equation 2.23. ■

Example 2.36 (Generalized Student's t distribution)**.** The Student's t distribution defined in Equation 2.21 has zero mean and variance equal to $v/(v-2)$ where v is the degrees of freedom parameter. We can define a Student's t random variable with non-zero mean μ and variance σ^2 from a Student's t random variable X as follows:

$$Y = \mu + \sigma\left(\frac{v - 2}{v}\right)^{1/2} X = \mu + \sigma Z_v,$$

where $Z_v = \left(\frac{v-2}{v}\right)^{1/2} X$. The random variable Z_v is called a *standardized Student's t* random variable with v degrees of freedom since $E[Z_v] = 0$ and $\mathrm{var}(Z_v) = \left(\frac{v-2}{v}\right)\mathrm{var}(X) = \left(\frac{v-2}{v}\right)\left(\frac{v}{v-2}\right) = 1$. Then

$$\begin{aligned} E[Y] &= \mu + \sigma E[Z] = \mu, \\ \mathrm{var}(Y) &= \sigma^2 \mathrm{var}(Z) = \sigma^2. \end{aligned}$$

We use the notation $Y \sim t_v(\mu, \sigma^2)$ to denote that Y has a *generalized Student's t* distribution with mean μ, variance σ^2, and degrees of freedom v. ■

Example 2.37 (Location scale return model for asset returns)**.** Let r denote the monthly continuously compounded return on an asset with $E[r] = \mu$ and $\text{var}(R) = \sigma^2$. If r has a normal distribution or a general Student's t distribution, then r can be expressed as:

$$r = \mu_r + \sigma_r \cdot \varepsilon,$$

where $\varepsilon \sim N(0, 1)$ or $\varepsilon \sim t_v(0, 1)$. The random variable ε can be interpreted as representing the random news arriving in a given month that makes the observed return differ from its expected value μ. The fact that ε has mean zero means that news, on average, is neutral. The value of σ_r represents the typical size of a news shock. The bigger is σ_r, the larger is the impact of a news shock and vice-versa. The distribution of ε impacts the likelihood of extreme new events (like stock market crashes). If $\varepsilon \sim N(0, 1)$ then extreme news events are not very likely. If $\varepsilon \sim t_v(0, 1)$ with v small (but bigger than 4) then extreme news events may happen from time-to-time. ■

2.1.8 Value-at-Risk: An introduction

As an example of working with linear functions of a normal random variable, and to illustrate the concept of *Value-at-Risk* (VaR), consider an investment of \$10,000 in Microsoft stock over the next month. Let R denote the monthly simple return on Microsoft stock and assume that $R \sim N(0.05, (0.10)^2)$. That is, $E[R] = \mu_R = 0.05$ and $\text{var}(R) = \sigma_R^2 = (0.10)^2$. Let W_0 denote the investment value at the beginning of the month and W_1 denote the investment value at the end of the month. In this example, $W_0 = \$10,000$. Consider the following questions:

(i) What is the probability distribution of end-of-month wealth, W_1?
(ii) What is the probability that end-of-month wealth is less than \$9,000, and what must the return on Microsoft be for this to happen?
(iii) What is the loss in dollars that would occur if the return on Microsoft stock is equal to its 5% quantile, $q_{.05}$? That is, what is the monthly 5% VaR on the \$10,000 investment in Microsoft?

To answer (i), note that end-of-month wealth, W_1, is related to initial wealth W_0 and the return on Microsoft stock R via the linear function:

$$W_1 = W_0(1 + R) = W_0 + W_0 R = \$10,000 + \$10,000 \cdot R.$$

Using the properties of linear functions of a random variable we have:

$$E[W_1] = W_0 + W_0 E[R] = \$10,000 + \$10,000(0.05) = \$10,500,$$

and,

$$\begin{aligned}\text{var}(W_1) &= (W_0)^2 \text{var}(R) = (\$10,000)^2 (0.10)^2, \\ \text{sd}(W_1) &= (\$10,000)(0.10) = \$1,000.\end{aligned}$$

Further, since R is assumed to be normally distributed it follows that W_1 is normally distributed:

$$W_1 \sim N(\$10,500, (\$1,000)^2).$$

To answer (ii), we use the above normal distribution for W_1 to get:

$$\Pr(W_1 < \$9,000) = 0.067.$$

To find the return that produces end-of-month wealth of \$9,000, or a loss of \$10,000-\$9,000=\$1,000, we solve

$$R = \frac{\$9,000 - \$10,000}{\$10,000} = -0.10.$$

If the monthly return on Microsoft is -10% or less, then end-of-month wealth will be \$9,000 or less. Notice that $R = -0.10$ is the 6.7% quantile of the distribution of R:

$$\Pr(R < -0.10) = 0.067.$$

Question (iii) can be answered in two equivalent ways. First, we use $R \sim N(0.05, (0.10)^2)$ and solve for the 5% quantile of Microsoft Stock using Equation 2.23[5]:

$$\begin{aligned}
&\Pr(R < q_{.05}^R) = 0.05 \\
&\qquad \Rightarrow q_{.05}^R = \mu_R + \sigma_R \cdot z_{.05} = 0.05 + 0.10 \cdot (-1.645) = -0.114.
\end{aligned}$$

That is, with 5% probability the return on Microsoft stock is -11.4% or less. Now, if the return on Microsoft stock is -11.4% the loss in investment value that occurs with 5% probability is

$$\text{VaR}_{0.05} = W_0 \cdot q_{.05}^R = \$10,000 \cdot (0.114) = -\$1,144.$$

Since losses are typically reported as positive numbers, $\text{VaR}_{0.05} = \$1,144$ is the 5% VaR over the next month on the \$10,000 investment in Microsoft stock.

For the second method, use $W_1 \sim N(\$10,500, (\$1,000)^2)$ and solve for the 5% quantile of end-of-month wealth directly:

$$\Pr(W_1 < q_{.05}^{W_1}) = 0.05 \Rightarrow q_{.05}^{W_1} = \$8,856.$$

This corresponds to a loss of investment value of \$10,000-\$8,856=\$1,144. Hence, if W_0 represents the initial wealth and $q_{.05}^{W_1}$ is the 5% quantile of the distribution of W_1 then the 5% VaR is:

$$\text{VaR}_{.05} = W_0 - q_{.05}^{W_1} = \$10,000 - \$8,856 = \$1,144.$$

The following is the usual definition of VaR.

[5]Using R, $z_{.05} = \text{qnorm}(0.05) = -1.645$.

Definition 2.14 (Value-at-Risk)**.** If W_0 represents the initial wealth in dollars and q_α^R is the $\alpha \times 100\%$ quantile of distribution of the simple return R for small values of α, then the $\alpha \times 100\%$ VaR is defined as:

$$\mathrm{VaR}_\alpha = -W_0 \cdot q_\alpha^R. \tag{2.24}$$

In words, VaR_α represents the dollar loss that could occur with probability α. By convention, it is reported as a positive number (hence the multiplication by minus one as $q_\alpha^R < 0$ for small values of α).

2.1.8.1 Value-at-Risk calculations for continuously compounded returns

The above calculations illustrate how to calculate VaR using the normal distribution for simple returns. However, as argued in Example 31, the normal distribution may not be appropriate for characterizing the distribution of simple returns and may be more appropriate for characterizing the distribution of continuously compounded returns. Let R denote the simple monthly return, let $r = \ln(1+R)$ denote the continuously compounded return and assume that $r \sim N(\mu_r, \sigma_r^2)$. The $\alpha \times 100\%$ monthly VaR on an investment of $\$W_0$ is computed using the following steps:

1. Compute the $\alpha \cdot 100\%$ quantile, q_α^r, from the normal distribution for the continuously compounded return r:
$$q_\alpha^r = \mu_r + \sigma_r z_\alpha,$$
where z_α is the $\alpha \cdot 100\%$ quantile of the standard normal distribution.
2. Convert the continuously compounded return quantile, q_α^r, to a simple return quantile using the transformation:
$$q_\alpha^R = e^{q_\alpha^r} - 1$$
3. Compute VaR using the simple return quantile Equation 2.24.

Step 2 works because quantiles are preserved under monotonically increasing transformations of random variables.

Example 2.38 (Computing VaR from simple and continuously compounded returns using R)**.** Let R denote the simple monthly return on Microsoft stock and assume that $R \sim N(0.05, (0.10)^2)$. Consider an initial investment of $W_0 = \$10,000$. To compute the 1% and 5% VaR values over the next month use:

```
mu.R = 0.05
sd.R = 0.10
w0 = 10000
q.01.R = mu.R + sd.R*qnorm(0.01)
q.05.R = mu.R + sd.R*qnorm(0.05)
```

```r
VaR.01 = -q.01.R*w0
VaR.05 = -q.05.R*w0
VaR.01
#> [1] 1826
VaR.05
#> [1] 1145
```

Hence with 1% and 5% probability the loss over the next month is at least \$1,826 and \$1,145, respectively.

Let r denote the continuously compounded return on Microsoft stock and assume that $r \sim N(0.05, (0.10)^2)$. To compute the 1% and 5% VaR values over the next month use:

```r
mu.r = 0.05
sd.r = 0.10
q.01.R = exp(mu.r + sd.r*qnorm(0.01)) - 1
q.05.R = exp(mu.r + sd.r*qnorm(0.05)) - 1
VaR.01 = -q.01.R*w0
VaR.05 = -q.05.R*w0
VaR.01
#> [1] 1669
VaR.05
#> [1] 1082
```

Notice that when $1+R = e^r$ has a log-normal distribution, the 1% and 5% VaR values (losses) are slightly smaller than when R is normally distributed. This is due to the positive skewness of the log-normal distribution. ■

2.2 Bivariate Distributions

So far we have only considered probability distributions for a single random variable. In many situations, we want to be able to characterize the probabilistic behavior of two or more random variables simultaneously. For example, we might want to know about the probabilistic behavior of the returns on a two asset portfolio. In this section, we discuss bivariate distributions and important concepts related to the analysis of two random variables.

2.2.1 Discrete random variables

Let X and Y be discrete random variables with sample spaces S_X and S_Y, respectively. The likelihood that X and Y takes values in the joint sample space $S_{XY} = S_X \times S_Y$ (Cartesian product of the individual sample spaces) is determined by the *joint probability distribution* $p(x, y) = \Pr(X = x, Y = y)$. The function $p(x, y)$ satisfies:

- (i) $p(x, y) > 0$ for $x, y \in S_{XY}$;
- (ii) $p(x, y) = 0$ for $x, y \notin S_{XY}$;
- (iii) $\sum_{x,y \in S_{XY}} p(x, y) = \sum_{x \in S_X} \sum_{y \in S_Y} p(x, y) = 1$.

Example 2.39 (Bivariate discrete distribution for stock returns)**.** Let X denote the monthly return (in percent) on Microsoft stock and let Y denote the monthly return on Starbucks stock. For simplicity suppose that the sample spaces for X and Y are $S_X = \{0, 1, 2, 3\}$ and $S_Y = \{0, 1\}$ so that the random variables X and Y are discrete. The joint sample space is the two dimensional grid $S_{XY} = \{(0, 0), (0, 1), (1, 0), (1, 1), (2, 0), (2, 1), (3, 0), (3, 1)\}$. Table **?@tbl-Table-bivariateDistn** illustrates the joint distribution for X and Y. From the table, $p(0, 0) = \Pr(X = 0, Y = 0) = 1/8$. Notice that the sum of all the entries in the table sum to unity. ■

Table 2.3: Discrete bivariate distribution for Microsoft and Starbucks stock prices.

		Y		
	%	**0**	**1**	$\Pr(X)$
	0	1/8	0	1/8
X	1	2/8	1/8	3/8
	2	1/8	2/8	3/8
	3	0	1/8	1/8
	$\Pr(Y)$	**4/8**	**4/8**	**1**

2.2.1.1 Marginal Distributions

The joint probability distribution tells us the probability of X and Y occurring together. What if we only want to know about the probability of X occurring, or the probability of Y occurring?

Example 2.40 (Find $\Pr(X = 0)$ and $\Pr(Y = 1)$ from joint distribution)**.** Consider the joint distribution in Table Table 2.3. What is $\Pr(X = 0)$ regardless of the value of Y? Now X can occur if $Y = 0$ or if $Y = 1$ and since these two events are mutually exclusive we have that

$$\Pr(X = 0) = \Pr(X = 0, Y = 0) + \Pr(X = 0, Y = 1) = 0 + 1/8 = 1/8.$$

Notice that this probability is equal to the horizontal (row) sum of the probabilities in the table at $X = 0$. We can find $\Pr(Y = 1)$ in a similar fashion:

$$\begin{aligned}\Pr(Y = 1) &= \Pr(X = 0, Y = 1) + \Pr(X = 1, Y = 1) + \Pr(X = 2, Y = 1) + \Pr(X = 3, Y = 1) \\ &= 0 + 1/8 + 2/8 + 1/8 = 4/8.\end{aligned}$$

This probability is the vertical (column) sum of the probabilities in the table at $Y = 1$. ■

Definition 2.15 (Marginal distribution)**.** The probability $\Pr(X = x)$ is called the *marginal probability* of X and is given by:

$$\Pr(X = x) = \sum_{y \in S_Y} \Pr(X = x, Y = y). \tag{2.25}$$

Similarly, the marginal probability of $Y = y$ is given by:

$$\Pr(Y = y) = \sum_{x \in S_X} \Pr(X = x, Y = y). \tag{2.26}$$

Example 2.41 (Marginal probabilities of discrete bivariate distribution)**.** The marginal probabilities of $X = x$ are given in the last column of Table Table 2.3, and the marginal probabilities of $Y = y$ are given in the last row of Table Table 2.3. Notice that these probabilities sum to 1. For future reference we note that $E[X] = 3/2, \mathrm{var}(X) = 3/4, E[Y] = 1/2$, and $\mathrm{var}(Y) = 1/4$. ■

2.2.1.2 Conditional Distributions

For random variables in Table Table 2.3, suppose we know that the random variable Y takes on the value $Y = 0$. How does this knowledge affect the likelihood that X takes on the values 0, 1, 2 or 3? For example, what is the probability that $X = 0$ *given that* we know $Y = 0$? To find this probability, we use Bayes' law and compute the *conditional probability*:

$$\Pr(X = 0|Y = 0) = \frac{\Pr(X = 0, Y = 0)}{\Pr(Y = 0)} = \frac{1/8}{4/8} = 1/4.$$

The notation $\Pr(X = 0|Y = 0)$ is read as "the probability that $X = 0$ given that $Y = 0$" . Notice that $\Pr(X = 0|Y = 0) = 1/4 > \Pr(X = 0) = 1/8$. Hence, knowledge that $Y = 0$ increases the likelihood that $X = 0$. Clearly, X *depends* on Y.

Now suppose that we know that $X = 0$. How does this knowledge affect the probability that $Y = 0$? To find out we compute:

$$\Pr(Y = 0|X = 0) = \frac{\Pr(X = 0, Y = 0)}{\Pr(X = 0)} = \frac{1/8}{1/8} = 1.$$

Notice that $\Pr(Y=0|X=0)=1>\Pr(Y=0)=1/2$. That is, knowledge that $X=0$ makes it certain that $Y=0$.

The conditional probability that $X=x$ given that $Y=y$ (provided $\Pr(Y=y)\neq 0$) is,

$$p(x|y)=\Pr(X=x|Y=y)=\frac{\Pr(X=x,Y=y)}{\Pr(Y=y)}=\frac{p(x,y)}{p(y)}, \tag{2.27}$$

and the conditional probability that $Y=y$ given that $X=x$ (provided $\Pr(X=x)\neq 0$) is,

$$p(y|x)=\Pr(Y=y|X=x)=\frac{\Pr(X=x,Y=y)}{\Pr(X=x)}=\frac{p(x,y)}{p(x)}. \tag{2.28}$$

Example 2.42 (Conditional distributions)**.** For the bivariate distribution in Table Table 2.3, the conditional probabilities along with marginal probabilities are summarized in Tables Table 2.4 and Table 2.5. Notice that the marginal distribution of X is centered at $x=3/2$ whereas the conditional distribution of $X|Y=0$ is centered at $x=1$ and the conditional distribution of $X|Y=1$ is centered at $x=2$. ■

Table 2.4: Conditional probability distribution of X from bivariate discrete distribution.

x	$\Pr(X=x)$	$\Pr(X\|Y=0)$	$\Pr(X\|Y=1)$
0	1/8	2/8	0
1	3/8	4/8	2/8
2	3/8	4/8	4/8
3	1/8	0	2/8

Table 2.5: Conditional distribution of Y from bivariate discrete distribution.

y	$\Pr(Y=y)$	$\Pr(Y\|X=0)$	$\Pr(Y\|X=1)$	$\Pr(Y\|X=2)$	$\Pr(Y\|X=3)$
0	1/2	1	2/3	1/3	0
1	1/2	0	1/3	2/3	1

2.2.1.3 Conditional expectation and conditional variance

Just as we defined shape characteristics of the marginal distributions of X and Y, we can also define shape characteristics of the conditional distributions of $X|Y=y$ and $Y|X=x$. The most important shape characteristics are the *conditional expectation* (*conditional mean*) and the *conditional variance*. The conditional mean of $X|Y=y$ is denoted by $\mu_{X|Y=y}=E[X|Y=y]$, and the conditional mean of $Y|X=x$ is denoted by $\mu_{Y|X=x}=E[Y|X=x]$.

Definition 2.16 (Conditional expectation)**.** For discrete random variables X and Y, the conditional expectations are defined as:

$$\mu_{X|Y=y} = E[X|Y=y] = \sum_{x \in S_X} x \cdot \Pr(X=x|Y=y), \tag{2.29}$$

$$\mu_{Y|X=x} = E[Y|X=x] = \sum_{y \in S_Y} y \cdot \Pr(Y=y|X=x). \tag{2.30}$$

Similarly, the conditional variance of $X|Y=y$ is denoted by $\sigma^2_{X|Y=y} = \text{var}(X|Y=y)$ and the conditional variance of $Y|X=x$ is denoted by $\sigma^2_{Y|X=x} = \text{var}(Y|X=x)$.

Definition 2.17 (Conditional variance)**.** For discrete random variables X and Y, the conditional variances are defined as:

$$\sigma^2_{X|Y=y} = \text{var}(X|Y=y) = \sum_{x \in S_X} (x - \mu_{X|Y=y})^2 \cdot \Pr(X=x|Y=y), \tag{2.31}$$

$$\sigma^2_{Y|X=x} = \text{var}(Y|X=x) = \sum_{y \in S_Y} (y - \mu_{Y|X=x})^2 \cdot \Pr(Y=y|X=x). \tag{2.32}$$

The conditional volatilities are defined as:

$$\sigma_{X|Y=y} = \sqrt{\sigma^2_{X|Y=y}}, \tag{2.33}$$

$$\sigma_{Y|X=x} = \sqrt{\sigma^2_{Y|X=x}}. \tag{2.34}$$

Then conditional mean is the center of mass of the conditional distribution, and the conditional variance and volatility measure the spread of the conditional distribution about the conditional mean.

Example 2.43 (Compute conditional expectation and conditional variance)**.** For the random variables in Table Table 2.3, we have the following conditional moments for X:

$$\begin{aligned}
E[X|Y=0] &= 0 \cdot 1/4 + 1 \cdot 1/2 + 2 \cdot 1/4 + 3 \cdot 0 = 1, \\
E[X|Y=1] &= 0 \cdot 0 + 1 \cdot 1/4 + 2 \cdot 1/2 + 3 \cdot 1/4 = 2, \\
\text{var}(X|Y=0) &= (0-1)^2 \cdot 1/4 + (1-1)^2 \cdot 1/2 + (2-1)^2 \cdot 1/2 + (3-1)^2 \cdot 0 = 1/2, \\
\text{var}(X|Y=1) &= (0-2)^2 \cdot 0 + (1-2)^2 \cdot 1/4 + (2-2)^2 \cdot 1/2 + (3-2)^2 \cdot 1/4 = 1/2.
\end{aligned}$$

Compare these values to $E[X] = 3/2$ and $\text{var}(X) = 3/4$. Notice that $\text{var}(X|Y) = 1/2 < \text{var}(X) = 3/4$ so that conditioning on information reduces variability. Also, notice that as y increases, $E[X|Y=y]$ increases.

For Y, similar calculations gives:

$$\begin{aligned}
&E[Y|X=0] = 0, E[Y|X=1] = 1/3, E[Y|X=2] = 2/3, E[Y|X=3] = 1, \\
&\text{var}(Y|X=0) = 0, \text{var}(Y|X=1) = 0.2222, \text{var}(Y|X=2) = 0.2222, \text{var}(Y|X=3) = 0.
\end{aligned}$$

Compare these values to $E[Y] = 1/2$ and $\text{var}(Y) = 1/4$. Notice that as x increases $E[Y|X=x]$ increases. ■

2.2.1.4 Conditional expectation and the regression function

Consider the problem of predicting the value Y given that we know $X = x$. A natural predictor to use is the conditional expectation $E[Y|X = x]$. In this prediction context, the conditional expectation $E[Y|X = x]$ is called the *regression function*. The graph with $E[Y|X = x]$ on the vertical axis and x on the horizontal axis gives the *regression line*. The relationship between Y and the regression function may be expressed using the trivial identity:

$$
\begin{aligned}
Y &= E[Y|X = x] + Y - E[Y|X = x] \\
&= E[Y|X = x] + \varepsilon,
\end{aligned}
$$

where $\varepsilon = Y - E[Y|X]$ is called the *regression error*.

Example 2.44 (Regression line for bivariate discrete distribution)**.** For the random variables in Table Table 2.3, the regression line is plotted in Figure **?@fig-figProbReviewDiscreteRegression**. Notice that there is a linear relationship between $E[Y|X = x]$ and x. When such a linear relationship exists we call the regression function a *linear regression*. Linearity of the regression function, however, is not guaranteed. It may be the case that there is a non-linear (e.g., quadratic) relationship between $E[Y|X = x]$ and x. In this case, we call the regression function a *non-linear regression*. ■

2.2.1.5 Law of Total Expectations

Before Y is observed, $E[X|Y]$ is a random variable because its value depends on the value of Y which is a random variable. Similarly, $E[Y|X]$ is also a random variable. Since $E[X|Y]$ and $E[Y|X]$ are random variables we can consider their mean values. To illustrate, for the random variables in Table **?@tbl-Table-bivariateDistn** notice that:

$$
\begin{aligned}
E[X] &= \sum_{y \in S_Y} E[X|Y = y] \cdot \Pr(Y = y) \\
&= E[X|Y = 0] \cdot \Pr(Y = 0) + E[X|Y = 1] \cdot \Pr(Y = 1) \\
&= 1 \cdot 1/2 + 2 \cdot 1/2 = 3/2,
\end{aligned}
$$

and,

$$
\begin{aligned}
E[Y] &= \sum_{x \in S_X} E[Y|X = x] \cdot \Pr(X = x) \\
&= E[Y|X = 0] \cdot \Pr(X = 0) + E[Y|X = 1] \cdot \Pr(X = 1) \\
&+ E[Y|X = 2] \cdot \Pr(X = 2) + E[Y|X = 3] \cdot \Pr(X = 3) = 1/2.
\end{aligned}
$$

This result is known as the *law of total expectations*.

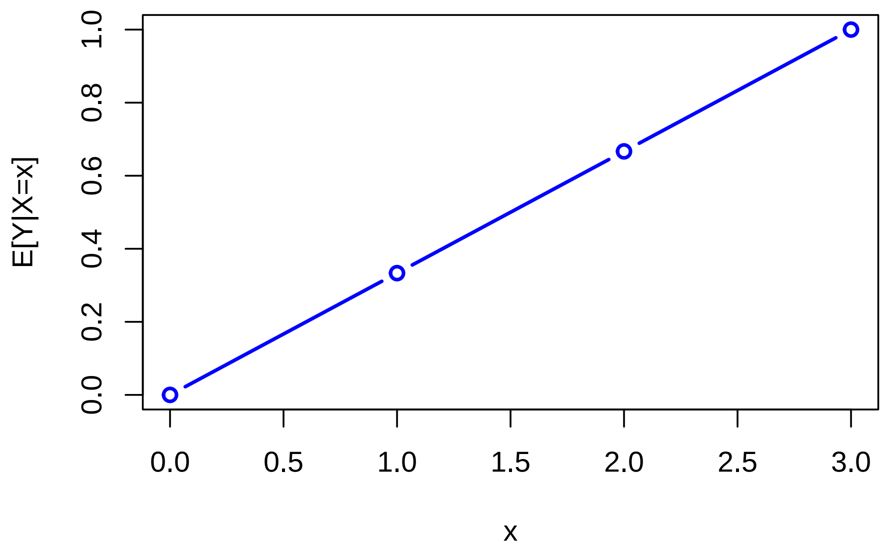


Figure 2.11: Regression function $E[Y|X = x]$ from discrete bivariate distribution.

Proposition 2.2. *For two random variables X and Y (discrete or continuous) we have,*

$$E[X] = E[E[X|Y]],$$
$$E[Y] = E[E[Y|X]],$$

where the expectation on the first line is taken with respect to the random variable Y, and the the expectation on the second line is taken with respect to the random variable X.

In words, the law of total expectations says that the mean of the conditional mean is the unconditional mean.

2.2.2 Bivariate distributions for continuous random variables

Let X and Y be continuous random variables defined over the real line. We characterize the joint probability distribution of X and Y using the joint probability function (pdf) $f(x, y)$ such that $f(x, y) \geq 0$ and,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)\ dx\ dy = 1.$$

The three-dimensional plot of the joint probability distribution gives a probability surface whose total volume is unity. To compute joint probabilities of $x_1 \leq X \leq x_2$ and $y_1 \leq Y \leq y_2$, we need to find the volume under the probability surface over the grid where the intervals $[x_1, x_2]$ and $[y_1, y_2]$ overlap. Finding this volume requries solving the double integral:

$$\Pr(x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2) = \int_{x_1}^{x_2}\int_{y_1}^{y_2} f(x, y)\ dx\ dy.$$

Example 2.45 (Standard bivariate normal distribution)**.** A *standard bivariate normal* pdf for X and Y has the form:

$$f(x, y) = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)},\ -\infty \leq x,\ y \leq \infty \tag{2.35}$$

and has the shape of a symmetric bell (think Liberty Bell) centered at $x = 0$ and $y = 0$. To find $\Pr(-1 < X < 1, -1 < Y < 1)$ we must solve:

$$\int_{-1}^{1}\int_{-1}^{1} \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}\ dx\ dy,$$

which, unfortunately, does not have an analytical solution. Numerical approximation methods are required to evaluate the above integral. The function `pmvnorm()` in the R package **mvtnorm** can be used to evaluate areas under the bivariate standard normal surface. To compute $\Pr(-1 < X < 1, -1 < Y < 1)$ use:

```r
pmvnorm(lower=c(-1, -1), upper=c(1, 1))
#> [1] 0.466
#> attr(,"error")
#> [1] 1e-15
#> attr(,"msg")
#> [1] "Normal Completion"
```

Here, $\Pr(-1 < X < 1, -1 < Y < 1) = 0.4661$. The attribute `error` gives the estimated absolute error of the approximation, and the attribute `message` tells the status of the algorithm used for the approximation. See the online help for `pmvnorm` for more details. ■

2.2.2.1 Marginal and conditional distributions

The marginal pdf of X is found by integrating y out of the joint pdf $f(x,y)$ and the marginal pdf of Y is found by integrating x out of the joint pdf $f(x,y)$:

$$f(x) = \int_{-\infty}^{\infty} f(x,y)\ dy, \tag{2.36}$$

$$f(y) = \int_{-\infty}^{\infty} f(x,y)\ dx. \tag{2.37}$$

The conditional pdf of X given that $Y = y$, denoted $f(x|y)$, is computed as

$$f(x|y) = \frac{f(x,y)}{f(y)}, \tag{2.38}$$

and the conditional pdf of Y given that $X = x$ is computed as,

$$f(y|x) = \frac{f(x,y)}{f(x)}. \tag{2.39}$$

The conditional means are computed as

$$\mu_{X|Y=y} = E[X|Y=y] = \int x \cdot p(x|y)\ dx, \tag{2.40}$$

$$\mu_{Y|X=x} = E[Y|X=x] = \int y \cdot p(y|x)\ dy, \tag{2.41}$$

and the conditional variances are computed as,

$$\sigma^2_{X|Y=y} = \mathrm{var}(X|Y=y) = \int (x - \mu_{X|Y=y})^2 p(x|y)\ dx, \tag{2.42}$$

$$\sigma^2_{Y|X=x} = \mathrm{var}(Y|X=x) = \int (y - \mu_{Y|X=x})^2 p(y|x)\ dy. \tag{2.43}$$

Example 2.46 (Conditional and marginal distributions from bivariate standard normal)**.** Suppose X and Y are distributed bivariate standard normal. To find the marginal distribution of X we use **?@eq-marginal-x-continuous** and solve:

$$f(x) = \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}\ dy = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2}\ dy = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}.$$

Hence, the marginal distribution of X is standard normal. Similiar calculations show that the marginal distribution of Y is also standard normal. To find the conditional distribution of $X|Y=y$ we use Equation 2.38 and solve:

$$\begin{aligned} f(x|y) = \frac{f(x,y)}{f(x)} &= \frac{\frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2}} \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x^2+y^2)+\frac{1}{2}y^2} \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \\ &= f(x) \end{aligned}$$

So, for the standard bivariate normal distribution $f(x|y) = f(x)$ which does not depend on y. Similar calculations show that $f(y|x) = f(y)$. ■

2.2.3 Independence

Let X and Y be two discrete random variables. Intuitively, X is independent of Y if knowledge about Y does not influence the likelihood that $X = x$ for all possible values of $x \in S_X$ and $y \in S_Y$. Similarly, Y is independent of X if knowledge about X does not influence the likelihood that $Y = y$ for all values of $y \in S_Y$. We represent this intuition formally for discrete random variables as follows.

Definition 2.18 (Independence of discrete random variables)**.** Let X and Y be discrete random variables with sample spaces S_X and S_Y, respectively. X and Y are independent random variables if and only if (iff):

$$\begin{aligned} \Pr(X = x|Y = y) = \Pr(X = x),\ \text{for all } x \in S_X, y \in S_Y \\ \Pr(Y = y|X = x) = \Pr(Y = y),\ \text{for all } x \in S_X, y \in S_Y \end{aligned}$$

Example 2.47 (Check independence of bivariate discrete random variables)**.** For the data in Table Table 2.3, we know that $\Pr(X = 0|Y = 0) = 1/4 \neq \Pr(X = 0) = 1/8$ so X and Y are not independent. ■

Proposition 2.3 (Independence of discrete random variables)**.** *Let X and Y be discrete random variables with sample spaces S_X and S_Y, respectively. X and Y are independent iff:*

$$\Pr(X = x, Y = y) = \Pr(X = x) \cdot \Pr(Y = y), \text{ for all } x \in S_X, y \in S_Y.$$

Intuition for the above result follows from:

$$\begin{aligned}
\Pr(X = x|Y = y) \quad &= \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)} = \frac{\Pr(X = x) \cdot \Pr(Y = y)}{\Pr(Y = y)} \\
&= \Pr(X = x), \\
\Pr(Y = y|X = x) \quad &= \frac{\Pr(X = x, Y = y)}{\Pr(X = x)} = \frac{\Pr(X = x) \cdot \Pr(Y = y)}{\Pr(X = x)} \\
&= \Pr(Y = y),
\end{aligned}$$

which shows that X and Y are independent.

For continuous random variables, we have the following definition of independence.

Definition 2.19 (Independence of continuous random variables)**.** Let X and Y be continuous random variables with marginal pdfs $f(x)$ and $f(y)$, and conditional pdfs $f(x|y)$ and $f(y|x)$. Then X and Y are independent iff:

$$\begin{aligned}
f(x|y) &= f(x), \text{ for } -\infty < x, y < \infty, \\
f(y|x) &= f(y), \text{ for } -\infty < x, y < \infty.
\end{aligned}$$

As with discrete random variables, we have the following result for continuous random variables.

Proposition 2.4 (Independence of continuous random variables)**.** *Let X and Y be continuous random variables with marginal pdfs $f(x)$ and $f(y)$, and joint pdf $f(x, y)$. Then X and Y are independent iff:*

$$f(x, y) = f(x)f(y)$$

This result is extremely useful in practice because it gives us an easy way to compute the joint pdf for two independent random variables: we simply compute the product of the marginal distributions.

Example 2.48 (Constructing the bivariate standard normal distribution)**.** Let $X \sim N(0, 1)$, $Y \sim N(0, 1)$ and let X and Y be independent. Then,

$$f(x, y) = f(x)f(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}.$$

This result is a special case of the general bivariate normal distribution to be discussed in the next sub-section. ■

A useful property of the independence between two random variables is the following Proposition.

Proposition 2.5 (Property of independent random variables)**.** *If X and Y are independent then $g(X)$ and $h(Y)$ are independent for any functions $g(\cdot)$ and $h(\cdot)$.*

For example, if X and Y are independent then X^2 and Y^2 are also independent.

2.2.4 Covariance and correlation

Let X and Y be two discrete random variables. Figure Figure 2.12 displays several bivariate probability scatterplots (where equal probabilities are given on the dots). In panel (a) we see no linear relationship between X and Y. In panel (b) we see a perfect positive linear relationship between X and Y and in panel (c) we see a perfect negative linear relationship. In panel (d) we see a positive, but not perfect, linear relationship; in panel (e) we see a negative, but not perfect, linear relationship. Finally, in panel (f) we see no systematic linear relationship but we see a strong nonlinear (parabolic) relationship. The *covariance* between X and Y measures the *direction* of the linear relationship between the two random variables. The *correlation* between X and Y measures the *direction* and the *strength* of the linear relationship between the two random variables.

Let X and Y be two random variables with $E[X] = \mu_X$, $\text{var}(X) = \sigma_X^2, E[Y] = \mu_Y$ and $\text{var}(Y) = \sigma_Y^2$.

Definition 2.20 (Covariance)**.** The covariance between two random variables X and Y is given by:

$$\begin{aligned}
\sigma_{XY} &= \text{cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] \\
&= \sum_{x\in S_X}\sum_{y\in S_Y}(x-\mu_X)(y-\mu_Y)\Pr(X=x,Y=y) \text{ for discrete } X,Y, \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x-\mu_X)(y-\mu_Y)p(x,y)\;dx\;dy \text{ for continuous } X,Y.
\end{aligned}$$

Definition 2.21 (Correlation)**.** The correlation between two random variables X and Y is given by:

$$\rho_{XY} = \text{cor}(X,Y) = \frac{\text{cov}(X,Y)}{\sqrt{\text{var}(X)\text{var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X\sigma_Y}.$$

The correlation coefficient, ρ_{XY}, is a scaled version of the covariance.

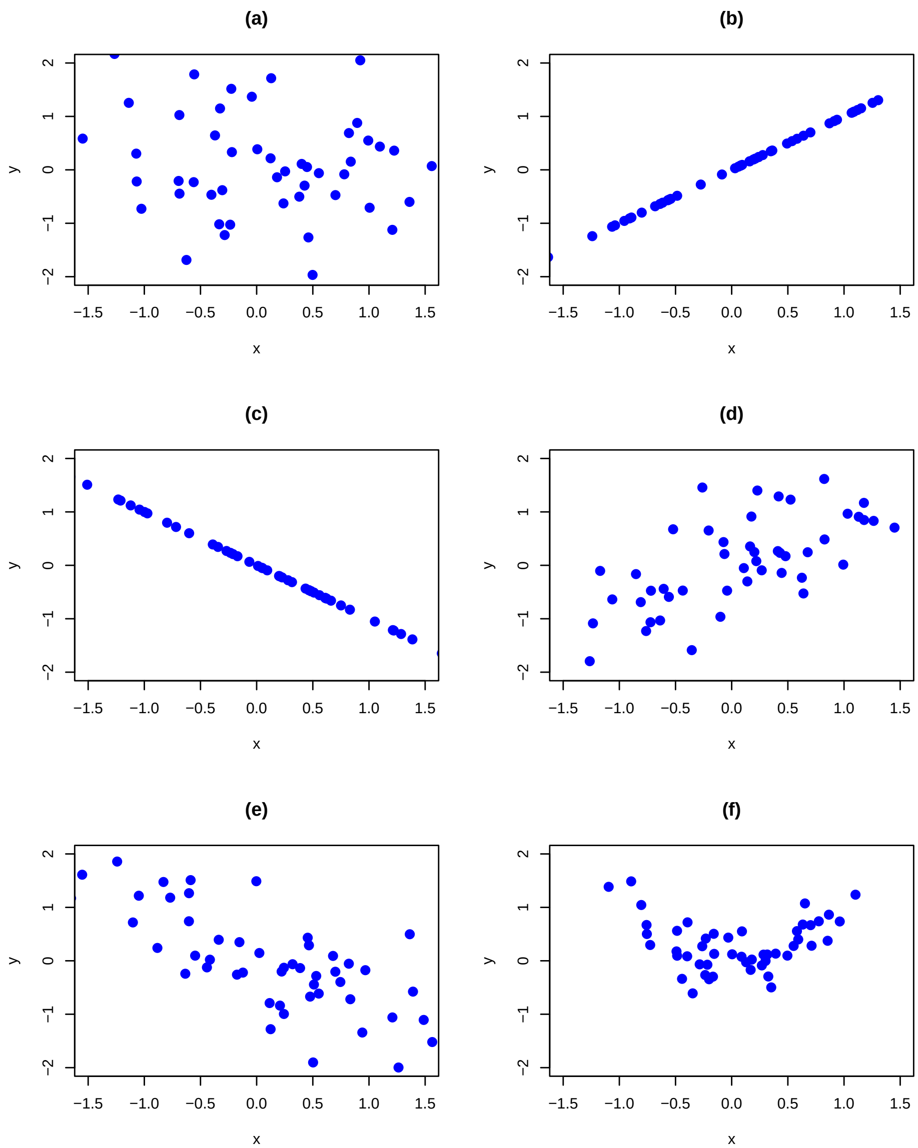


Figure 2.12: Probability scatterplots illustrating dependence between X and Y.

To see how covariance measures the direction of linear association, consider the probability scatterplot in Figure Figure 2.13. In the figure, each pair of points occurs with equal probability. The plot is separated into quadrants (right to left, top to bottom). In the first quadrant (black circles), the realized values satisfy $x < \mu_X, y > \mu_Y$ so that the product $(x - \mu_X)(y - \mu_Y) < 0$. In the second quadrant (blue squares), the values satisfy $x > \mu_X$ and $y > \mu_Y$ so that the product $(x - \mu_X)(y - \mu_Y) > 0$. In the third quadrant (red triangles), the values satisfy $x < \mu_X$ and $y < \mu_Y$ so that the product $(x - \mu_X)(y - \mu_Y) > 0$. Finally, in the fourth quadrant (green diamonds), $x > \mu_X$ but $y < \mu_Y$ so that the product $(x - \mu_X)(y - \mu_Y) < 0$. Covariance is then a probability weighted average all of the product terms in the four quadrants. For the values in Figure Figure 2.13, this weighted average is positive because most of the values are in the second and third quadrants.

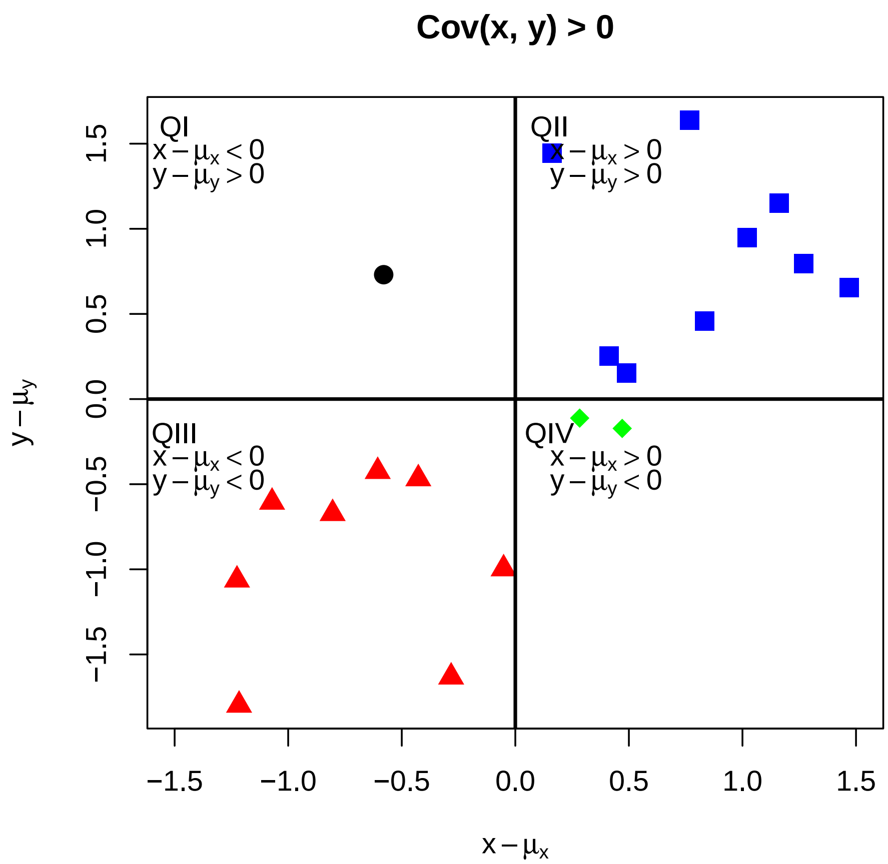


Figure 2.13: Probability scatterplot of discrete distribution with positive covariance. Each pair (X, Y) occurs with equal probability.

Example 2.49 (Calculate covariance and correlation for discrete random variables)**.** For the data in Table Table 2.3, we have:

$$\sigma_{XY} = \text{cov}(X,Y) = (0-3/2)(0-1/2)\cdot 1/8 + (0-3/2)(1-1/2)\cdot 0$$
$$+\cdots+(3-3/2)(1-1/2)\cdot 1/8 = 1/4$$
$$\rho_{XY} = \text{cor}(X,Y) = \frac{1/4}{\sqrt{(3/4)\cdot(1/2)}} = 0.577$$

■

2.2.4.1 Properties of covariance and correlation

Let X and Y be random variables and let a and b be constants. Some important properties of $\text{cov}(X,Y)$ are summarized in the following Proposition:

Proposition 2.6 (Properties of covariance)**.**

1. $\text{cov}(X,X) = \text{var}(X)$
2. $\text{cov}(X,Y) = \text{cov}(Y,X)$
3. $\text{cov}(X,Y) = E[XY] - E[X]E[Y] = E[XY] - \mu_X\mu_Y$
4. $\text{cov}(aX,bY) = a\cdot b\cdot \text{cov}(X,Y)$
5. *If X and Y are independent then $\text{cov}(X,Y) = 0$ (no association $\Longrightarrow$ no linear association). However, if $\text{cov}(X,Y) = 0$ then X and Y are not necessarily independent (no linear association $\nRightarrow$ no association).*
6. *If X and Y are jointly normally distributed and $\text{cov}(X,Y) = 0$, then X and Y are independent.*

The first two properties are intuitive. The third property results from expanding the definition of covariance:

$$\begin{aligned}
\text{cov}(X,Y) &= E[(X-\mu_X)(Y-\mu_Y)] \\
&= E\left[XY - X\mu_Y - \mu_X Y + \mu_X\mu_Y\right] \\
&= E[XY] - E[X]\mu_Y - \mu_X E[Y] + \mu_X\mu_Y \\
&= E[XY] - 2\mu_X\mu_Y + \mu_X\mu_Y \\
&= E[XY] - \mu_X\mu_Y
\end{aligned}$$

The fourth property follows from the linearity of expectations:

$$\begin{aligned}
\text{cov}(aX,bY) &= E[(aX-a\mu_X)(bY-b\mu_Y)] \\
&= a\cdot b\cdot E[(X-\mu_X)(Y-\mu_Y)] \\
&= a\cdot b\cdot \text{cov}(X,Y)
\end{aligned}$$

The fourth property shows that the value of $\text{cov}(X,Y)$ depends on the scaling of the random variables X and Y. By simply changing the scale of X or Y we can make $\text{cov}(X,Y)$ equal to any value that we want. Consequently, the numerical value of $\text{cov}(X,Y)$ is not informative about the strength of the linear association between X and Y. However, the sign of $\text{cov}(X,Y)$ is informative about the direction of linear association between X and Y.

The fifth property should be intuitive. Independence between the random variables X and Y means that there is no relationship, linear or nonlinear, between X and Y. However, the lack of a linear relationship between X and Y does not preclude a nonlinear relationship.

The last result illustrates an important property of the normal distribution: lack of covariance implies independence.

Some important properties of $\text{cor}(X,Y)$ are:

Proposition 2.7.

1. $-1 \leq \rho_{XY} \leq 1$
2. *If* $\rho_{XY} = 1$ *then* X *and* Y *are perfectly positively linearly related. That is,* $Y = aX + b$ *where* $a > 0$.
3. *If* $\rho_{XY} = -1$ *then* X *and* Y *are perfectly negatively linearly related. That is,* $Y = aX + b$ *where* $a < 0$.
4. *If* $\rho_{XY} = 0$ *then* X *and* Y *are not linearly related but may be nonlinearly related.*
5. $\text{cor}(aX, bY) = \text{cor}(X,Y)$ *if* $a > 0$ *and* $b > 0$*;* $\text{cor}(X,Y) = -\text{cor}(X,Y)$ *if* $a > 0, b < 0$ *or* $a < 0, b > 0$.

TBD: discuss the results briefly

2.2.5 Bivariate normal distributions

Let X and Y be distributed bivariate normal. The joint pdf is given by:

$$
\begin{aligned}
f(x,y) = &\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} \times \\
&\exp\left\{ -\frac{1}{2(1-\rho_{XY}^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - \frac{2\rho_{XY}(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} \right] \right\}
\end{aligned}
$$

where $E[X] = \mu_X$, $E[Y] = \mu_Y$, $\text{sd}(X) = \sigma_X$, $\text{sd}(Y) = \sigma_Y$, and $\rho_{XY} = \text{cor}(X,Y)$. The correlation coefficient ρ_{XY} describes the dependence between X and Y. If $\rho_{XY} = 0$ then the pdf collapses to the pdf of the standard bivariate normal distribution. In this case, X and Y are independent random variables since $f(x,y) = f(x)f(y)$.

It can be shown that the marginal distributions of X and Y are normal: $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim N(\mu_Y, \sigma_Y^2)$. In addition, it can be shown that the conditional distributions $f(x|y)$ and $f(y|x)$ are also normal with means given by:

$$\mu_{X|Y=y} = \alpha_X + \beta_X \cdot y,$$
$$\mu_{Y|X=x} = \alpha_Y + \beta_Y \cdot x,$$

where,

$$\alpha_X = \mu_X - \beta_X \mu_X,\ \beta_X = \sigma_{XY}/\sigma_Y^2,$$
$$\alpha_Y = \mu_Y - \beta_Y \mu_X,\ \beta_Y = \sigma_{XY}/\sigma_X^2,$$

and variances given by,

$$\sigma_{X|Y=y}^2 = \sigma_X^2 - \sigma_{XY}^2/\sigma_Y^2,$$
$$\sigma_{X|Y=y}^2 = \sigma_Y^2 - \sigma_{XY}^2/\sigma_X^2.$$

Notice that the conditional means (regression functions) are linear functions of x and y, respectively.

Example 2.50 (Expressing the bivariate normal distribution using matrix algebra)**.** The formula for the bivariate normal distribution Equation 2.44 is a bit messy. We can greatly simplify the formula by using matrix algebra. Define the 2×1 vectors $\mathbf{x} = (x, y)'$ and $\mu = (\mu_X, \mu_y)'$, and the 2×2 matrix:

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_Y^2 \end{pmatrix}$$

Then the bivariate normal distribution Equation 2.44 may be compactly expressed as:

$$f(\mathbf{x}) = \frac{1}{2\pi \det(\Sigma)^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)} \tag{2.44}$$

where,

$$\det(\Sigma) = \sigma_X^2\sigma_Y^2 - \sigma_{XY}^2 = \sigma_X^2\sigma_Y^2 - \sigma_X^2\sigma_Y^2\rho_{XY}^2 = \sigma_X^2\sigma_Y^2(1-\rho_{XY}^2)$$
$$\Sigma^{-1} = \frac{1}{\det(\Sigma)} \begin{pmatrix} \sigma_Y^2 & -\sigma_{XY} \\ -\sigma_{XY} & \sigma_X^2 \end{pmatrix}$$

■

Example 2.51 (Plotting the bivariate normal distribution)**.** The R package **mvtnorm** contains the functions `dmvnorm()`, `pmvnorm()`, and `qmvnorm()` which can be used to compute the bivariate normal pdf, cdf and quantiles, respectively. Plotting the bivariate normal distribution over a specified grid of x and y values in R can be done with the `persp()` function. First, we specify the parameter values for the joint distribution. Here, we choose $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$ and $\rho = 0.5$. We will use the `dmvnorm()` function to evaluate the joint pdf at these values. To do so, we must specify a covariance matrix of the form:

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_Y^2 \end{pmatrix} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}.$$

In R this matrix can be created using:

```
sigma = matrix(c(1, 0.5, 0.5, 1), 2, 2)
```

Next we specify a grid of x and y values between -3 and 3:

```
x = seq(-3, 3, length=25)
y = seq(-3, 3, length=25)
```

To evaluate the joint pdf over the two-dimensional grid we can use the `outer()` function:

```
# function to evaluate bivariate normal pdf on grid
bv.norm <- function(x, y, sigma) {
  z = cbind(x,y)
  return(dmvnorm(z, sigma=sigma))
}
# use outer function to evaluate pdf on 2D grid of x-y values
fxy = outer(x, y, bv.norm, sigma)
```

To create the 3D plot of the joint pdf, use the `persp()` function:

```
persp(x, y, fxy, theta=60, phi=30, expand=0.5, ticktype="detailed",
      zlab="f(x,y)",col="grey")
```

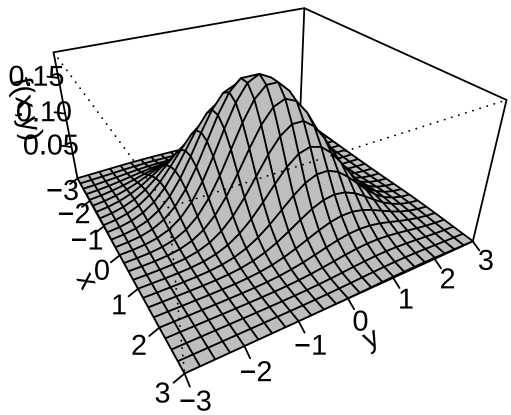


Figure 2.14: Bivariate normal pdf with $\mu_X = \mu_Y = 0$, $\sigma_X \sigma_Y = 1$ and $\rho = 0.5$.

The resulting plot is given in Figure Figure 2.14. ■

2.2.6 Expectation and variance of the sum of two random variables

Proposition 2.8 (Expectation and variance of the sum of two random variables)**.** *Let X and Y be two random variables with well defined means, variances and covariance and let a and b be constants. Then the following results hold:*

$$
\begin{aligned}
E[aX + bY] &= aE[X] + bE[Y] = a\mu_X + b\mu_Y \\
\mathrm{var}(aX + bY) &= a^2 \mathrm{var}(X) + b^2 \mathrm{var}(Y) + 2 \cdot a \cdot b \cdot \mathrm{cov}(X, Y) \\
&= a^2 \sigma_X^2 + b^2 \sigma_Y^2 + 2 \cdot a \cdot b \cdot \sigma_{XY}
\end{aligned}
$$

The first result states that the expected value of a linear combination of two random variables is equal to a linear combination of the expected values of the random variables. This result indicates that the expectation operator is a linear operator. In other words, expectation is additive. The second result states that variance of a linear combination of random variables is not a linear combination of the variances of the random variables. In particular, notice that covariance comes up as a term when computing the variance of the sum of two (not independent) random variables. Hence, the variance operator is not, in general, a linear operator. That is, variance, in general, is not additive.

It is instructive to go through the derivation of these results. Let X and Y be discrete random variables. Then,

$$\begin{aligned}
E[aX+bY] &= \sum_{x\in S_X}\sum_{y\in S_y}(ax+by)\Pr(X=x,Y=y) \\
&= \sum_{x\in S_X}\sum_{y\in S_y} ax\Pr(X=x,Y=y) + \sum_{x\in S_X}\sum_{y\in S_y} bx\Pr(X=x,Y=y) \\
&= a\sum_{x\in S_X} x\sum_{y\in S_y}\Pr(X=x,Y=y) + b\sum_{y\in S_y} y\sum_{x\in S_X}\Pr(X=x,Y=y) \\
&= a\sum_{x\in S_X} x\Pr(X=x) + b\sum_{y\in S_y} y\Pr(Y=y) \\
&= aE[X]+bE[Y] = a\mu_X + b\mu_Y.
\end{aligned}$$

The result for continuous random variables is similar. Effectively, the summations are replaced by integrals and the joint probabilities are replaced by the joint pdf.

Next, let X and Y be discrete or continuous random variables. Then, using the definition of variance we get:

$$\begin{aligned}
\mathrm{var}(aX+bY) &= E[(aX+bY-E[aX+bY])^2] \\
&= E[(aX+bY-a\mu_X-b\mu_Y)^2] \\
&= E[(a(X-\mu_X)+b(Y-\mu_Y))^2] \\
&= a^2E[(X-\mu_X)^2]+b^2E[(Y-\mu_Y)^2]+2\cdot a\cdot b\cdot E[(X-\mu_X)(Y-\mu_Y)] \\
&= a^2\mathrm{var}(X)+b^2\mathrm{var}(Y)+2\cdot a\cdot b\cdot \mathrm{cov}(X,Y).
\end{aligned}$$

2.2.6.1 Linear combination of two normal random variables

The following proposition gives an important result concerning a linear combination of normal random variables.

Proposition 2.9 (Linear combination of normal random variables)**.** *Let* $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim N(\mu_Y, \sigma_Y^2)$, $\sigma_{XY} = \mathrm{cov}(X,Y)$ *and* a *and* b *be constants. Define the new random variable* Z *as:*

$$Z = aX + bY.$$

Then,

$$Z \sim N(\mu_Z, \sigma_Z^2),$$

where $\mu_Z = a\mu_X + b\mu_Y$ *and* $\sigma_Z^2 = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}$.

This important result states that a linear combination of two normally distributed random variables is itself a normally distributed random variable. The proof of the result relies on the change of variables theorem from calculus and is omitted. Not all random variables have the nice property that their distributions are closed under addition.

Example 2.52 (Portfolio of two assets)**.** Consider a portfolio of two stocks A (Amazon) and B (Boeing) with investment shares x_A and x_B with $x_A + x_B = 1$. Let R_A and R_B denote the simple monthly returns on these assets, and assume that $R_A \sim N(\mu_A, \sigma_A^2)$ and $R_B \sim N(\mu_B, \sigma_B^2)$. Furthermore, let $\sigma_{AB} = \rho_{AB}\sigma_A\sigma_B = \mathrm{cov}(R_A, R_B)$. The portfolio return is $R_p = x_A R_A + x_B R_B$, which is a linear function of two random variables. Using the properties of linear functions of random variables, we have:

$$
\begin{aligned}
\mu_p &= E[R_p] = x_A E[R_A] + x_B E[R_B] = x_A\mu_A + x_B\mu_B \\
\sigma_p^2 &= \mathrm{var}(R_p) = x_A^2\mathrm{var}(R_A) + x_B^2\mathrm{var}(R_B) + 2x_A x_B \mathrm{cov}(R_A, R_B) \\
&= x_A^2\sigma_A^2 + x_B^2\sigma_B^2 + 2x_A x_B\sigma_{AB}.
\end{aligned}
$$

■

2.3 Multivariate Distributions

Multivariate distributions are used to characterize the joint distribution of a collection of N random variables $X_1, X_2, \ldots, X_N$ for $N > 1$. The mathematical formulation of this joint distribution can be quite complex and typically makes use of matrix algebra. Here, we summarize some basic properties of multivariate distributions without the use of matrix algebra. In chapter **?@sec-Matrix-Algebra-Review**, we show how matrix algebra can greatly simplify the description of multivariate distributions.

2.3.1 Discrete random variables

Let $X_1, X_2, \ldots, X_N$ be N discrete random variables with sample spaces $S_{X_1}, S_{X_2}, \ldots, S_{X_N}$. The likelihood that these random variables take values in the joint sample space $S_{X_1} \times S_{X_2} \times \cdots \times S_{X_N}$ is given by the joint probability function:

$$
p(x_1, x_2, \ldots, x_N) = \Pr(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N).
$$

For $N > 2$ it is not easy to represent the joint probabilities in a table like Table Table 2.3 or to visualize the distribution.

Marginal distributions for each variable X_i can be derived from the joint distribution as in Equation 2.25 by summing the joint probabilities over the other variables $j \neq i$. For example,

$$
p(x_1) = \sum_{x_2 \in S_{X_2}, \ldots, x_N \in S_{X_N}} p(x_1, x_2, \ldots, x_N).
$$

With N random variables, there are numerous conditional distributions that can be formed. For example, the distribution of X_1 given $X_2 = x_2, \ldots, X_N = x_N$ is determined using:

$$\Pr(X_1 = x_1 | X_2 = x_2, \ldots, X_N = x_N) = \frac{\Pr(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N)}{\Pr(X_2 = x_2, \ldots, X_N = x_N)}.$$

Similarly, the joint distribution of X_1 and X_2 given $X_3 = x_3, \ldots, X_N = x_N$ is given by:

$$\begin{aligned} &\Pr(X_1 = x_1, X_2 = x_2 | X_3 = x_3, \ldots, X_N = x_N) \\ &= \frac{\Pr(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N)}{\Pr(X_3 = x_3, \ldots, X_N = x_N)}. \end{aligned}$$

2.3.2 Continuous random variables

Let $X_1, X_2, \ldots, X_N$ be N continuous random variables each taking values on the real line. The joint pdf is a function $f(x_1, x_2, \ldots, x_N) \geq 0$ such that:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \ldots, x_N)\ dx_1\ dx_2 \cdots\ dx_N = 1.$$

Joint probabilities of $x_{11} \leq X_1 \leq x_{12}, x_{21} \leq X_2 \leq x_{22}, \ldots, x_{N1} \leq X_N \leq x_{N2}$ are computed by solving the integral equation:

$$\int_{x_{11}}^{x_{12}} \int_{x_{21}}^{x_{22}} \cdots \int_{x_{N1}}^{x_{N2}} f(x_1, x_2, \ldots, x_N)\ dx_1\ dx_2 \cdots\ dx_N. \tag{2.45}$$

For most multivariate distributions, the integral in Equation 2.45 cannot be solved analytically and must be approximated numerically.

The marginal pdf for x_i is found by integrating the joint pdf with respect to the other variables. For example, the marginal pdf for x_1 is found by solving:

$$f(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \ldots, x_N)\ dx_2 \cdots\ dx_N.$$

Conditional pdf for a single random variable or a collection of random variables are defined in the obvious way.

2.3.3 Independence

A collection of N random variables are independent if their joint distribution factors into the product of all of the marginal distributions:

$$\begin{aligned} p(x_1, x_2, \ldots, x_N) &= p(x_1)p(x_2) \cdots p(x_N) \text{ for } X_i \text{ discrete}, \\ f(x_1, x_2, \ldots, x_N) &= f(x_1)f(x_2) \cdots f(x_N) \text{ for } X_i \text{ continuous}. \end{aligned}$$

In addition, if N random variables are independent then any functions of these random variables are also independent.

2.3.4 Dependence concepts

In general, it is difficult to define dependence concepts for collections of more than two random variables. Dependence is typically only defined between pairwise random variables. Hence, covariance and correlation are also useful concepts when dealing with more than two random variables.

For N random variables $X_1, X_2, \ldots, X_N$, with mean values $\mu_i = E[X_i]$ and variances $\sigma_i^2 = \mathrm{var}(X_i)$, the pairwise covariances and correlations are defined as:

$$
\begin{aligned}
\mathrm{cov}(X_i, X_j) &= \sigma_{ij} = E[(X_i - \mu_i)(X_j - \mu_i)], \\
\mathrm{cov}(X_i, X_j) &= \rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j},
\end{aligned}
$$

for $i \neq j$. There are $N(N-1)/2$ pairwise covariances and correlations. Often, these values are summarized using matrix algebra in an $N \times N$ covariance matrix Σ and an $N \times N$ correlation matrix $\mathbf{C}$:

$$
\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1N} & \sigma_{2N} & \cdots & \sigma_N^2 \end{pmatrix}, \tag{2.46}
$$

$$
\mathbf{C} = \begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1N} \\ \rho_{12} & 1 & \cdots & \rho_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1N} & \rho_{2N} & \cdots & 1 \end{pmatrix}. \tag{2.47}
$$

These matrices are formally defined in chapter **?@sec-Matrix-Algebra-Review**.

2.3.5 Linear combinations of N random variables

Many of the results for manipulating a collection of random variables generalize in a straightforward way to the case of more than two random variables. The details of the generalizations are not important for our purposes. However, the following results will be used repeatedly throughout the book.

Proposition 2.10 (Linear combination of random variables)**.** *Let $X_1, X_2, \ldots, X_N$ denote a collection of N random variables (discrete or continuous) with means μ_i, variances σ_i^2 and covariances σ_{ij}. Define the new random variable Z as a linear combination:*

$$
Z = a_1 X_1 + a_2 X_2 + \cdots + a_N X_N
$$

where $a_1, a_2, \dots, a_N$ are constants. Then the following results hold:

$$
\begin{aligned}
\mu_Z = E[Z] &= a_1 E[X_1] + a_2 E[X_2] + \cdots + a_N E[X_N] \\
&= \sum_{i=1}^{N} a_i E[X_i] = \sum_{i=1}^{N} a_i \mu_i.
\end{aligned}
$$

$$
\begin{aligned}
\sigma_Z^2 = \mathrm{var}(Z) &= a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \cdots + a_N^2\sigma_N^2 \\
&\quad + 2a_1a_2\sigma_{12} + 2a_1a_3\sigma_{13} + \cdots + a_1a_N\sigma_{1N} \\
&\quad + 2a_2a_3\sigma_{23} + 2a_2a_4\sigma_{24} + \cdots + a_2a_N\sigma_{2N} \\
&\quad + \cdots + \\
&\quad + 2a_{N-1}a_N\sigma_{(N-1)N} \\
&\quad = \sum_{i=1}^{N} a_i^2\sigma_i^2 + 2\sum_{i=1}^{N}\sum_{j\neq i} a_i a_j \sigma_{ij}.
\end{aligned}
$$

The derivation of these results is very similar to the bivariate case and so is omitted. In addition, if all of the X_i are normally distributed then Z is also normally distributed with mean μ_Z and variance σ_Z^2 as described above.

The variance of a linear combination of N random variables contains N variance terms and $N(N-1)$ covariance terms. For $N = 2, 5, 10$ and 100 the number of covariance terms in $\mathrm{var}(Z)$ is 2, 20, 90 and $9,900$, respectively. Notice that when N is large there are many more covariance terms than variance terms in $\mathrm{var}(Z)$.

The expression for $\mathrm{var}(Z)$ is messy. It can be simplified using matrix algebra notation, as explained in detail in chapter **?@sec-Matrix-Algebra-Review**. To preview, define the $N \times 1$ vectors $\mathbf{X} = (X_1, \dots, X_N)'$ and $\mathbf{a} = (a_1, \dots, a_N)'$. Then $Z = \mathbf{a}'\mathbf{X}$ and $\mathrm{var}(Z) = \mathrm{var}(\mathbf{a}'\mathbf{X}) = \mathbf{a}'\Sigma\mathbf{a}$, where Σ is the $N \times N$ covariance matrix, which is much more compact than Equation 2.48.

Example 2.53 (Square-root-of-time rule for multi-period continuously compounded returns)**.** Let r_t denote the continuously compounded monthly return on an asset at times $t = 1, \dots, 12$. Assume that $r_1, \dots, r_{12}$ are independent and identically distributed (iid) $N(\mu, \sigma^2)$. Recall, the annual continuously compounded return is equal to the sum of twelve monthly continuously compounded returns: $r_A = r(12) = \sum_{t=1}^{12} r_t$. Since each monthly return is normally distributed, the annual return is also normally distributed. The mean of $r(12)$ is:

$$\begin{aligned}
E[r(12)] &= E\left[\sum_{t=1}^{12} r_t\right] \\
&= \sum_{t=1}^{12} E[r_t] \text{ (by linearity of expectation)}, \\
&= \sum_{t=1}^{12} \mu \text{ (by identical distributions)}, \\
&= 12 \cdot \mu.
\end{aligned}$$

Hence, the expected 12-month (annual) return is equal to 12 times the expected monthly return. The variance of $r(12)$ is:

$$\begin{aligned}
\text{var}(r(12)) &= \text{var}\left(\sum_{t=1}^{12} r_t\right) \\
&= \sum_{t=1}^{12} \text{var}(r_t) \text{ (by independence)}, \\
&= \sum_{j=0}^{11} \sigma^2 \text{ (by identical distributions)}, \\
&= 12 \cdot \sigma^2,
\end{aligned}$$

so that the annual variance is also equal to 12 times the monthly variance[6]. Hence, the annual standard deviation is $\sqrt{12}$ times the monthly standard deviation: $\text{sd}(r(12)) = \sqrt{12}\sigma$ (this result is known as the *square-root-of-time rule*). Therefore, $r(12) \sim N(12\mu, 12\sigma^2)$. ■

2.3.6 Covariance between linear combinations of random variables

Consider the linear combinations of two random variables:

$$\begin{aligned}
Y &= aX_1 + bX_2, \\
Z &= cX_3 + dX_4,
\end{aligned}$$

[6] This result often causes some confusion. It is easy to make the mistake and say that the annual variance is $(12)^2 = 144$ time the monthly variance. This result would occur if $r_A = 12r_t$, so that $\text{var}(r_A) = (12)^2\text{var}(r_t) = 144\text{var}(r_t)$.

where a, b, c and d are constants. The covariance between Y and Z is,

$$\begin{aligned}
\operatorname{cov}(Y, Z) &= \operatorname{cov}(aX_1 + bX_2, cX_3 + dX_4) \\
&= E[((aX_1 + bX_2) - (a\mu_1 + b\mu_2))((cX_3 + dX_4) - (c\mu_3 + d\mu_4))] \\
&= E[(a(X_1 - \mu_1) + b(X_2 - \mu_2))(c(X_3 - \mu_3) + d(X_4 - \mu_4))] \\
&= acE[(X_1 - \mu_1)(X_3 - \mu_3)] + adE[(X_1 - \mu_1)(X_4 - \mu_4)] \\
&+ bcE[(X_2 - \mu_2)(X_3 - \mu_3)] + bdE[(X_2 - \mu_2)(X_4 - \mu_4)] \\
&= ac\operatorname{cov}(X_1, X_3) + ad\operatorname{cov}(X_1, X_4) + bc\operatorname{cov}(X_2, X_3) + bd\operatorname{cov}(X_2, X_4).
\end{aligned}$$

Hence, covariance is additive for linear combinations of random variables. The result above extends in an obvious way to arbitrary linear combinations of random variables.

2.4 Further Reading: Review of Random Variables

This material in this chapter is a selected survey of topics, with applications to finance, typically covered in an introductory probability and statistics course using calculus. A good textbook for such a course is (DeGroot and Schervish 2011). Chapter 1 of (Carmona 2014) and the appendix of (Ruppert and Matteson 2015) give a slightly more advanced review than is presented here. Two slightly more advanced books, oriented towards economics students, are (Amemiya 1994) and (Goldberger 1991). Good introductory treatments of probability and statistics using R include (Dalgaard 2002) and (Verzani 2005). (Pfaff 2013) discusses some useful financial asset return distributions and their implementation in R.

2.5 Exercises

Exercise 2.1. Suppose X is a normally distributed random variable with mean 0.05 and variance $(0.10)^2$. That is, $X \sim N(0.05,\,(0,10)^2)$. Use the `pnorm()` and `qnorm()` functions in R to compute the following:

1. $\Pr(X \geq 0.10)$, $\Pr(X \leq -0.10)$.
2. $\Pr(-0.05 \leq X \leq 0.15)$.
3. 1% and 5% quantiles, $q^R_{0.01}$ and $q^R_{0.05}$.
4. 95% and 99% quantiles, $q^R_{0.95}$ and $q^R_{0.99}$.

Exercise 2.2. Let X denote the monthly return on Microsoft Stock and let Y denote the monthly return on Starbucks stock. Assume that $X \sim N(0.05,\,(0.10)^2)$ and $Y \sim N(0.025,\,(0.05)^2)$.

1. Using a grid of values between -0.25 and 0.35, plot the normal curves for X and Y. Make sure that both normal curves are on the same plot.
2. Plot the points $(\sigma_X, \mu_X) = (0.10, 0.05)$ and $(\sigma_Y, \mu_Y) = (0.05, 0.025)$ in an x-y plot.
3. Comment on the risk-return tradeoffs for the two stocks.

Exercise 2.3. Let X_1, X_2, Y_1, and Y_2 be random variables. Show that

$$\begin{aligned}\mathrm{Cov}(X_1 + X_2, Y_1 + Y_2) =& \mathrm{Cov}(X_1, Y_1) + \mathrm{Cov}(X_1, Y_2) + \\ & \mathrm{Cov}(X_2, Y_1) + \mathrm{Cov}(X_2, Y_2).\end{aligned}$$

Exercise 2.4. Let X be a random variable with $\mathrm{var}(X) < \infty$. Show that $\mathrm{var}(X) = E[(X - \mu_X)^2] = E[X^2] - \mu_X^2$.

Exercise 2.5. Let X and Y be a random variables such that $\mathrm{cov}(X, Y) < \infty$.

1. If $Y = aX + b$ where $a > 0$ show that $\rho_{XY} = \mathrm{cor}(X, Y) = 1$.
2. If $Y = aX + b$ where $a < 0$ show that $\rho_{XY} = \mathrm{cor}(X, Y) = -1$.

Exercise 2.6. Let R denote the simple monthly return on Microsoft stock and let W_0 denote initial wealth to be invested over the month. Assume that $R \sim N(0.04,\,(0.09)^2)$ and that $W_0 = \$100,000$. Determine the 1% and 5% Value-at-Risk (VaR) over the month on the investment. That is, determine the loss in investment value that may occur over the next month with 1% probability and with 5% probability.

Exercise 2.7. Let R_t denote the simple monthly return and assume $R_t \sim iid\, N(\mu, \sigma^2)$. Consider the 2-period return $R_t(2) = (1 + R_t)(1 + R_{t-1}) - 1$.

1. Assuming that $\text{cov}(R_t, R_{t-1}) = 0$, show that $E[R_t R_{t-1}] = \mu^2$. Hint: use $\text{cov}(R_t, R_{t-1}) = E[R_t R_{t-1}] - E[R_t]E[R_{t-1}]$.
2. Show that $E[R_t(2)] = (1 + \mu^2) - 1$.
3. Is $R_t(2)$ normally distributed? Justify your answer.

Exercise 2.8. Let r denote the continuously compounded monthly return on Microsoft stock and let W_0 denote initial wealth to be invested over the month. Assume that $r \sim iid\, N(0.04, (0.09)^2)$ and that $W_0 = \$100,000$.

1. Determine the 1% and 5% value-at-risk (VaR) over the month on the investment. That is, determine the loss in investment value that may occur over the next month with 1% probability and with 5% probability. (Hint: compute the 1% and 5% quantile from the Normal distribution for r and then convert the continuously compounded return quantile to a simple return quantile using the transformation $R = e^r - 1$.)
2. Let $r(12) = r_1 + r_2 + \cdots + r_{12}$ denote the 12-month (annual) continuously compounded return. Compute $E[r(12)]$, $\text{var}(r(12))$ and $\text{sd}(r(12))$. What is the probability distribution of $r(12)$?
3. Using the probability distribution for $r(12)$, determine the 1% and 5% value-at-risk (VaR) over the year on the investment.

Exercise 2.9. Let R_{VFINX} and R_{AAPL} denote the monthly simple returns on VFINX (Vanguard S&P 500 index) and AAPL (Apple stock). Suppose that $R_{VFINX} \sim i.i.d.\, N(0.013, (0.037)^2)$, $R_{AAPL} \sim i.i.d.\, N(0.028, (0.073)^2)$.

1. Sketch the normal distributions for the two assets on the same graph. Show the mean values and the ranges within 2 standard deviations of the mean. Which asset appears to be more risky?
2. Plot the risk-return tradeoff for the two assets. That is, plot the mean values of each asset on the y-axis and the standard deviations on the x-axis. Comment on the relationship.
3. Let $W_0 = \$1,000$ be the initial wealth invested in each asset. Compute the 1% monthly Value-at-Risk values for each asset. (Hint: $q^Z_{0.01} = -2.326$).
4. Continue with the above question, state in words what the 1% Value-at-Risk numbers represent (i.e., explain what 1% Value-at-Risk for a one month \$1,000 investment means).
5. The normal distribution can be used to characterize the probability distribution of monthly simple returns or monthly continuously compounded returns. What are two problems with using the normal distribution for simple returns? Given these two problems, why might it be better to use the normal distribution for continuously compounded returns?

Exercise 2.10. In this question, you will examine the chi-square and Student's t distributions. A chi-square random variable with n degrees of freedom, denoted $X \sim \chi^2_n$, is defined as $X = Z_1^2 + Z_2^2 + \cdots + Z_n^2$ where $Z_1, Z_2, \ldots, Z_n$ are $iid\, N(0, 1)$ random variables. Notice that X only takes positive values. A Student's t random variable with n degrees of freedom, denoted

$t \sim t_n$, is defined as $t = Z/\sqrt{X/n}$, where $Z \sim N(0,1)$, $X \sim \chi_n^2$ and Z is independent of X. The Student's t distribution is similar to the standard normal except that it has fatter tails.

1. On the same graph, plot the probability curves of chi-squared distributed random variables with 1, 2, 5 and 10 degrees of freedom. Use different colors and line styles for each curve. Hint: In R the density of the chi-square distribution is computed using the function `dchisq()`.
2. On the same graph, plot the probability curves of Student's t distributed random variables with 1, 2, 5 and 10 degrees of freedom. Also include the probability curve for the standard normal distribution. Use different colors and line styles for each curve. Hint: In R the density of the chi-square distribution is computed using the function `dt()`.

Exercise 2.11. Consider the following joint distribution of X and Y:

X/Y	1	2	3
1	0.1	0.2	0
2	0.1	0	0.2
3	0	0.1	0.3

1. Find the marginal distributions of X and Y. Using these distributions, compute $E[X]$, $\mathrm{var}(X)$, $\mathrm{sd}(X)$, $E[Y]$, $\mathrm{var}(Y)$ and $\mathrm{sd}(Y)$.
2. Compute $\mathrm{cov}(X,Y)$ and $\mathrm{cor}(X,Y)$.
3. Find the conditional distributions $\Pr(X|Y = y)$ for $y = 1,2,3$ and the conditional distributions $\Pr(Y|X = x)$ for $x = 1,2,3$.
4. Using the conditional distributions $\Pr(X|Y = y)$ for $y = 1,2,3$ and $\Pr(Y|X = x)$ for $x = 1,2,3$ compute $E[X|Y = y]$ and $E[Y|X = x]$.
5. Using the conditional distributions $\Pr(X|Y = y)$ for $y = 1,2,3$ and $\Pr(Y|X = x)$ for $x = 1,2,3$ compute $\mathrm{var}[X|Y = y]$ and $\mathrm{var}[Y|X = x]$.
6. Are X and Y independent? Fully justify your answer.

Exercise 2.12. Let X and Y be distributed bivariate normal with $\mu_X = 0.05$, $\mu_Y = 0.025$, $\sigma_X = 0.10$, and $\sigma_Y = 0.05$.

1. Using R package **mvtnorm** function `rmvnorm()`, simulate 100 observations from the bivariate normal distribution with $\rho_{XY} = 0.9$. Using the `plot()` function create a scatterplot of the observations and comment on the direction and strength of the linear association. Using the function `pmvnorm()`, compute the joint probability $Pr(X \leq, Y \leq 0)$.
2. Using R package **mvtnorm** function `rmvnorm()`, simulate 100 observations from the bivariate normal distribution with $\rho_{XY} = -0.9$. Using the `plot()` function create a scatterplot of the observations and comment on the direction and strength of the linear association. Using the function `pmvnorm()`, compute the joint probability $Pr(X \leq, Y \leq 0)$.

3. Using R package **mvtnorm** function `rmvnorm()`, simulate 100 observations from the bivariate normal distribution with $\rho_{XY} = 0$. Using the `plot()` function create a scatterplot of the observations and comment on the direction and strength of the linear association. Using the function `pmvnorm()`, compute the joint probability $Pr(X \leq, Y \leq 0)$.

References

Amemiya, T. 1994. *Introduction to Statistics and Econometrics*. Cambridge, MA.: Harvard University Press.

Ang, A. 2014. *Asset Management: A Systematic Approach to Factor Investing*. New York: Oxford University Press.

Bacon, C. 2008. *Practical Portfolio Performance Measurement and Attribution, Second Edition*. Chichester, England: John Wiley & Sons.

Carmona, R. 2014. *Statistical Analysis of Financial Data in r, Second Edition*. New York: Springer.

Christopherson, J. A., D. R. Carino, and W. E. Ferson. 2009. *Portfolio Performance Measurement and Benchmarking*. McGraw-Hill.

Dalgaard, P. 2002. *Introductory Statistics with r*. Springer.

DeGroot, M. H., and M. J. Schervish. 2011. *Probability and Statistics, 4th Edition*. Pearson.

Goldberger, A. S. 1991. *A Course in Econometrics*. Cambridge, MA.: Harvard University Press.

Pfaff, B. 2013. *Financial Risk Modelling and Portfolio Optimization with r*. Chichester, UK.: Wiley.

Qian, E. E. 2019. *Portfolio Rebalancing*. Boca Raton, FL: Chapman; Hall/CRC.

Ruppert, D., and D. S. Matteson. 2015. *Statistics and Data Analysis for Financial Engineering with r Examples*. New York: Springer.

Verzani, J. 2005. *Using r for Introductory Statistics*. Boca Raton: Chapman & Hall/CRC.